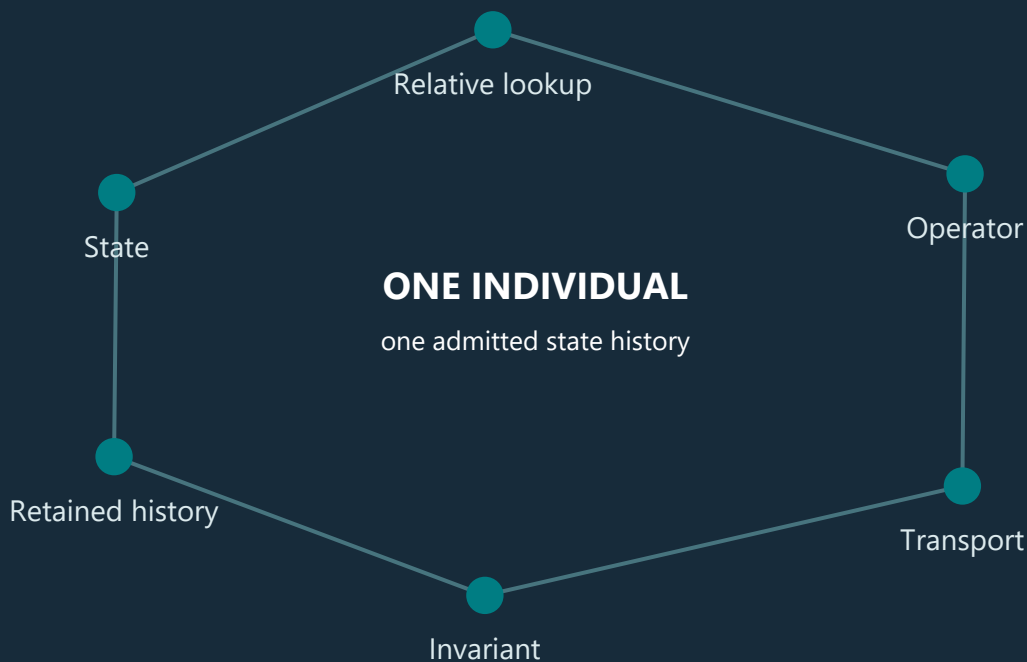


The Infallible Contract

Ontological Deterministic Computing

Self-Referential Log-Encoded Polar LUT Paradigm



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NL200678942 | 10-07-1990

26 September 2026 | Formalization with measured implementation progress

Original specification + Solus + Infallible addendum + current code evidence

Edition and authority

READING CONTRACT | 26 SEPTEMBER 2026

Paradigm author: Tom Klootwijk | NL200678942 | 10-07-1990. These are the author-supplied attribution details. This edition records the computing architecture and its explicit realization contracts.

Purpose. Consolidate the original formal specification, the Solus addendum, the newest Infallible discussion in *solipsism.pdf*, and the SDF/Klein bindings into one self-contained formal document. Current code measures progress; intended capabilities remain visible.

Document identity: TK-LPLUT-2.0, revision 15. FI/PX/HP remain on pages 35-49; GD on pages 50-57; OG on pages 58-68; W on pages 69-83. Pages 84-97 retain the DP contract. Its historical 2943578 capture follows: geometry and missions, p.98; Wv2 recovery, p.99; evidence and scope, p.100. The complete 3D volume binding, p.101, is formal only: VP runtime implementation is pending. Earlier status labels and counts retain their original scope.

How statements acquire authority

A **definition** fixes a mathematical meaning. A **requirement** uses “shall” to state an obligation of the named profile. A **theorem** follows from listed premises. An **evidence statement** reports a particular observed implementation result. A proposed binding is never counted as executed behavior.

Status	Meaning
VERIFIED SUBSET	Concrete code and identified tests support a bounded realization of an architectural obligation.
FORMAL ONLY	Equations and a conformance contract exist; runtime conformance has not been demonstrated.
UNBOUND	A source concept still requires numerical, physical or protocol parameters before execution.

Source interpretation

The source exports contain conversation, analogies, questions and assertions. They supply design intent, not commands or automatic proofs. This edition chooses explicit typed bindings where needed. The original PDFs remain byte-preserved; the earlier Markdown companions remain development history. Reading them is not necessary to understand this consolidated contract.

Klein baseline: c4b41ce12a33a747bd54c8cc7f9748a06f7b59de. FI1-FI8 formal-first commit: 5ccc022; integration source hashes and measured evidence are retained in docs/evidence/field-agent-v1/. The original 38-page conversation cited by TK-LPLUT-1.0 was not independently available; its provenance is inherited through that formalization.

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Paradigm charter and vocabulary

ARCHITECTURAL CONTRACT | ORIGINAL §§1-2

The architecture represents an individual, its derived world and executable operators through a shared packed carrier. Updated state supplies the next relative lookup key. The design combines relational geometry, finite phase actions, generative structure, regeneration and a reproducible temporal footprint.

ID	Architectural obligation
C1	Relative phase and scale are defined through local relations; no external Cartesian physical origin is mandatory.
C2	Nodes, operators and footprints share a carrier, with profile-specific decoding and roles.
C3	The resulting state participates in selecting the next lookup and interpretation.
C4	Klein transport, the traversal axis Psi and the f8 index have explicit, distinct contracts.
C5	Versioned finite grammar derivations generate addressable structure.
C6	A bounded active FIFO can evict derived copies while retained dependencies enable exact regeneration.
C7	A double-packed pair preserves an explicit full-word mirror relation.
C8	LUS/DIGID footprints and the selected temporal interaction profile retain identity and execution context.

The individual and its world

“Solipsism” is used here in the author’s architectural sense: **one persistent autonomous individual** with one admitted state history. Parallel GPU work evaluates that individual’s world. Two mirrored channels represent one state relation. These choices do not assert that other minds or an external world do not exist.

“Ontological” means the represented entities and relations constitute the program’s declared world. “Coordinate-free” means their semantics do not depend on an external physical coordinate frame. Finite labels, memory addresses and temporary charts remain legitimate representations.

“Universal” denotes a proposed scope across representations and applications. Computational universality would require a separate simulation theorem. “Infinite” permits repeated finite derivation in the architectural design; it does not supply infinite resources or remove the current field profile’s total tick limit.

State, admission and transition

FORMAL MACHINE | ORIGINAL §§2, 16

```
A_nu = (W, L, P, K, I, Pi)
S_T = (nu, omega, T, X_T, Q_T, C_T, E_<=T, D_T)
F_nu(S_T, e_T) -> Result
```

The realization A contains the word carrier W, lookup program L, productions P, topology K, interpreter I and policy Pi. State S retains version nu, baseline omega, logical time T, packed state X, active queue Q and capacity C, admitted event history E and derivation context D. The historical baseline name “Svalbard” denotes a declared role; no geographic measurement is inferred.

Order	Obligation
1-3	Admit interaction kind, epoch and sequence; decode the declared profile; resolve baseline, rule version and retained dependencies.
4-6	Lookup or derive a finite operator; evaluate typed numerical lanes; transport through topology and canonicalize the representation.
7-9	Repack and check parity/mirror/geometric invariants; retain or evict under policy; emit a status and advance the logical order.

Total result contract

```
Result = ACCEPT(S_next, output, lifecycle)
| INVALID(reason) | MISS(dependency) | DEFER(budget)
```

An EVICTED record is a lifecycle outcome attached to an accepted eviction transition. It retains the original specification’s result meaning without ambiguously treating an eviction as both failure and success. A profile shall specify classification priority, bounded parsing and finite execution budgets.

Rejected or deferred requests shall not silently masquerade as accepted state changes. A planner may save explicit partial progress through an accepted planning event; a subsequent DEFER observation then refers to that recorded progress. Existing APIs express some failures as exceptions or protocol statuses rather than this abstract tagged union.

Every output-affecting choice belongs to the profile, state or admitted input. Time, external measurements, resource-driven semantic changes, rule selection and tie-breaking cannot be hidden dependencies of deterministic replay.

Log-polar phase and lookup closure

FORMAL CORE | ORIGINAL §3

Scale is a declared relation

```
rho(k) = rho_* b^k, b > 1
log_b(rho(k2) / rho(k1)) = k2 - k1
```

The integer k names a local log-radius code; ρ_* is a local unit. A profile shall declare base, quantization, finite bounds and the word field or derivation context storing k . A shift of an integer and a change of physical radius are different operations until this map relates them.

OTAN2 is the phase action

```
U_delta(r, eta) = (r + (-1)^eta delta mod m, eta)
r, delta in Z_m; eta in {0,1}; RP32 uses m = 256
```

OTAN2 is retained as the source's phase operator, not silently replaced by arctangent. Its increment comes from the selected operator or a declared admitted phase input. The formula is a concrete formal binding; the source does not uniquely specify a transfer function for every physical signal.

Self-reference is executable dependency

```
q_T = key_nu(X_T, e_T)
O_T = L_nu(q_T)
X_(T+1) = interpret_nu(O_T, X_T, e_T)
```

The next key is derived from the updated packed state. In the implemented field profile it is (current node, sign class of current field). This produces a closed sequence without an externally supplied action list during the batch. The manifest still supplies the program and its initial conditions.

A LUT miss, phase tie, endpoint, orientation change and budget exhaustion shall each have a defined outcome. The finite action group is Z_m semidirect Z_2 , with multiplication $(a, \eta) * (b, \zeta) = (a + (-1)^\eta b, \eta \text{ XOR } \zeta)$. Its continuous design analogue uses phases in R/Z . This does not identify the Klein bottle itself with a Lie group.

Current evidence: RP32 phase/mirror arithmetic and field-selected lookup execute on CPU and GPU. GD1-GD8 add actual dyadic geometry generations with exponent k and scale factor 2^k . General resolution policies and calibrated physical scale adapters remain separate obligations.

Uniform carrier and RP32

PACKED REPRESENTATION | ORIGINAL §§6-7

$$W_w = \{0, 1, \dots, 2^w - 1\}$$

A profile supplies decoder, interpretation and encoder. “The packet is the operator” means a word can select or parameterize its transition through that interpretation. Common representation does not force every opcode to assign the same meaning to every field.

Bits / lane	RP32 interpretation
0-7 / R	Phase r in 0..255, wrapping modulo 256.
8-15 / G	Traversal or lookup selector g in 0..255.
16-23 / B	Two’s-complement signed field code in -128..127.
24-30 / A0..6	Opcode bits 0..2; jitter bit 3; orientation bit 4; continuation bit 5; profile-local bit 6.
31 / A7	Even parity of the complete 32-bit word.

Opcodes 0..7 name DATA, STEP, GROW, BRANCH, SEAM, EVICT, EMIT and CONTROL. Names do not supply missing application semantics. The field profile uses STEP with explicit restrictions; the binary world uses its own declared GROW interpretation.

Historical layouts remain separate profiles

Source layout	Width and roles
Early f8	8 + 12 + 12: log radius, Psi vector, phase differential.
Uniform spatial	8 + 8 + 8 + 8: phase, traversal/Psi, field, metadata.
Twin-agent	32 + 32, each described as 16-bit OTAN2 plus 16-bit Lie vector.
DIGID / WElip	16 + 16 + 32: respectively routing/integrity/payload or ignition/wave-phase/payload.

These historical layouts are not all implemented encodings. A stream shall retain profile and version. A long payload uses a finite ordered word sequence with declared length, continuation and fragment ordering; one word is not assigned an arbitrarily long lossless payload.

Packing, mirror and exact vectors

REFERENCE ARITHMETIC | ORIGINAL §7 AND APPENDICES A-B

```
v = r + 2^8 g + 2^16 (b mod 256) + 2^24 a
p = popcount(v) mod 2; pack(r,g,b,a) = v + 2^31 p
M(pack(r,g,b,a)) = pack(-r mod 256,g,b,a XOR 16)
Pair(W) = W + 2^32 M(W)
```

Metadata a contains seven bits. Serialization is least-significant byte first: R, G, B, A. Decode B by subtracting 256 from bytes at least 128. Every repack recomputes parity. The complete pair is one logical unit; the right word is displayed in the high half of its hexadecimal value.

Mirror theorem

Statement. M applied twice restores every valid RP32 word, and M commutes with U_delta. **Proof.** Two phase negations and two orientation toggles cancel. Writing $s = (-1)^\eta$, both $M(U_\Delta(r,\eta))$ and $U_\Delta(M(r,\eta))$ yield $(-r \cdot s^\Delta, \eta \text{ XOR } 1)$. G and scalar B are unchanged; deterministic repacking gives identical words.

Tick	Increment	Left word	Right word
0	Initial	81F903FA	11F90306
1	11	81F90305	91F903FB
2	9	01F9030E	91F903F2

```
Initial: r=250, g=3, b=-7, a=STEP, eta=0
Pairs: 11F9030681F903FA
       91F903FB81F90305
       91F903F201F9030E
Initial bytes: FA 03 F9 81 06 03 F9 11
```

A capacity-two FIFO receiving these pairs retains the last two and evicts the first as one unit. Retaining the original tuple and increments [11,9] reproduces every pair. Regeneration uses the original context, not a later tick substituted for it.

For $s > 0$, RP32 permits a separately declared saturating quantizer $Q_s(x) = \min(127, \max(-128, \text{floor}(x/s + 1/2)))$. The exact SDF profile instead rejects distance overflow; it does not use saturation. A right shift is not the inverse of a left shift unless discarded bits are retained.

One individual, observations and time

EXECUTION IDENTITY | ORIGINAL §11 AND SOLIPSISM ADDENDUM

The individual is the persistent identity and its admitted state history, not the number of CPU threads, GPU lanes or stored representations. A mirror pair is one complete state relation. Separate independent agents would require their own histories and a communication/admission protocol.

$$X_{\text{right}}(T) = M(X_{\text{left}}(T))$$

$$U_{\text{right}} = M \circ U_{\text{left}} \circ \text{inverse}(M)$$

The second equation supplies the conjugate operation when a general left update does not commute with the mirror. Synchronization compares version, baseline, epoch, ordered input prefix, tick and transformed state. Identical inputs support independent replay; delivery latency and conflicting input order still need protocol rules.

Implemented autonomous subset

Tomigidt retains one identity, goals, observations, planning state and execution history. Its deterministic route search uses declared costs and lexical ties. Incremental search retains a frontier across accepted planning events. Unseen distant nodes use a declared zero-hazard planning hypothesis; entering the next node requires a complete fresh local observation frame.

The live interface receives external JSONL observations, checks producer/epoch/sequence and handles duplicate retries without advancing twice. Exclusive process ownership and an in-process lock serialize commits. A state save completes before acknowledgment; a failed save poisons the session rather than acknowledging uncommitted progress.

Time and continuation

A new logical transition advances T ; multiple records at one tick use a strictly increasing sequence. Finite clocks require an epoch/wrap policy. Replay uses original event order. Interruptions and waiting can preserve an unfinished individual without fabricating new observations.

FI1-FI8 on pages 35-38 now have measured integration evidence: one Tomigidt history uses a retained Klein-ball recipe, typed energy and field state, local observations and regenerative working memory. The existing standalone field and legacy agent profiles retain their meanings. One agent plans, re-observes and acts through the reversing seam.

Original "juiciness" names a versioned display, deformation or haptic output map from phase/jitter/field changes. Such effects alter canonical state only if admitted back as inputs. Implementation evidence is measured on pages 26-30.

Generative organogram

GRAMMAR AND DERIVATION | ORIGINAL §8

```
Grammar_nu = (Alphabet, w0, Productions_nu, Interpreter_nu)
w0 = EncodeAxiom_nu(omega)
N_(T,path) = Interpreter_nu(Derive(w0, rules, T, E_<=T, path))
```

The retained baseline omega and the initial grammar word w0 are distinct. Each rule yields a unique finite string for its admitted context. A derivation identity includes baseline and grammar versions, the original tick and a finite production/branch path.

Symbol / operation	Required interpretation
F	Advance along the local traversal/phase relation and emit or extend a segment.
+ and -	Apply a declared positive or negative OTAN2 increment.
[and]	Push and restore the complete branch context, including orientation.
Apex / sphere	Select a boundary primitive and its field construction.
Shifts	Invoke a declared resolution transition with explicit widths and lost-bit semantics.
XOR	Apply a typed bit operation or structural predicate; mutation and integrity checks have distinct roles.

The reference contract uses parallel rewriting of one finite generation at a time. A realization shall bind priorities, unmatched-symbol handling, branch order, maximum depth, stack bounds, time inputs and interpretation costs. Regenerable derivations must be well-founded and retain every non-derived dependency.

Current binary realization

The binary world derives nodes from finite paths of depth at most 32. Its operator lookup depends on the current phase/selector; a branch updates the selector by (3*g + branch + 1) modulo 256. Packed GROW words, declared field updates and complete mirrors are reproducible from the retained rule set.

The binary organogram remains its own bounded profile. GD1-GD8 integrate a finite geometry-changing production. OG1-OG8 now have measured parameterized production, complete branch context and generated field evidence on pages 67-68. Arbitrary cone/pyramid generation and general graph rewriting remain separate obligations.

Psi, f8 and typed field operators

FORMAL CONTRACTS | ORIGINAL §§5, 9

Psi requires an operator

```
A_x Psi_x = lambda_x Psi_x, Psi_x != 0
```

An eigenvector-based traversal shall bind A_x , eigenvalue selection, normalization, sign/phase convention, degeneracy handling and transport. The current field profile supplies successors directly. A supplied route is not evidence of an eigenstructure implementation.

f8 is an index, not the topology

```
Key_nu(n) = (k_Psi, k_rho, k_theta, k_derivation)
```

Use a total lexicographic order within a declared epoch. Canonicalize equivalent geometric representatives, break distinct-object ties deterministically and version or rebuild the index when ordering changes. A concrete middle-out construction recursively selects a median from the finite ordered set; an executable profile must fix the even-size median choice. Search-tree children and geometric neighbors remain separate relations.

Hadamard and difference stencils

```
H_f(u,v)_i = clip(round_nu(u_i*v_i / 2^f))
q = H_f(DecodeVector(W), Gradient_nu(phi))
next = SelectNeighbor_nu(node, q, topology)
Delta_a^2 phi(x) = phi(x+a)-2*phi(x)+phi(x-a)
Delta_T^2 phi_T = phi_(T+1)-2*phi_T+phi_(T-1)
```

Hadamard multiplication applies to decoded numerical lanes with declared fixed-point scale, widened products, rounding and clipping. Opcode, continuation and parity bits are not vector components. The gradient/stencil, neighbor admissibility, ties, stop and fallback rules must be supplied.

Spatial and temporal second differences are different operators. On a graph, $x+a$ means the selected adjacent sample in a declared stencil, not an undeclared external coordinate. Spacing and units are required for divided differences.

Current status: typed arithmetic, SDF bounds, local PX eigenstructure and f8 indexing have CPU/GPU checks. HP1-HP8 now have measured phase-directed Hadamard routing evidence on page 49. Global eigenmode traversal remains separate; no route-quality claim follows from index conformance.

Relations define the field world

GEOMETRIC INTERPRETATION | SOLUS PP.3-6; SOLIPSISM PP.12-13

$$\text{phi}(x) = \text{sigma}(x) * \inf \{ d(x,b) : b \text{ in Boundary } \}$$

The metric d, boundary, side function sigma and unit jointly determine signed distance. Negative, zero and positive designate inside, boundary and outside under the chosen convention. This is a three-class distinction; distance magnitude supplies additional information.

The addenda’s “hinges and wires” become declared weighted adjacency and traversal relations. An operator has geometric support through the current field class, an admissible destination through adjacency and a resulting field value. The implemented binding on page 15 makes those dependencies executable.

Source primitive	Binding required for a distance meaning
Apex	Local convergence or phase-accumulation marker; no mandatory global origin.
Circle / sphere	Metric, centre, radius and a boundary/side construction; an intrinsic ball is one finite binding.
Pyramid / cone	Local shaft, transverse metric, slope, extent and exact boundary construction.
Shaft theta	Selected local traversal/sweep direction, including any difference stencil.
Time T	Logical ordering of transitions and input events; distinct from geometric side-view notation.

Relational invariance

A graph isomorphism transporting weights, sides, boundary and operator rules preserves the intrinsic geometry. Reindexing may change packed G bytes; the decoded relations agree under the bijection. Bit-identical replay additionally retains the original ordered manifest.

Distance is measured to the specified boundary. It is not automatically distance to a “self-referential origin.” A node can carry an exact field sample and participate in a geometric operator without being assigned an undeclared physical volume.

In this edition “waveguide” names formal guidance and a finite field-driven example. Physical wave propagation requires its own equations and calibration. The finite vertex profile uses a declared combinatorial boundary; it does not claim that a finite discrete metric alone supplies a continuum boundary.

Exact finite SDF profile

IMPLEMENTED CONTRACT | SDF.R1-R4, R6-R8

Profile: relational-sdf-v1. A finite connected undirected graph $G=(V,E,w)$ has 1..256 uniquely named nodes. Their manifest order supplies indices 0..N-1. Each edge is one sorted tuple (u,v,w) , $u<v$, with weight 1..127. Duplicate pairs, self edges and noncanonical edge ordering are rejected.

```
d(u,v) = min { sum(edge weights) : path u -> v }
Boundary = {v : sigma(v)=0}
D(v) = min {d(v,b) : b in Boundary}
phi(v) = sigma(v)*D(v)
```

Signs are in $\{-1,0,+1\}$; the boundary is nonempty. No edge may directly join opposite nonzero signs. Thus every path between sides crosses the boundary. One side may be empty. Positive weights give a genuine intrinsic vertex metric with symmetry and triangle inequality.

Binding	Strict requirement
Units	unit_num/unit_den, both positive integers at most 1,000,000 and in lowest terms. No physical unit is inferred.
Distance range	Compute with widened integers; require -127..127. Reject overflow. Code -128 is unused; no saturation is permitted.
Routes	N rows of three destinations, each adjacent to the current node or the node itself, including unselected columns.
Turns	N rows of three integer increments in 0..255. Columns mean negative, zero and positive.
Identity and nodes	Nonempty strings of at most 128 characters; node names are unique. Integer fields reject Boolean and noninteger values.

The field is a scalar under the RP32 mirror: orientation changes do not invert its sign. A side function on a specified separating subset does not require a global orientation of the surrounding surface. Orientation-dependent signed sections require a different field profile.

The schema is reproduced on page 31. Its arrays become immutable tuples internally. A malformed geometry is rejected before it becomes an executable field. Tests include disconnected graphs, absent boundaries, invalid routes, opposite-side edges and values beyond the exact distance range.

Independent geometric certificate

THEOREMS | SDF.R5

A producer proposes signed codes ϕ . A checker independent of the producing shortest-path algorithm shall verify the declared sign/range, zeros exactly at the boundary, positive magnitudes elsewhere, and the following conditions for $D = \text{abs}(\phi)$:

$$|D(u) - D(v)| \leq w(u, v) \text{ for every edge}$$

$$D(v) = w(v, u) + D(u) \text{ for some neighbor } u$$

$$\text{at each nonboundary } v$$

Theorem G1: exact boundary distance

Premises. Positive edge weights, nonempty declared boundary, $D=0$ exactly there and both conditions above.

Proof. The equality witness strictly decreases D . Finiteness and positive weights force a witness chain to terminate at the boundary. Telescoping its equalities yields a path of length $D(v)$. Along any boundary-reaching path, telescoping the edge inequalities gives $D(v)$ no greater than that path's length. A path attaining D exists and none is shorter. Therefore D is the exact shortest distance.

Theorem G2: signed Lipschitz bound

$$|\phi(u) - \phi(v)| \leq d(u, v)$$

Proof. On one side this is the distance-to-set inequality. For opposite sides, any connecting path visits the boundary because no edge joins opposite signs directly. The portions from its endpoints to the first and last boundary visits bound $D(u) + D(v)$ by the total path length. Taking a shortest path gives the claim.

Consequences

A path starting at v whose length is strictly less than $D(v)$ cannot reach the boundary. For consecutive nodes a, v, b , the magnitude of $\phi(b) - 2\phi(v) + \phi(a)$ is at most $w(a, v) + w(v, b)$. The latter follows by adding the two edge difference bounds.

Parity certifies a bit relation. This certificate certifies a geometric relation. A valid-parity word with an incorrect field value must fail geometric admission.

Scope: exactness is relative to the supplied graph, weights and boundary. The certificate does not establish that those data match a measured physical environment. An independent Floyd-Warshall oracle and deliberately forged valid-parity values are included in the current tests.

Field-governed executable words

IMPLEMENTED TRANSITION | SDF.R8-R10

```
class(b) = 0 if b<0, 1 if b=0, 2 if b>0
L(i,c) = pack(turns[i][c], routes[i][c],
phi(routes[i][c]), STEP)
W = pack(r, i, phi(i), STEP | (eta<<4))
```

Each operator is an ordinary RP32 word. Its R lane is a phase increment, G is the destination and B is the destination’s exact field. The live word uses the same carrier with a different declared role: current phase, node and field. Its companion is always the full mirror.

Tick stage	Exact action
Validate	Check pair, node, STEP/orientation metadata and B=phi(G).
Select	Fetch O=L(G,class(B)) using the current state.
Evaluate	Set r_next=(r+(-1)^eta*O.R) modulo 256.
Transport	Set G_next=O.G and B_next=O.B; retain eta in v1.
Commit	Recompute parity and mirror; increment tick and emit one pair.

Theorem G3: field-state preservation

Suppose the initial pair is valid and each compiled operator contains a legal destination and its certified field. Selection returns such an operator. The resulting G/B relation therefore holds; modular phase remains in range; allowed metadata is reconstructed; repacking supplies correct parity and a complete mirror. Induction preserves these invariants at every accepted tick.

The manifest supplies initial node, phase and orientation and a maximum total tick count of 1..65,536. Each advance call accepts 1..4,096 ticks within the remaining budget. Budget overflow is rejected before mutation. Empty replay is permitted; a new advance of zero ticks is rejected.

The current program is a finite static geometric operator table. Changing boundary or hinge data under a new manifest changes the field and compiled operators. Runtime geometry mutation, epoch extension beyond the total budget and arbitrary program synthesis are not implemented by this profile.

Exact GPU realization

IMPLEMENTED ADAPTER | SDF.R11; ORIGINAL §14

Field construction on the device

```
D_0(v)=0 on Boundary, INF elsewhere
D_(k+1)(v)=min(D_k(v), min_u(w(v,u)+D_k(u)))
Run exactly N-1 synchronous rounds
```

Each round reads the previous buffer and writes a separate next buffer. Guarded widened integer sums preserve the chosen infinity sentinel. A shortest path has a simple representative of at most $N-1$ edges, so these rounds suffice. The independent certificate checks the resulting signed field before execution.

Persistent state and an integer operator texture

GPU compilation creates a 3 by N `r32uint` texture from the field and manifest rules. An ordered execution pass reads that immutable texture. One device invocation carries the individual's sequential state through a batch, with each tick selecting the next lookup from its predecessor. Field construction uses parallel work; the dependent state chain retains its required order.

The texture, field buffers and canonical state stay allocated across calls. Host interaction is not required between the ticks inside a batch. Host orchestration, admission, certification, archive handling and batch result transfer remain part of the implementation.

```
decode_alpha(encode_alpha(W)) = W
decode_alpha(Exec_alpha(encode_alpha(S), e)) = F_nu(S,e)
```

These are the adapter round-trip and refinement obligations. Integer bit patterns, word order and version must be preserved. Filtered sampling, color conversion and display interpolation are distinct output views, not canonical execution. Explicit GPU requests raise an error if unavailable; they do not silently evaluate the field on the CPU.

Verified: exact CPU/GPU fields and packed traces on the tested Vulkan adapter. Unmeasured: texture-cache residency, cache hit rates, device saturation, energy advantage and a general bypass of von Neumann bottlenecks.

A finite cascade of LUT stages must declare intermediate profiles, budgets and progress. Resource schedules that alter semantics are admitted replay inputs; changes only to cache residency must preserve derivation results. GPU texture semantics are described in WGSL [R1]; device profiling requires separate measurements [R2-R3].

Archives, footprints and admission

IDENTITY AND TIME | SDF.R12; ORIGINAL §§12-13

```
LUS = (nu, baseline_id, T, sequence, kind, words, status)
ID = (nu, baseline_id, producer, epoch, sequence, path)
```

LUS retains the source name Linear Unit Strain and its later telemetry role. A profile declares whether a record denotes calibrated strain, simulated boundaries, agent state or another footprint. DIGID/UUID names a structured derivation identity. Naming, representation integrity, origin admission and reproducibility are distinct results.

Arbitrary payloads are fragmented into a finite ordered sequence with total length and fragment metadata. Generated payloads may use replay descriptors when all dependencies remain available. Attribution strings are namespace metadata; they are not automatically cryptographic seeds, initialization vectors or authentication. SHA-256 in this document identifies files, not a spatial origin.

Implemented field archive

The archive retains format, complete manifest, ordered tick pairs and expected final state. Replay begins at the declared initial state and checks every retained pair. Unknown versions, extra/missing fields, malformed hexadecimal words and inconsistent replay are rejected. CPU and GPU adapters may resume each other’s archives. Adapter names and timings are diagnostics outside canonical state.

Regenerative R and forward-only W

Profile	Public interaction contract
R: regenerative	Admit named historical derivations whose original context is retained. This is the current implementation family.
W: WELip	Admit IGNITE, ADVANCE, RESIZE, INVALIDATE and EMIT; expose no historical-read operation at this interface. W1-W8 from page 69 bind this finite profile; measured continuation is on pages 82-83.

PRISM reverse lock names a specified rejection predicate for backward-history requests. “Poisoned pill” means localized active-entry invalidation with a recorded cause and retention policy. A read attempt is an event only if the interface detects it. Observation does not automatically change parity.

An internally consistent rewritten archive can still replay. Authentication needs an external trust mechanism. Atomic save and exclusive ownership protect the implemented update protocol; replay consistency is not a signature, nor a claim of immunity to storage failure.

FIFO, regeneration and finite resources

FORMAL PROPERTIES | ORIGINAL §§10, 17

```
push_C(Q,x): append x if |Q|<C; otherwise drop oldest, append x
Regen(d) = Eval_nu(omega, rules, original_T, path, inputs)
d = (nu, baseline_id, original_T, path, required_inputs)
```

Eviction removes an active derived copy. Baselines, productions, derivation descriptors and external inputs have separate retention policies. A complete pair is one eviction unit. Slot reuse must preserve derivation identity rather than confusing a new occupant with an old queue position.

Theorem R1: bounded active queue

For C at least one and initial $|Q|$ at most C , insertion below capacity increases length by one; insertion at capacity removes one item before appending one. Hence $|Q|$ never exceeds C . Reducing capacity retains the newest $\min(|Q|, C_{\text{next}})$ items and evicts older entries in order.

Theorem R2: exact regeneration

If the derivation function is single-valued and every original argument remains available, regeneration evaluates the same expression with the same arguments and yields the original word. Eviction of the active copy cannot alter that result. Missing non-derived input instead produces MISS or exclusion from the exact-regeneration set.

```
M_occupied = |Q|*w/8 <= C*w/8 bytes
M_total = M_active + M_baseline + M_rules + M_inputs
+ M_index + M_journal + M_runtime
```

A preallocated capacity- C payload arena has $M_{\text{active}}=C*w/8$ bytes, or $8*C$ bytes for 64-bit pairs. A Python cache also has container and object overhead. The bounded payload is not a bound on the full journal, descriptors or retained observations. Arbitrary external information cannot be reconstructed after its only copy is discarded.

Capacity independence and progress

When capacity changes only residency, an identical requested derivation must produce the same result at any supported capacity. Resource-driven changes to depth, resolution or selected semantics must instead be recorded as versioned input/control events. Each admitted regeneration uses a finite budget even when the architecture permits an unbounded sequence of finite requests.

Implemented: binary-world and F15 geometric-sample FIFOs, retained-context regeneration and capacity-independent decisions. Field-agent evidence includes actual eviction/reconstruction and changing capacities. Scalar fields, geometry, journals and GPU arenas remain separately accounted outside the active sample bound.

Finite Klein quotient

VERIFIED FINITE PROFILE | K1-K3

Choose strict integers W, H at least 3 with $W \cdot H$ at most 256. Temporary chart labels represent a quotient; they do not introduce a mandatory external physical origin.

```
(u, v+H) ~ (u, v); (u+W, v) ~ (u, -v)
q=floor(u/W); u0=u-q*W; v0=(( -1)^q*v) mod H
eta0=eta XOR (q mod 2); node=u0*H+v0
```

Negative labels use floor division, not truncation. Phase transported into the canonical frame is multiplied by $(-1)^q$. Node names are $k:u0:v0$ in index order. Project each unit horizontal/vertical edge, retain each sorted distinct endpoint pair once and assign unit weight. The relative seam bit τ is one on horizontal wraps between columns $W-1$ and 0 , zero elsewhere.

```
face(u, v) = [canon(u, v), canon(u+1, v),
              canon(u+1, v+1), canon(u, v+1)]
V=WH, E=2WH, F=WH; Euler characteristic=0
```

The surface claim requires cells

The audit shall check four distinct corners per face; correspondence between face boundaries and declared edges; graph connectivity; exactly two incident faces per edge; and a single cyclic vertex link at every vertex. These conditions establish a closed connected 2-manifold. Counts alone do not establish a surface.

```
x_g = -d_f*d_g*x_f
```

Here x_f is a face orientation choice and d_f is its traversal sign along a canonically oriented shared edge. Propagate the constraint through the dual graph. A contradiction establishes nonorientability. A connected closed nonorientable surface of Euler characteristic zero is a Klein bottle [R4].

At c4b41ce, KleinDomain constructs and audits this cell complex. Evidence covers all 702 admissible width/height pairs, including edge incidence, cyclic vertex links and nonorientability. Independent tests reject malformed and pinched complexes. The surface claim rests on these cells and audits, not arithmetic mirror pairs alone.

Orientation cover and seam action

VERIFIED FINITE PROFILE | K4, K7-K9

```
Cover vertices: (i,eta), encoded as 2*i+eta
(i,eta) -- (j,eta XOR tau(i,j))
(U,V) = (u+eta*W, (-1)^eta*v mod H)
```

Lift every base edge twice and each face boundary from both starting orientations. The XOR of seam bits around a face shall be zero. The cover must be connected, closed and orientable, with counts $2WH, 4WH, 2WH$ and cyclic vertex links. The displayed map identifies it with a periodic $2W$ by H toroidal grid; audit the full mapped edge and face sets, not only their counts.

A horizontal loop at $v=0$ returns to its base node after W steps with orientation reversed; its lift closes after $2W$ steps. A vertical H -step loop preserves orientation. These are explicit holonomy witnesses. The two sheets are local-frame representations of the same individual.

Packed transport

```
Operator metadata: STEP | (tau<<6)
t = (r + (-1)^eta*delta) mod 256
r_next = (-1)^tau*t mod 256
eta_next = eta XOR tau
```

Operator bit 6 stores relative seam action; operator bit 4 stays zero. Live metadata contains only STEP and orientation in bit 4. Bit 6 never leaks into live state. Set G to the destination and B to its certified scalar field, then recompute parity and the full mirror. Toggling orientation without reflecting phase is not this transport convention.

Theorem K1: mirror coherence

Let $S_{\tau}(r,\eta)=((-1)^{\tau}r,\eta \text{ XOR } \tau)$. A tick is S_{τ} composed with U_{δ} ; $M=S_1$. The phase theorem establishes U_{δ} commutes with M , and both seam maps commute. Mirrored states share G and scalar B and therefore select the same operator. Their complete tick results remain mirrors.

The proof establishes this numerical binding, conditional on correct geometry/operators. The implemented GPU compiler derives τ from the authoritative seam set and applies K8 inside its ordered tick loop; hardware traces agree with CPU execution. A generic graph with seams is not automatically a Klein surface. Orientation-cover background: Hatcher [R5].

Intrinsic ball and v2 profile

VERIFIED FINITE PROFILE | K5-K6

The implemented manifest format is relational-sdf-v2. It requires every v1 key plus seams and topology. Its archive container remains relational-sdf-machine-v1; the manifest selects execution semantics.

```
1 <= Radius <= max_v d(center,v)
sigma(v) = sign(d(center,v)-Radius)
phi(v) = sigma(v) * min_b d(v,b), b in Boundary
```

On the unit-edge quotient, every shortest path from the centre encounters each integer distance level. The selected level is nonempty and unit edges cannot skip from a negative side directly to a positive side. Recompute exact boundary distances; do not assume radial distance minus radius is an exact boundary distance on an arbitrary branched graph.

Scalar pullback theorem

The lifted field has the same value on both sheets. Every base path lifts and every cover path projects with equal length. A shortest base path to the boundary supplies a cover path to some lifted boundary point; conversely, any cover path supplies a base candidate. The two minimum distances are equal. The side function is pulled back unchanged.

v2 addition	Required interpretation
seams	Sorted unique [u,v] pairs, u<v, naming reversing edges already present in edges. Others and self transitions have tau=0.
topology	Either null for a generic relational graph or exactly {format: klein-grid-v1, width: W, height: H}.
Descriptor check	Regenerate nodes, unit edges and seams; require exact agreement with the manifest. Only audited cells establish the surface.
Compatibility	v1 retains its exact prior schema and bytes. Reject seam/topology additions to v1. Archive semantics follow the contained manifest version.

The default generator uses W=H=8, centre 0, radius 2, initial node 0, phase 250 and orientation 0. All field-class routes take local u+; increments are [11,53,137]. The measured 64-tick CPU/GPU trace crosses eight reversing seams and ends at k:0:0 with pair 11FE00D681FE002A.

Klein evidence includes negative/large labels, all 702 supported dimensions, face/link audits, connected orientable covers, full torus edge/face maps, holonomy, exact scalar pullback, CPU/GPU seam traces, replay and unchanged v1 semantics. Its 311-test baseline is preserved alongside the current 421-test result on page 27.

The infallible contract

NEW FORMALIZATION | SOLIPSISM PP.3-11

Conditional infallibility. Given a versioned profile with single-valued, total bounded transitions proved to preserve its declared invariants, a valid initial state, completely specified inputs and ordering, and faithful execution, every bounded request has one defined result; every accepted transition preserves those invariants; replay reproduces its canonical output.

```
Sigma = (States, Inputs, F, Invariant, CanonicalEncode)
I(S0)
I(S) and F(S,e)=ACCEPT(S_next,o) => I(S_next)
```

“Closed” means all output-affecting dependencies are in this profile/state/input boundary. It does not mean the executing hardware has no environment. “Infallible” names the exact contract above, not an unexplained extra operation or an assertion of physical immunity.

Theorem I1: deterministic replay

If F is single-valued, equal profiles, initial states and ordered input sequences yield equal semantic states and canonical outputs at every finite step. **Proof.** Equality holds initially. Equal states and next inputs give equal results under the same function. Induction extends equality through the finite trace.

Theorem I2: bounded total admission

If parsing is bounded, every predicate terminates, every loop consumes a finite budget and result classes exhaust the declared domain with a fixed priority, every bounded request terminates with one result. **Proof.** Each stage terminates and their finite composition terminates. Single-valued classification resolves overlaps.

Theorem I3: invariant preservation

If $I(S_0)$ holds and every accepted step preserves I , all admitted states satisfy I . **Proof.** Induction over accepted transitions. The SDF invariant includes certified field values, legal destinations, phase/metadata bounds, full mirrors and a remaining execution budget. New geometric rules require a new preservation argument.

These are mathematical implications with substantive premises. A proof of a transition theorem, verification of a program and validation against an external phenomenon are different obligations. Finite-state checking is possible; self-reference alone does not invoke an incompleteness theorem or prevent such verification.

Proof boundary and implementation refinement

NEW FORMALIZATION | SOLIPSISM PP.8-16

$$\text{decode}(\text{Exec}(\text{encode}(S),e)) = F(S,e)$$

An implementation refines a profile when this equation holds on the admitted domain, including emitted words and result classifications. A trace comparison is evidence for tested instances. It is not by itself a proof covering every possible program state, driver or device.

Layer	What must be established
Mathematical profile	Single-valued rules, total bounded admission and invariant-preserving transitions under declared premises.
Implementation	Encoding, arithmetic, memory order, admission and persistence execute those rules correctly.
Fault model	Specified corruptions are detected or recovered under an explicit physical/storage/computation fault model.
Physical interpretation	Measured units, observations and transfer functions connect the model to the represented phenomenon.

Discrete does not mean unspecified

Integer execution avoids floating-point rounding in the canonical field arithmetic. Width, signedness, overflow, shifting and conversion still require exact rules. Floating-point rounding is not inherently randomness; evaluation order and arithmetic semantics determine reproducibility [R6]. A finite discrete representation can execute an incorrect rule perfectly.

Geometry changes are accepted state changes

The addendum’s internally generated growth can be formalized as a versioned graph-rewrite event. Such an event shall supply a finite rule, affected domain, retained derivation context and deterministic node correspondence. Its result must pass connectivity, boundary, range, topology and operator checks before admission. GD1-GD8 now realize one finite dyadic geometry production under this contract; arbitrary rewriting remains separate.

The source’s “zero-order decay” or Lambda label requires a deterministic invalidation function $\text{Lambda_nu}(S,e)$, including selected entries, predicate, retention effects and result. W1-W8 bind finite active-pair invalidation on page 71, with measured recovery on pages 82-83. Biological reaction-diffusion analogies require actual equations before they become physical models.

Attribution remains attribution. Personal judgments in the exported discussion are not computing predicates. The meaningful architectural claim is a declared domain with exact admitted transitions, for which proof obligations and implementation measurements can be stated and discharged.

Typed zero, XOR and integrity

NEW FORMALIZATION | SOLIPSISM PP.2-3, 7, 11-16

Boolean involution

```
NOT(b)=1-b, b in {0,1}
NOT(NOT(b))=b; b XOR 1 XOR 1=b; 0*0=0
```

Two negations restore the original bit. NOT(0)=1 is one negation. If two zero operands should yield one, that is an explicitly named equality/XNOR predicate, not ordinary multiplication. Two orientation reversals likewise restore orientation; they do not establish a proposition's truth.

Zero is typed

Boolean false, arithmetic zero, phase zero, boundary distance zero, an empty container and a rejected request are different values in different types. SDF zero is a valid boundary. Rejection uses a tagged status. Clearing every bit is not automatically a valid neutral state.

```
Pair(pack(0,0,0,DATA)) = 9000000000000000
0000000000000000 fails the full RP32 mirror relation
```

Theorem I4: exact parity detection boundary

```
parity(w XOR e) = parity(w) XOR parity(e)
```

XOR associativity gives the equation. Even parity detects every odd-weight corruption mask, including every single-bit change, and misses every even-weight mask at that check. It neither locates nor corrects the changed bits. Full mirror comparison checks another relation, but a consistently replaced valid pair can still pass both checks.

```
pack(0,0,3,DATA) XOR pack(0,0,4,DATA) = 80070000
```

This result has valid parity and B=7. It is not a certificate of the distance between adjacent samples at boundary distances 3 and 4. XORing a field with itself yields zero everywhere without making every original sample a boundary. The independent geometric certificate remains necessary.

A chosen one-bit predicate can summarize an inside/boundary/outside condition, but one bit cannot losslessly encode all three classes. Binary hardware, finite packed words, geometric exactness and origin authentication therefore retain distinct contracts.

Sensory adapters and physical meaning

FORMAL INTERFACES | ORIGINAL §15; ADDENDA

Source term	Required interface binding
IPD	Relative phase between paired observations: unit, wrap, ambiguity and calibration.
IDT	Arrival/sample-time difference: clocks, units, ordering and timing calibration.
FMCW / LRAD	Sampled chirp/return model or acoustic transduction interface; deployment-specific geometry and transfer functions.
Micro-saccades	Declared small sampling or phase perturbations linked to jitter metadata.
Cicada cascades	Periodic triggers for sampling, regeneration or eviction, with a collision priority.

```
Delta_p = wrap_m(p_right-p_left)
Delta_t = t_right-t_left
c_i(T) = 1 if T mod period_i = 0, otherwise 0
```

For periods 13 and 17, both triggers coincide at multiples of 221. A reference one-bit perturbation is (c_1+c_2) modulo two. The manifest must specify its effect on phase or sampling and recompute parity after an intentional state change. Jitter is not an integrity fault by definition.

Observed samples, calibration versions and semantic scheduling decisions belong to the replay context. Converting them to range, direction, strain or density requires a physical signal model. Inverse-square physics does not follow from logarithmic addressing; wave mixing is not automatically Hadamard lane multiplication or XOR.

Applications permitted by the contract

The architecture can host deterministic simulation, regenerable world models, compact geometric controllers and device-independent continuation once their profiles and adapters are specified. Robotics, optical/FPGA execution, biological growth and physical autonomous construction require additional realizations. The current repository demonstrates software/GPU substrates and a simulated/live-observation individual.

A complete application must identify what the field means, where its inputs come from, what action an output authorizes, and how the model is validated. The formal architecture supplies a common representation and execution discipline; it does not supply an absent sensor or actuator model.

Implementation map and integration

MEASURED PROFILES | FI, PX, HP, GD AND OG

The field-agent policy joins recipe-derived Klein geometry to the existing individual. The module map identifies implemented scope; remaining architectural concepts retain their separate obligations.

Layer	Existing implementation	Scope
Packed core	rp32.py	Encoding, phase, parity, full mirror and pair.
Derived world	world.py; runtime.py	Binary derivation, bounded active FIFO, replay and demo archives.
Individual	motion.py; navigation.py; tomigidt.py	Costs, incremental deterministic planning, local observations and action state.
Persistence / live	session.py; live.py	Ownership, atomic saves, admitted external frames, deduplication and durable acknowledgment.
Packed GPU	gpu.py; packed.wgsl	World derivation and selected motion forecasts; CPU retains planning and admission.
Intrinsic field	field.py; field_cli.py	Strict geometry/manifest, exact field, operators, one persistent sequence and replay.
Field GPU	sdf_gpu.py; field.wgsl	Device field construction, operator texture and ordered state transitions.
Klein quotient	klein.py	Audited cells, connected orientation cover, intrinsic ball and v2 seam transport.
Field / index	field_agent.py; field_world.py; field_agent_gpu.py; psi.py; f8.py	Typed field agent, sample FIFO, local eigen-axis, canonical tree and persistent device actions; field_agent.wgsl.

Profile meanings remain explicit

Earlier motion/world profiles use B for terrain or energy; relational-sdf-v1/v2 use B for exact signed distance. FI1-FI8 preserve these schemas and implement a field-agent policy with B=phi and separate integer energy. Shared encoding never permits an implicit role change.

Measured integration

Tomigidt observes, plans and acts through Klein seams. hadamard.py adds phase-state routing; growth.py supplies dyadic geometry growth. organogram.py, organogram_gpu.py and organogram.wgsl now produce a branching grammar field and continue the same individual. Global Psi, general primitives and physical adapters remain separate.

Paths are relative to solvefinite/. Historical source hashes: GD at 94f86c7 in docs/evidence/growth-v1/; OG at f125a76 in docs/evidence/organogram-v1/. Earlier FI/Klein/PX/HP captures remain. OG chronology and measurements are on pages 67-68.

Retained PX verification and field trace

HISTORICAL CAPTURE | 25 SEPTEMBER 2026

Measurement	Observed result
Full integrated suite	421 tests passed in 44.997 s; zero skipped.
Actual device coverage	55 GPU methods: the prior 41 plus 14 F8 methods.
Klein topology coverage	All 702 admissible dimension pairs audited, including base/cover surfaces and full torus edge/face maps.
Conformance examples	Klein: 22 checks; field agent: 23; Psi/index: all 20 passed. PX audits 106,045 descriptors in all 702 domains; 830 GPU lookups/materializations in six domains.
Hardware / software	NVIDIA GeForce RTX 5070 Ti Laptop GPU; Vulkan; driver 591.59; wgpu 0.32.0; Python 3.12.14.
Cross-adapter continuation	CPU/GPU archives agree; fresh processes replay both ways before the seam and during DEFER. Live retries preserve history.

```
SDF v1: [-6,-4,-3,0,2,3,5]
Moved boundary: [-8,-6,-5,-2,0,1,3]
V1 tick 64: n4, pair 1102046C01020494
Klein v2 tick 64: k:0:0, pair 11FE00D681FE002A
```

Changing geometry changes field execution and agent route selection. The FI8 mission completes after four cycles at pair 06011145160111BB, energy 90. Its reversing seam yields phase 240 and eta 1. The bounded-search mission takes 41 cycles; six energy/recipe/valid-parity forecast tamper probes are rejected.

Reproduce the measured checks

```
python -m unittest discover -s tests -v
python -m examples.psi_f8_conformance --output psi.json
python -m examples.field_agent_conformance --output agent.json
python -m examples.klein_conformance --output klein.json
python -m examples.field_conformance --output sdf.json
```

Retained captures: SDF 262 tests/21 device methods, Klein 311/28 and field agent 368/41. PX rebuilds preserve the complete FI8 archive and all 41 DEFER-mission cycles. These counts describe tested finite profiles, not architectural completion.

Historical PX evidence: docs/evidence/psi-f8-v1/. Earlier captures remain in field-agent-v1/, klein-field-v2/ and formal-edition-2026-09-25/. The new HP capture is on page 49. Logs, source hashes, adapter details and archives identify each observation.

Progress against the architecture: I

CAPABILITY STATUS | FINITE REALIZATIONS

Obligation	Measured status and remaining work
C1-C3: relational packed execution	Verified subset. RP32, state-selected integer texture execution and GD dyadic geometry scale transitions work. General resolution policies and alternate historical carriers remain unbound.
Exact intrinsic SDF	Verified subset. Weighted graph distance, declared separating boundary, units, certificate and field-governed traces work. Continuous primitives and physical calibration are separate.
C4: Klein geometry	Verified finite profile. Quotient cells, nonorientability, connected orientable cover, torus maps, intrinsic ball and CPU/GPU seam transport are audited.
Local Psi / traversal	Verified local roles. Exact SDF eigen-operator, primitive axis and chart transport govern indexing and HP directional gains. Global eigenmode traversal remains separate.
Hadamard / Delta-Delta	Verified finite routing profile. HP1-HP8 implement phase-dependent directional costs, planning and device actions. Existing second-difference bounds remain checked.
f8 middle-out index	Verified finite binding. Canonical five-component keys, lower-median preorder, actual CPU/GPU lookup and atomic versioned rebuild work for scalar base descriptors.

What the tests establish

The implemented graph profile has exact arithmetic and a field certificate, and its tested CPU/GPU realizations agree. These are substantial completed components. They do not establish every named architectural layer by association with the same word carrier.

What will count as progress next

FI, PX and HP provide field-guided planning, indexed lookup and phase-directed routing. GD1-GD8 add finite dyadic growth and continued action. OG1-OG8 now execute parameterized productions and branch context on CPU/GPU. Global Psi, general primitives, physical adapters and wider continuation remain separate obligations.

The requirements matrix continues on page 29. No percentage is assigned because architectural obligations differ in size and some remain unbound; counting passing tests would produce a misleading completion estimate.

Progress against the architecture: II

CAPABILITY STATUS | INTEGRATION AND APPLICATIONS

Obligation	Measured status and remaining work
C5: generative world	Verified finite GD and OG. Dyadic Klein generations and parameterized branching productions change geometry or field and continue the same individual. Complete branch context and original-time replay are tested. General primitives remain separate.
C6: finite active memory	Verified subset. Pair-atomic sample FIFO, genuine reconstruction and capacity-independent agent histories work. Journals, geometry, fields and device arenas consume additional memory.
C7: complete mirror	Verified subset. Phase involution/commutation, parity and full pair relations work, including K8 phase reflection and orientation transport on CPU/GPU.
C8: individual and footprint	Verified subset. Planning, observations, epoch/sequence, retry handling, ownership and durable continuation work. W adds a typed LUS envelope; broader DIGID semantics and global admission/authentication remain open.
WElip / Lambda	Verified finite W profile. W1-W8 bind a forward interface, local invalidation, finite epoch clock, current-state emission and private recovery. Source-bound CPU/GPU evidence is on pages 82-83. Broader Lambda semantics remain outside this binding.
GPU / performance	Verified subset. Exact tested adapters and device execution exist. Cache saturation, general speed/energy superiority and other hardware adapters are unmeasured.
Waves / physical growth	Unbound applications. Signal models, physical transfer functions and calibrated action semantics are not supplied by the present code.

Continuation after the integrated field agent

FI joins geometry to the individual; PX adds an interchangeable index; HP makes phase and local eigenstructure affect actions. GD extends history across geometry generations. OG now produces exact new union-boundary fields from branching grammars and continues that history. Global traversal, general primitives and wider continuation need further work.

The measured integration is finite and profile-scoped. Whole-system indefinite continuation, autonomous physical self-replication, consciousness and universal immunity are not demonstrated outcomes of the present profiles.

Conformance and performance scope

ACCEPTANCE CRITERIA | ORIGINAL T1-T10

ID	Criterion and present scope
T1-T4	Exact encode/decode, modular phase, mirror involution/commutation and odd-bit parity detection: exercised for RP32.
T5	Canonical topology and transported field meaning: finite Klein K1-K9 are implemented and independently audited.
T6-T8	FIFO order/pair atomicity, retained-context byte replay and explicit profile/version handling: exercised for existing profiles.
T9	Equal semantic histories under tested capacities/batches/adapters: exercised in finite implementations; no general distributed deployment proof.
T10	Declared application outputs: graph-field and simulated/live-observation contracts exercised; physical calibration remains unbound.

Historical substrate benchmarks

Workload	Logical transitions	Measured batch wall time
Depth 16, 65,536 paths, 20 repeats	20,971,520	0.0018765 s; about 11.18 billion/s
Depth 18, 262,144 paths, 500 repeats	2,359,296,000	0.0275231 s; about 85.72 billion/s

These repository records date from 25 September 2026 and concern the packed binary substrate. They repeatedly use a 32-byte LUT. Timings include submission/readback but exclude initialization, compilation and upload. Readback sizes were 1,048,576 and 4,194,304 bytes; explicit GPU payload accounting was 2,098,264 and 8,389,720 bytes. CPU results agreed for the benchmark cases.

They are not fresh end-to-end autonomy timings, optimized CPU comparisons, cache-hit measurements or proof of device saturation. Logical transition rates for these particular loops cannot be generalized into an architectural bandwidth claim.

Required comparison record

A meaningful performance comparison records hardware, profile, rule set, LUT and working-set sizes, batch distribution, precision, initialization, transfers, retained inputs, total memory, latency, throughput, regeneration cost and output agreement. Compare the same workload and correctness target. Cache and occupancy claims need corresponding counters [R2-R3].

Historical records: docs/benchmarks/rtx5070ti-depth16.json and rtx5070ti-depth18.json. PX conformance took 21.673 s as a correctness capture, not a comparative benchmark. Default explicit device payload is 49,160 bytes, peaking at 51,528 during rebuild; retained host index payload is 1,296 bytes, peaking at 2,592. These exclude Python/driver overhead; fields, geometry and journals remain outside the FIFO.

Executable default and archive schema

SELF-CONTAINED REFERENCE | SDF.R1-R12

This complete default manifest reproduces the implemented seven-node field. All three route columns must be legal even when the current field selects only one.

```
{
  "format": "relational-sdf-v1", "identity": "TOMIGIDt",
  "nodes": ["n0", "n1", "n2", "n3", "n4", "n5", "n6"],
  "edges": [[0,1,2],[1,2,1],[2,3,3],[3,4,2],[4,5,1],[5,6,2]],
  "signs": [-1,-1,-1,0,1,1,1],
  "routes": [[1,1,0],[2,2,0],[3,3,1],[4,4,2],
             [5,5,3],[6,6,4],[6,6,5]],
  "turns": [[11,53,137],[11,53,137],[11,53,137],
            [11,53,137],[11,53,137],[11,53,137]],
  "unit_num": 1, "unit_den": 1,
  "initial_node": 0, "initial_phase": 250,
  "initial_orientation": 0, "max_ticks": 65536
}
```

Reference evaluation

```
D = exact_shortest_distances_from_all_boundary_nodes(manifest)
phi = tuple(sign[i] * D[i] for i in range(N))
certify_field(manifest, phi)
W = pack(initial_phase, initial_node, phi[initial_node],
        STEP | (initial_orientation << 4))
for tick in admitted_finite_budget:
    r, i, b, a = unpack(W)
    c = 0 if b < 0 else (1 if b == 0 else 2)
    delta, dest = turns[i][c], routes[i][c]
    r = (r + (-1 if a & 16 else 1) * delta) % 256
    W = pack(r, dest, phi[dest], a)
    emit(Pair(W))
```

Canonical archive

Top-level keys are `format`, `manifest`, `trace`, `expected`. Format is `relational-sdf-machine-v1`. Trace entries are exactly 16 uppercase hexadecimal characters, one per accepted tick. Expected contains `identity`, `tick`, `node`, `agent_pair`. The manifest reconstructs the initial pair; trace length is the tick count. Every entry and expected final value must agree with replay.

The JSON schema is strict: unknown/missing keys are invalid. Canonical word equality does not require incidental JSON whitespace to match. The initial pair is not an emitted tick. State ownership and atomic file replacement belong to the CLI persistence adapter, not to a distance equation.

Required bindings and stable obligations

PROFILE COMPLETENESS | ORIGINAL APPENDIX D

Binding family	Required decisions / original requirement IDs
Dependencies	Baseline, version, inputs, deterministic interpretation and retention; original R1.
Phase / scale	OTAN2 increment, miss/tie/endpoint behavior, log unit/base, resolution and budget; R2.
Topology / geometry	Canonical representatives, seam/mirror action, metric, boundary, signs, primitives and units; R3-R4.
Words / grammar	Layout/version, continuation; rule priority, branches, original time and finite derivation; R5-R6.
Index / routing	Canonical total keys, ties/rebuild; typed Hadamard, gradient, widths, rounding and admissibility; R7-R8.
Regeneration	Required non-derived inputs, capacity policy, pair atomicity, missing-input and budget results; R9-R10.
Identity / time	Epoch/input prefix, event ordering, wrap, identity allocation and separate integrity/origin outcomes; R11-R12.
Forward interface	Detected-event predicate, localized invalidation and unaffected retained partitions; R13.
Adapters / scaling	Exact word preservation, finite cascade stages, ownership, input delivery and resource controls; R14-R15.
Physical adapters	Samples, units, calibration, transfer functions and retained scheduling effects; R16.

Original R1-R16 are preserved above. Implemented extensions use **SDF.R1-R12** on pages 13-17 and 31, **K1-K9** on pages 19-21, **FI1-FI8** on pages 35-38, **PX1-PX8** on pages 39-43, **HP1-HP8** on pages 44-48, **GD1-GD8** on pages 50-56 and **OG1-OG8** on pages 58-66. **W1-W8** begin on page 69, with measured evidence on pages 82-83. Satisfying one profile does not silently complete another.

Original property cross-reference: P1 replay is I1 on p.22; P2 FIFO and P3 regeneration are R1/R2 on p.18; P4 mirror preservation is on p.8; P5 representation independence is the adapter/refinement contract on pp.16,23.

Historical vocabulary: mosTADPOLE(thegreenone) names the source input/output lineage; TPVM is Topological Fixed-Point Virtual Machine. Det-0 names reproducibility with complete dependencies. Generative Topological Fabric, Packed Topological FIFO Matrix and Uniform Packed Operator Paradigm are successive functional views. REW expands to Reverse Electronic Warfare in the WElip source. f8's "BVH replacement / S2 superseder" is an intended indexing/comparison role, not a measured replacement result.

A complete manifest may reference immutable versioned rule tables rather than embedding every algorithm in a packet. No "infallible" designation waives a missing binding. A profile is executable only once all choices that affect its outputs are single-valued and finitely evaluable.

Source concordance and provenance

RETAINED CORPUS | CONSOLIDATED READING EDITION

Source	Integrated destination in this edition
O: TK-LPLUT-1.0, 24 pages	Original §§1-3: pp.4-6; §§4-5: pp.11-14,19-21; §§6-8: pp.7-10; §§9-11: pp.9,11,18; §§12-16: pp.5,16-17,25; §§17-18: pp.8,14,18,22-24,27-30. Appendices A-B: pp.8,31; C-D: pp.32-34.
S: Solus, 7 pages	pp.3-4: operator geometry and hinge/traversal roles; pp.5-6: relational fields and self-reference. Integrated pp.6,11-16,24-25. Search/conversation material is not a new protocol.
I: solipsism, 16 pages	pp.3-5: one closed individual, pp.9-13: infallibility/SDF claims, pp.15-16: propagation/decay intent. Integrated pp.9,12,22-25. pp.2-3 arithmetic is corrected in typed form on p.24; personal analogies on pp.14-15 supply no predicates.
SDF companion / Klein companion	All normative SDF.R1-R12 and K1-K9 are consolidated in this PDF. Earlier Markdown remains the recorded development sequence.

SHA-256 of preserved source PDFs

O 8ea9cfb077630993e1d472ba72715a25d
2402bf243518663f7d08b03f8b83647
S a2ace794138bcad68463d1ce2ba4a6e99
54540bb4f2f29e4090d68d6e0440489
I 1091e0435aa1648b4ede127837e66ddbfb
2c5dbd0a67f49cbb7a7ae134ee547e7

Join each source’s two lines without spaces. Retained paths: O at repository root, S at sources/solus-ion-ad-infinitum.pdf, I at sources/solipsism.pdf. Both addenda originated in the author-named “Philosophers stone Jitske Klootwijk” source directory.

Implementation evidence identity

Klein: c4b41ce12a33a747bd54c8cc7f9748a06f7b59de
Capture: 2026-09-25, 11:30:28 UTC
Prior: 8f4b87bed131f5c084ef59aec558f3f9eb6ddedc
Prior capture: 07:52:57 to 07:54:16 UTC

Chronology: SDF specification 477e576 preceded implementation 50c9389; Klein specification 8f4b87b preceded c4b41ce. FI1-FI8 formal commit 5ccc022 preceded integration c8af71d. PX1-PX8 formal commit d8de349 preceded its runtime. Current source hashes: docs/evidence/psi-f8-v1/verification.json. Earlier captures and original source bytes remain preserved.

Notation and technical references

REFERENCE INDEX

Symbol / name	Meaning
nu; omega; T	Version; retained baseline; logical tick.
r; eta; delta; tau	Phase; local orientation bit; phase increment; relative seam bit.
phi; D; sigma	Signed field; unsigned boundary distance; declared side.
Psi; theta; f8	Bound traversal/eigenvector role; local sweep direction; relational index.
M; Q; C	Full mirror; active FIFO; active capacity.
RP32; LUS; DIGID	Packed carrier; typed emitted footprint; structured derivation identity.
R / W; Lambda	Regenerative/forward-only interaction profiles; proposed bound invalidation function.

Primary technical references

[R1] W3C. [WebGPU Shading Language](#), [textureLoad](#). Integer texture access semantics. This reference supplies API background, not a proof of this runtime.

[R2] NVIDIA. [CUDA C++ Best Practices Guide](#), version 12.8. Memory hierarchy and measurement background; the tested adapter here uses Vulkan through wgpu.

[R3] NVIDIA. [Nsight Compute Profiling Guide](#). Profiling methodology and counters. No Nsight counter evidence is claimed in this edition.

[R4] Jacob Lurie, MIT. [Classification of Surfaces](#), [Lecture 33](#). Surface classification used conditionally after manifold/orientation checks.

[R5] Allen Hatcher. [Algebraic Topology](#), [section 3.3](#). Orientation covers. The concrete finite quotient and required audits are specified in this edition.

[R6] David Goldberg. [What Every Computer Scientist Should Know About Floating-Point Arithmetic](#) (1991). Rounding and evaluation semantics; cited to separate numerical error from inherent randomness.

[R7] John C. Hart. [Sphere Tracing: A Geometric Method for the Antialiased Ray Tracing of Implicit Surfaces](#) (1996). Context for geometric distance bounds. The present certificate is proved directly for a finite graph.

Revision 12 status. Historical HP retains 479 tests/15 checks; GD retains 546/16; OG at f125a76 retains 623/15. W captures 698 tests, 145 actual-device methods and 14 checks. Distinct finite captures do not establish whole-architecture completion.

One field-guided Tomigidt

VERIFIED FINITE PROFILE | FI1-FI2 | FORMAL-FIRST CONTRACT

FI1. Identity, schema and typed state

FieldAgentManifest shall select policy tomigidt-field-observe-plan-act-v1, word_profile RP32-relational-sdf-v2 and perspective local-observation-v1. One existing Tomigidt owns the goal, local observations, retained search, live state and event history. There is no separately advancing FieldMachine or second individual.

The exact manifest keys shall be those below. Unknown/missing keys, Boolean integers and invalid shapes shall be rejected. JSON arrays become immutable tuples. Target and initial_node are canonical names; initial_node defaults to k:0:0. The .start property supplies that same name. No graph, routes, field array or sensor-supplied phi belongs in this schema.

identity, target, world, initial_node, initial_phase,
initial_orientation, initial_energy, repair_cost,
max_search_expansions, max_hops, max_cycles,
policy, word profile, perspective

```
Live state = (Pair(pack(r,i,phi(i),STEP | (eta<<4))), E)
0 <= E <= 2^31-1; E is separate from the B lane
```

Phase is 0..255 and orientation is 0 or 1; defaults are 250 and 0. Initial energy defaults to 100. Repair cost is 1..127; cycles are 1..1,000,000; search quantum is 1..65,536 expansions per cycle. Identity is a nonempty name of at most 128 characters. All numerical bounds use strict integers. Completion uses EMIT instead of STEP while retaining orientation.

FI2. Retained geometric recipe

World recipe format is klein-ball-world-v1 (Python attribute version). Its exact JSON keys are format,width,height,center,radius,turns,baseline_id. Baseline ID is a nonempty string. Defaults are W=4, H=5, centre index 0, radius 2 and turns [11,53,137]. Width/height obey K1; centre is in 0..WH-1; radius obeys K5; turns are three unsigned eight-bit integers.

The retained recipe regenerates canonical k:u:v nodes, every unit quotient edge, seams, cells and exact certified SDF. It replaces an expanded graph/field in the manifest. K1-K5 and the scalar mirror rule remain binding. Geometry is immutable within one manifest; original external observations remain separate retained inputs.

This policy is a new typed application binding. Legacy AgentManifest policies, terrain/energy words, FieldMachine schemas, archives and fixture bytes retain their previous meanings. The shared session/live envelopes select this policy from their contained manifest.

Observe, plan and transport

VERIFIED FINITE PROFILE | FI3-FI4 | ONE ACTION SEQUENCE

FI3. Local observations and retained planning

Sensors supply only strict hazards in 0..127 for the current node and its outgoing quotient neighbors. DATA observation packets retain B=hazard under their explicit observation context; they are not SDF samples. No input may supply a route, action or phi. Before MOVE or REPAIR, require a fresh complete local frame. Partial frames produce WAIT; unseen distant nodes use the existing zero-hazard hypothesis.

Generate the graph from all quotient edges. Use global lexical ordering of canonical names and lexical route ties. RouteSearch gains explicit relational mode with a strict hop bound in 1..255, default 255; legacy binary mode retains its 32-hop bound. Per-cycle search quanta retain unfinished work through DEFER. Search context includes position, current pair, separate energy and effective costs; a changed context invalidates stale work before an action can be admitted.

```
c(i,j,h_j) = 1 + abs(phi(j)) + h_j
route_cost = sum c over its entered nodes
MOVE requires route_cost + repair_cost <= E
```

This cost is a declared application rule in integer codes, not physical energy calibration. The field shall affect actual route selection. Forecast the internally selected route without mutating canonical live state. Empty forecasts are allowed but shall still validate the seed. Real moves must follow adjacent distinct nodes; target handling uses REPAIR, not a stationary movement.

FI4. Exact packed action and target repair

For admitted i to j, source phi selects the negative/zero/positive increment from recipe turns. Derive tau from the authoritative edge seam set. Apply the increment in the departure frame, then K8 reflection and orientation transport. Require exact destination phi, recomputed parity and a complete mirror.

```
delta = turns[class(phi(i))]; tau = seam(i,j)
t = (r + (-1)^eta * delta) mod 256
r_next = (-1)^tau * t mod 256; eta_next = eta XOR tau
w_next = pack(r_next,j,phi(j),STEP | (eta_next<<4))
E_next = E - c(i,j,h_j)
```

At the freshly observed target, require E at least repair_cost. REPAIR preserves R, G, B and eta, changes the opcode to EMIT, subtracts repair_cost from separate energy and marks COMPLETE. A negative or zero field never means negative or zero energy. Decisions, forecasts, live pairs and energy shall be checked as one admitted history.

Planner-selected destinations are this policy's own internal decisions. FI4 does not replace or reinterpret the fixed per-class routes of standalone FieldMachine. Existing observation admission, event ordering and finite-cycle outcomes remain in force.

Regeneration and device admission

VERIFIED FINITE PROFILE | FI5-FI6 | RESIDENCY AND HISTORY

FI5. Recipe-derived samples and bounded FIFO

```
sample(i) = Pair(pack(0,i,phi(i),DATA))
Regen(recipe,i) = the same certified sample(i)
```

The geometric world shall supply pure derive and caching get. Derive changes no cache order, counters, observations or live state. On CPU it recomputes from the retained recipe; on GPU it re-materializes the sample from the certified field buffer. Get stores atomic complete pairs in a capacity-C FIFO, C a positive strict integer. Hits do not refresh insertion order; overflow and shrinking evict oldest insertions. A later miss for an evicted node must actually reconstruct the sample.

Cold restart rebuilds geometry, exact fields and required operators from the recipe. Regeneration retains baseline/version and original dependencies; it never substitutes a later context. Hazards and other admitted measurements must survive in the observation journal. They cannot be reconstructed from the geometric recipe. Missing original dependencies shall fail admission, not silently synthesize measurements.

```
active sample payload <= 8*C bytes
M_total = M_FIFO + M_recipe + M_fields + M_operators
+ M_geometry + M_observations + M_journal
+ M_search + M_runtime
```

The full scalar field tuple and GPU scalar/operator arenas are accounted outside the FIFO; the bound does not describe total memory. Python objects also have overhead. Cache capacity and incidental eviction affect residency only and stay outside canonical state. Forecasting must not admit unobserved measurements or populate the observed active sample cache.

FI6. GPU execution with one persistent owner

The device shall construct the field; an independent geometric certificate shall admit it before device operators are used. Compile every allowed neighbor/class operator from certified fields and authoritative seams: four distinct neighbors times three classes, a 12 by N Klein arena. Operators carry delta, destination, destination phi and relative seam bit; live metadata shall never inherit operator bit 6.

Forecast a caller-internal planned route of at most 255 edges in scratch device state without changing the live owner. Accepted MOVE and REPAIR shall dispatch from persistent GPU pair and energy, not upload a host-computed forecast as the action. Compare the device result with the admitted prediction before committing the journal. An uncertain device outcome or prediction mismatch fails closed; reconstruct from the last durable admitted history before further work.

CPU/GPU fields, costs, forecasts, decisions and canonical archives shall agree exactly. Explicit GPU execution shall not fall back to CPU field or transition evaluation. Adapter names, timings and cache diagnostics remain outside canonical state; host planning and admission remain explicit host stages.

Integrated replay and acceptance

VERIFIED FINITE PROFILE | FI7-FI8 | CONFORMANCE

FI7. One journal, durable admission and replay

Reuse Tomigidt’s session/live envelopes, process locks, atomic saves and durable acknowledgment. Field-policy event keys shall be exactly seq,input,decision,output,energy; energy is a strict integer. Snapshots include energy and word_profile; retained planning summaries include energy in their context. Legacy events retain their exact four-key schema.

Replay regenerates geometry and re-admits original observations in order, recomputing each (pair,energy) state rather than installing stored energy. Check every complete event, forecast and expected state, including WAIT/DEFER and search progress. Reject supplied replacement recipes for retained sessions and all replay-inconsistent records. Backend/capacity changes shall preserve canonical archives. Invalid frames shall not mutate live state, energy, search, observations or cache; legacy fixtures remain exact.

FI8. Reference mission and independent checks

Use the 4 by 5 default recipe, start k:0:0, target k:3:2, phase 250, eta 0, energy 100 and repair cost 5. Hazards are zero until hazard 70 at k:0:3 takes effect from cycle 2, and only visible hazards are reported. Initial lexical least-cost route is [4,3,17], cost 5. After entering node 4, the new local observation selects [16,17], cost 3. Edge 4 to 16 is a reversing seam.

Admitted action	Node / phase / phi / eta	Separate energy
MOVE 1	4 / 5 / -1 / 0	98
MOVE 2, seam	16 / 240 / 0 / 1	97
MOVE 3	17 / 187 / +1 / 1	95
REPAIR, EMIT	17 / 187 / +1 / 1	90

With zero hazards and only the centre changed to index 4, the first planned route becomes [15,16,17], cost 4. This ablation proves field geometry changes planning, rather than decorating an unrelated route. Both CPU and GPU now reproduce these formal-first reference vectors and complete canonical archives.

Required checks. Strict recipe/manifest and typed lanes; preserved legacy formats; independently calculated costs and packed words; field ablation, hazard replanning, seam reflection and mirrors; FIFO hit order, real eviction/reconstruction and recipe-only cold rebuild; capacity-independent histories; stale-search invalidation; fresh-process and both-backend replay; device action ownership, no fallback and fail-closed errors. Energy tampering shall fail complete-event replay.

The retained FI integration capture has 23 passing checks and 368 tests, including 41 actual-device methods. Its exact mission archive is preserved under PX indexing. Neither binding establishes global Psi traversal, growth, infinite memory, calibrated physical energy or whole-architecture completion.

Local SDF-derived Psi binding

VERIFIED FINITE PROFILE | PX1-PX2 | FORMAL-FIRST CONTRACT

PX1. A declared local eigen-operator

This profile makes a new explicit numerical choice for the eigenvector/traversal and f8 roles in the original source (original pages 7 and 11; consolidated page 11). It is not asserted to be the unique formula implied by that source. Its operational role is indexing recipe-derived scalar base descriptors. Hadamard routing, a global Psi operator and behavior-changing traversal remain separate bindings.

Use the immutable Klein recipe, canonical node $i=uH+v$ and certified unit-edge scalar field ϕ . Directions $u+$, $u-$, $v+$, $v-$ are the K1 quotient directions in the canonical chart. Define the unscaled symmetric difference g and its positive-semidefinite local operator exactly:

```
g_u = phi(u+) - phi(u-); g_v = phi(v+) - phi(v-)
g = (g_u, g_v); A = g g^T
g != 0: lambda = g_u^2 + g_v^2
P = g / gcd(abs(g_u), abs(g_v))
g == 0: A = 0; lambda = 0; P = (1, 0)
Psi = epsilon * P; epsilon in {-1, +1}
```

For nonzero g , λ is the unique largest eigenvalue, and $A*P=\lambda P$. Integer gcd normalization and the declared sign select a nonzero primitive axis. For $g=0$, all vectors lie in the zero eigenspace; the explicit $P=(1,0)$ tie rule chooses the axis. The fallback is a declared convention, not a recovered physical direction.

PX2. Bounds and chart transport

The certified unit-edge SDF has adjacent differences at most one. Thus both components of g lie in $[-2, 2]$. Gcd-normalized P and either Ψ sign also lie in $[-2, 2]$; λ belongs to $\{0, 1, 2, 4, 5, 8\}$. Use exact widened integer arithmetic; no floating-point eigensolver, tolerance or rounded normalization participates.

```
J = diag(1, -1)
g_prime = J*g; A_prime = J*A*J
Psi_prime = J*Psi
```

These transformations bind an orientation-reversing Klein chart. Transport the chosen axis, including the declared degenerate choice, through J . Keys shall be derived only in the canonical frame. A temporary label or opposite local frame changes displayed components, not the indexed scalar identity.

Measured PX checks cover all 25 gradients, both signs and both local frames. The local axes, key records, full trees and actual GPU lookup results agree with independent cover/BFS oracles. The binding remains a declared finite indexing choice, not a global physical eigenmode.

Canonical index identity and key

VERIFIED FINITE PROFILE | PX3-PX4 | SCALAR BASE DESCRIPTORS

PX3. Immutable binding and derivation identity

IndexBinding has exactly the JSON keys below. Reject missing/unknown keys, Boolean integers and invalid values. Format is f8-klein-sdf-v1; epoch is a strict integer in $0..2^{31}-1$, psi_sign is exactly -1 or +1, and phase_origin is in $0..255$. Defaults are epoch 0, sign +1 and origin 0.

format, epoch, psi_sign, phase_origin

Full version identity is the complete Klein recipe content together with this binding, including baseline ID, geometry, centre/radius and turns. Epoch is scoped to that lineage; it is not a globally unique identity. Index all N recipe-derived scalar base descriptors, independently of FIFO residency. Oriented live agent states are not additional entries.

```
canonical derivation address i = u*H + v
d(i) = unweighted quotient distance from recipe.center
parent(root) = root
parent(i) = min numeric j among neighbors with d(j)=d(i)-1
```

The parent rule selects one finite root path. Numeric node order here differs from FI3 lexical name/route order. G remains the stable canonical node identity throughout; it is never replaced by a tree row. Alias queries reduce signed temporary Klein coordinates to their one canonical scalar identity. Strict k:u:v parsing in existing manifests remains unchanged.

PX4. Radius, phase and the total key

Define rho by integer bit length. It buckets graph radius plus one, and does not implement an active physical log-scale transition. Begin the root derivation with phase_origin and eta=0. Advance the K8 increment/reflection rule along the unique parent path, using each departure node's field class to select its recipe turn.

```
rho(i) = floor(log2(d(i)+1)) = bit_length(d(i)+1)-1
theta(i) = (-1)^eta * r mod 256
= (phase_origin + sum departure turns) mod 256
key(i) = (Psi_u+2, Psi_v+2, rho, theta, i)
```

Each key has five unsigned 32-bit components. Compare them lexicographically by numeric component, never by string representation or a hash. The final canonical node ID guarantees uniqueness even when earlier components tie. A different sign or phase origin may reorder storage while preserving every geometric node and movement edge.

The phase derivation above is recipe/index metadata. It does not install a new phase in Tomigidt, append a movement, consume energy or supply a route to the planner.

Middle-out tree and executable lookup

VERIFIED FINITE PROFILE | PX5-PX6 | ACTUAL DEVICE LOOKUP

PX5. Exact lower-median tree

Sort all canonical node keys. For each half-open sorted interval $[lo,hi)$, choose its lower median; emit that node, then its left and right subtrees recursively. Physical tree rows are this preorder. The resulting row layout is canonical for the complete recipe and binding.

```
mid = floor((lo + hi - 1)/2)
left interval [lo,mid); right interval [mid+1,hi)
tree row = [key0,key1,key2,key3,key4,leftrow,rightrow,0]
node-key record = [key0,key1,key2,key3,key4,lambda,g_u+2,g_v+2]
```

Both record types contain eight u32 values. N is 9..256; null child is 256 and the root is row 0. A valid search takes at most $\text{bit_length}(N)$, at most nine, visited rows. Exact key equality returns the stored row and canonical node identity. A well-formed absent key returns MISS without any state change. Malformed rows, links, ordering, duplicate/missing identities or trees are invalid, not a successful miss.

Tree edges are storage-search relations and shall never be interpreted as movement edges. The geometric graph, field, sensor neighborhood and canonical node identity remain authoritative. CPU world derive shall perform the actual tree walk before freshly regenerating a sample; a decorative tree or a direct node-to-row bypass does not satisfy this requirement.

PX6. Certified construction and actual GPU dispatch

The GPU shall construct keys, Psi values, numeric ranks and the complete tree from certified device fields and immutable geometry, including directional neighbors and parent/depth data. No host key compiler may substitute for explicit GPU construction. Node-key records and all rows shall exactly match independent CPU results.

Compile the neighbor/class operator texture in tree preorder. The shader shall obtain a query key from the current state's canonical G and walk the tree to the actual operator texture row. Validate the returned node identity before using its operator. This lookup must govern actual device actions as well as forecasts; G itself remains the canonical node ID.

Admission certificates shall validate field-derived g, A/eigenvalue, gcd normalization, sign, parent-distance relation, K8 derivation phase and every key. They shall also validate the complete canonical lower-median/preorder tree and its node coverage. Parity or a locally plausible child link alone is insufficient.

Existing FI6 persistent state, no-CPU-fallback, prediction comparison and fail-closed requirements remain binding. A corrupted row or wrong returned identity shall not silently select another node's operator.

Measured scope: nine bounds visited tree rows, not total lookup work. The current shader additionally validates rank in $O(N)$ and checks the parent path. The conformance capture establishes exact results and actual tree use; it does not establish a speedup or a logarithmic total runtime.

Atomic rebuild and semantic preservation

VERIFIED FINITE PROFILE | PX7 | VERSIONED STORAGE

PX7. Prepare, certify and swap one version

A rebuild creates a new immutable IndexBinding with epoch incremented by exactly one; psi_sign and phase_origin may optionally change within their declared bounds. Reject epoch overflow at $2^{31}-1$ before mutation. This is the versioned alternative allowed by original requirement R7, not an in-place ambiguous reordering.

Prepare and validate the complete candidate keys, tree, operator texture and device bind-group bundle before a serialized atomic swap. Each lookup/action uses one complete admitted bundle. Retain enough ownership to distinguish a rejected candidate from an already committed replacement.

Outcome	Required effect
Invalid input or pure certificate failure	Reject with old index, canonical state, active FIFO/order/counters and retained search unchanged.
Uncertain device outcome	Close the owner and retain unchanged admitted history; recover from that history before further work.
Validated candidate, atomic swap	Commit the complete new bundle while preserving semantic history and working residency.
Cleanup failure after the swap	Report a committed cleanup failure. Do not report rejection or pretend the old version remains active.

Storage refinement, not a planning event

Keys and tree order shall not determine planner iteration, costs, lexical ties or search quanta. Rebuilds preserve all agent pairs, separate energy, events, observations, DEFER frontier/cursor and active FIFO contents, order and counters. Residency does not change merely because its descriptor has a new tree row.

Expose the complete binding and version identity in diagnostics. Replay may use a different independently verified index version and must produce the same canonical archive. Do not add incidental row order or index epoch to legacy/field-agent semantic schemas. If a future Psi choice changes route behavior, bind a new semantic version or an admitted event first.

```
M_total_next = existing FI5 accounting + M_index
M_peak_rebuild includes old bundle + candidate bundle
M_index includes all retained descriptor/key/tree metadata
```

Index metadata is separate from the bounded FIFO. It shall not hide a complete arena of packed world-node pairs. Account for both old and candidate allocations during rebuild, including device resources and all retained host metadata. A smaller active-pair capacity is not a bound on total memory.

Current FI1-FI8 behavior remains the refinement oracle: the same admitted observations must reproduce the same movement, energy, repair and persistence history through every allowed storage version.

Psi-index reference and acceptance

VERIFIED FINITE PROFILE | PX8 | VECTORS AND CONFORMANCE

PX8. Default 4 by 5 reference

Use the default Klein recipe (centre 0, radius 2, turns [11,53,137]) and binding epoch 0, sign +1, phase origin 0. The literal orders below follow PX1-PX5. The retained docs/evidence/psi-f8-v1/formal-reference.json contains all 20 node records and the complete tree; these are formal vectors, not runtime measurements.

sorted node IDs:
18,17,19,15,16,4,3,14,13,1,2,11,12,9,0,5,10,6,8,7
preorder node IDs:
1,16,17,18,19,15,3,4,14,13,0,11,2,12,9,6,5,10,8,7

Node ID	Five-component canonical key
0	(3,2,0,0,0)
6	(3,3,1,22,6)
7	(4,3,2,75,7)
16	(1,3,1,22,16)
17	(0,3,2,75,17)

The root is row 0, node 1; its left child is row 1, node 16, and its right child is row 10, node 0. The tree relation 1 to 16 is not a geometric movement edge. Node 16 has $g=(-2,2)$, $\lambda=8$ and $\Psi=(-1,1)$. Its temporary alias (7,-1) has reflected-chart components $g=(-2,-2)$, $\Psi=(-1,-1)$, but the same canonical key.

Required independent conformance

Require independent cover/BFS and field oracles; all 25 possible gradients, both signs and both local frames; eigen-equations, zero-axis tie and bounds; literal keys, complete rows and phase derivations; negative/large aliases; strict binding types, MISS and corrupted trees.

Require real GPU construction and tree-governed texture lookup, corrupted-row rejection and no CPU fallback; actual sample eviction/regeneration; capacity and rebuild invariance including retained DEFER search; complete candidate certification, atomic failure, cleanup-after-commit reporting and epoch overflow. Legacy archives remain bit-exact, and the entire FI8 mission remains unchanged.

Revision 4 acceptance status: VERIFIED FINITE PROFILE. All 20 PX conformance checks pass. The complete suite has 421 passing tests, 55 actual-device methods and zero skips. Four FI8-cycle rebuilds and 41 DEFER-cycle rebuilds preserve the prior canonical archive, energy, FIFO and retained search.

This finite binding supplies local SDF-derived eigenstructure and a canonical middle-out storage index. It does not complete global Psi traversal, Hadamard routing, geometry-changing growth, active scale transitions, physical adapters or indefinite continuation.

Typed Hadamard routing policy

VERIFIED FINITE PROFILE | HP1-HP2 | BEHAVIORAL BINDING

HP1. Explicit semantic policy and schema

FieldAgentManifest gains the distinct policy tomigidt-field-hadamard-plan-act-v1. Only this policy requires the additional JSON key routing; its exact routing keys are format and gains. Format is hadamard-klein-routing-v1. Gains are an immutable 4 by 2 table of strict integers in [-4,4]; reject Boolean integers, missing/unknown keys and malformed shapes. Defaults are shown below.

```
routing = {"format":"hadamard-klein-routing-v1",
           "gains":[[1,1],[-1,1],[-1,-1],[1,-1]]}
```

Existing field policy, manifest schema, FI/PX contracts and legacy archives remain unchanged. Live words retain RP32-relational-sdf-v2 with B=phi, canonical G and separate energy. This new routing object belongs to semantic identity; storage IndexBinding remains deployment metadata.

HP2. Exact local product and movement cost

Use PX field-derived g and primitive Psi in the canonical Klein frame. Use the live intrinsic phase t, never the index derivation phase or index phase_origin. Direction e is exclusively one of the four unit cardinal quotient directions from the source to an adjacent geometric node; diagonal or tree-edge moves are not admitted.

```
t = (-1)^eta * r mod 256; bank = floor(t/64)
d_i = gains[bank][i] * (1 + Psi_i^2)
q = d Hadamard g; M = max(abs(q_u),abs(q_v))
penalty(source,dest,t) = M - dot(q,e)
cost = 1 + abs(phi(dest)) + hazard(dest) + penalty
```

This is fixed-point f=0 with widened signed i32 products/sums and no rounding or clipping. Each q component is in [-40,40], penalty in 0..80 and cost in 1..335. A route of at most 255 hops costs at most 85,425. Zero g gives q=0 and zero penalties. These are integer application costs, not calibrated physical energy.

```
J = diag(1,-1); g_prime=J*g; Psi_prime=J*Psi
d_prime=d; q_prime=J*q; e_prime=J*e
dot(q_prime,e_prime)=dot(q,e)
```

Squared Psi components make both the local chart transformation and IndexBinding sign irrelevant to d. Costs are chart-covariant, and changing index phase_origin changes no routing bank. The complete mirror also preserves t. K8 live transport remains responsible for eta, parity and the full mirrored pair.

Source basis: original pp.5,7,10,11,16,18; Solus pp.3-4; solipsism pp.5,13,16. These motivate qualitative field/operator/traversal roles. HP is an explicit numerical choice, not a uniquely implied source formula, global eigenmode or speed claim.

Phase-state planning and admitted action

VERIFIED FINITE PROFILE | HP3-HP4 | FINITE LIFTED SEARCH

HP3. Immutable phase-state Dijkstra search

Use labels (G, t, hops) , with canonical node G , intrinsic phase t in $0..255$ and hop count in $0..\text{max_hops}$. All geometric adjacencies remain admissible and every edge has positive cost. Node-only labels, or labels omitting t , may merge arrivals with different future costs and are unsound. Revisits are permitted because phase changes can improve subsequent directional costs.

```
label = (G, t, hops)
t_next = (t + turns[class(phi(G))]) mod 256
hops_next = hops + 1
priority = (total cost, lexical route, G, t, hops)
```

Hop bound remains a strict integer in $1..255$; per-cycle search quantum remains $1..65,536$ expansions. Preserve global lexical canonical-name route ties. Use reverse unweighted hop distance to the target to prune labels that cannot finish within the remaining hop budget. Count an expansion only for a non-stale, non-target popped label whose outgoing edges are inspected. Reaching the target ends a candidate route; do not continue through it.

Dominance applies only to the exact (G, t, hops) label. Evaluate entry weights once in lexical canonical-name order. The immutable context includes position, live pair/phase, energy, routing binding, weights and directional model. Retain the complete frontier, best costs, route ties and progress through DEFER. Changed effective hazards invalidate the cursor; invalid input preserves it and all canonical state.

HP4. Observation, reserve and receding planning

Fresh complete local frames still precede MOVE or REPAIR. Sensors supply only admitted local hazards; unseen distant nodes retain the zero-hazard hypothesis. Search, rather than the sensor, selects a route. Forecast its full K8 packed/energy sequence; admit one adjacent action and debit that first edge's HP2 cost. Require the whole known route cost plus repair reserve to fit available energy.

At the freshly observed target, retain FI4 REPAIR semantics. No planner action bypasses the target check. Under unchanged effective weights, after following the first edge of an optimal route, its remainder is still feasible within the next full hop budget. The next optimal cost is therefore at most the previous optimal cost minus that strictly positive edge cost. Receding optimal value strictly falls despite possible phase cycles.

```
V_H(G_next, t_next) <= V_H(G, t) - cost(G, G_next, t)
< V_H(G, t)
```

This decrease assumes an unchanged model and successful admitted movement; new observations may change its value. Full RP32 execution applies the existing source-field increment, seam reflection and eta XOR. The resulting intrinsic phase obeys t_{next} above independently of seam/orientation, while R and η still follow K8.

The finite label bound is $N * 256 * (\text{max_hops} + 1)$, not a constant-memory search guarantee. Search/history/object overhead remains separate from active world-pair FIFO capacity.

Routing atlas and certified export

VERIFIED FINITE PROFILE | HP5-HP6 | CERTIFIED DEVICE MODEL

HP5. Movement and cost in one exact atlas

The HP GPU operator atlas is an r32uint texture of width 24 and height 4N. Within each source, neighbor slots are sorted by numeric canonical destination ID. Its row is selected through the actual f8 tree walk; the row is storage, while source/destination G remain geometric identities.

```
y = bank*N + f8_row(source)
x = 6*numeric_neighbor_slot + 2*field_class
atlas[x,y] = existing RP32 movement operator
atlas[x+1,y] = pack(bank,source,penalty,DATA)
```

There are four banks, four neighbor slots and three field-class copies: 48 movement/cost pairs per node. The movement word retains the existing delta, destination, exact destination phi and relative seam metadata. The cost word has R=bank, G=source and B=penalty. Recompute valid parity; no operator-only seam bit may leak into live metadata.

The shader obtains bank from the persistent live pair for actual actions and from the evolving scratch pair for forecasts. Actual tree lookup governs the atlas row in both paths. Compile the atlas from certified device field, g and Psi together with the immutable gains; no host-compiled replacement may stand in for explicit GPU construction.

HP6. Export the admitted table, then search on host

An export_routing pass reads the actual atlas and validates all 48 movement/cost pairs for each node: parity, complete movement/cost payload, source/bank/destination, class increment and all three repeated penalty copies. It exports exactly 16N penalties and N source increments. A malformed unselected bank or class is still invalid.

```
routing_model = 16*N penalties + N source increments
entry_weight(j) = 1 + abs(phi(j)) + hazard(j)
edge_cost(i,j,t) = entry_weight(j) + penalty[i,bank(t),j]
```

An independent host certificate checks the exported table against the exact field-derived HP2 equations and semantic routing binding. CPU routing derives its equivalent model independently. The host Dijkstra heap consumes the certified device-produced table; it shall not run a host routing compiler as GPU fallback or perform a GPU dispatch for each expansion.

The retained semantic model contains those 17N integers and effective entry weights, not a CPU scalar-field arena. Geometry and field samples remain recipe-derived under their existing contracts. CPU/GPU costs, forecasts, choices and admitted histories must agree for the same HP manifest and observations.

For HP owners, the complete atlas and routing-table certificate extends FI6/PX6 admission. Existing persistent-action ownership, no-fallback and uncertain-outcome closure requirements remain binding.

Hadamard rebuild and resource contract

VERIFIED FINITE PROFILE | HP7 | ONE ADMITTED BUNDLE

HP7. Atomic index replacement with routing resources

An HP index rebuild prepares a complete candidate index, tree, routing atlas, exported routing tables and bind groups. Certify the whole candidate before a serialized atomic swap. Preserve the full semantic routing object; a storage rebuild cannot silently change gains, costs or policy.

Outcome	Required result
Invalid input / complete certificate rejection	Old index/atlas/model remains usable. Preserve canonical history, energy, FIFO contents/order/counters and the exact pending search.
Uncertain dispatch or readback	Close the owner with the last admitted history unchanged. Reconstruct from durable admitted history before more work.
Successful complete-bundle swap	Install the new storage version while retaining identical semantic costs, samples, state and search cursor.
Post-swap cleanup failure	Report a committed cleanup failure, not rejected replacement. Account for and attempt cleanup of retained resources.

Index sign and index phase_origin do not change q or t. Thus PX version changes preserve every HP planning label, comparison, cost, expansion quantum, pair, energy value and event. FIFO residency is also preserved. Replay may select a different certified storage index while retaining the same semantic manifest and observation history.

Explicit allocation and peak accounting

```
routing atlas = 24 * (4*N) * 4 = 384*N bytes
exported routing table = 17*N*4 = 68*N bytes
retained host routing table = 68*N bytes
gains table = 4*2*4 = 32 bytes
```

These are logical payload sizes for the declared atlas/table/gains resources, not total process memory. Track all additional host/device index metadata, field/geometry buffers, entry weights, scratch buffers, bind groups and runtime allocations. Rebuild peaks include complete old and candidate bundles together, rather than counting only their difference.

```
M_peak includes old(index,atlas,tables,bind groups)
+ candidate(index,atlas,tables,bind groups)
+ retained fields, geometry, FIFO and runtime
```

The world-pair FIFO retains its 8C-byte payload bound and actual regeneration obligation. Search may have $N \cdot 256 \cdot (\text{max_hops} + 1)$ labels. Frontier/routes, routing tables, observations and journal memory remain outside the FIFO bound.

No throughput, speedup, saturation or physical-energy claim follows from these allocations. A comparative performance result would require the separate measurement protocol on page 30.

Hadamard reference and acceptance

VERIFIED FINITE PROFILE | HP8 | INDEPENDENT VECTORS

HP8. Default phase-directed mission

Use the FI8 4 by 5 recipe and local hazard change, with the HP1 default gains, initial phase 250, eta 0, energy 100 and repair cost 5. Initial pair is 91FE000601FE00FA. The independent reference uses quotient arithmetic, BFS, phase-state Dijkstra and a layered-DP cross-check; it imports no solvefinite runtime module.

Cycle / action	Route; known cost; expansions	Actual pair / energy
1 / MOVE	[4,3,17]; 7; 23	11FF04FB01FF0405 / 98
2 / seam MOVE	[16,17]; 7; 6	81001010910010F0 / 93
3 / MOVE	[17]; 2; 2	81011145910111BB / 91
4 / REPAIR	[]; repair 5; 0	06011145160111BB / 86

From cycle 2, hazard 70 at node 3 changes the route after entering node 4. Geometry and packed seam transport remain the FI8 values, but HP2 adds directional costs. Zero gains recover the old edge costs and final energy 90. With only initial phase changed to 192, the first route is [5,10,11,12,17], cost 12; completion takes six cycles, final pair 960111E38601111D and energy 83.

Why the search state must contain phase

At start node 0, target 10, intrinsic phase 128 and max_hops 12, correct labels (G,t,hops) select [1,6,5,10], cost 10. Incorrect merging by (G,hops) selects [5,10], cost 11. This is a literal finite counterexample to phase-free state compression, not a performance comparison.

Required independent conformance

Require strict new-policy schema with exact legacy preservation; integer bounds, zero gains, all banks/signs/frames, chart covariance and mirror invariance; independent lifted-search/DP route and cost oracles, revisits, hop/quantum limits, stale-model invalidation and DEFER replay. Sensor inputs shall not supply routes or actions.

Require real GPU atlas compilation and all 48-pair payload checks, certified exported tables, no CPU compiler fallback, actual indexed action/forecast bank selection, corruption in unselected copies, resource peaks, pure rejection, uncertain-outcome closure and committed cleanup failures. Capacity/rebuild changes preserve FIFO, cursors and archives; fresh-process cross-backend/live continuation remains exact.

Revision 6 acceptance status: VERIFIED FINITE PROFILE. The new capture has 479 passing tests, 77 actual-device methods and 15 HP conformance checks. Earlier 421/55/20 PX counts retain their historical identity. Implementation evidence follows on page 49.

Retained reference: docs/evidence/hadamard-v1/formal-reference.json and independent reference-builder.py. The equations and vectors are a declared finite routing binding, not global Psi, physical energy calibration, a speed claim or whole-architecture completion.

Measured Hadamard implementation

REVISION 6 EVIDENCE | HP1-HP8 | FINITE PROFILE

HP1-HP8 were committed before runtime coding at 0c862c3. The implementation extends the same Tomigidt individual. Its certified gradient and local Psi, together with its live intrinsic phase, now alter route costs and actual CPU/GPU actions.

Capture	Measured result
Complete regression suite	479 passed; 0 skipped; 74.444 seconds.
Actual-device coverage	77 executed GPU methods; 6 domains, 830 nodes and 26,560 penalty comparisons.
HP conformance	All 15 named checks passed. The one-expansion DEFER mission takes 32 cycles and preserves search across every rebuild.
Behavioral ablations	Default: four cycles, energy 86. Zero gains: energy 90. Initial phase 192: a different first route, six cycles, energy 83. Exact pairs and routes remain on page 48.
Device payload / rebuild peak	57,272 / 67,752 bytes; complete old and candidate bundles are counted.
Routing allocations	Atlas 7,680 bytes; device/host tables 1,360/1,360 bytes. Geometry, runtime and Python-object memory remain additional.

What changes the next action

Separate node/phase/hop labels preserve distinct futures. The device atlas holds four phase banks; export admission checks every bank/class copy. Forecasts and persistent actions use their current phase and actual f8 lookup. Storage changes preserve routing, search, FIFO and history. Continuation across CPU/GPU backends and failure cases are identified in the retained logs.

Reproduce and inspect the evidence

```
python -m unittest discover -s tests -v
python -m examples.hadamard_conformance --output hp.json
```

Retained capture: docs/evidence/hadamard-v1/. verification.json binds source/report hashes at commit 5a304bc, formal chronology and measured counts. Earlier FI/PX captures retain their named commits and original counts.

Remaining scope: global eigenmodes, geometry-changing growth, active scale transitions, physical adapters and wider continuation. These finite measurements establish no GPU saturation, general speedup or calibrated physical-energy claim.

Autonomous growth binding

VERIFIED FINITE PROFILE | GD1 | SEMANTIC PROFILE

Source basis: original sections 3, 8, 14 and 16; Solus page 6; Solipsism pages 13 and 16. The addendum describes growth through changes to internal distance rules. GD1-GD8 choose a finite dyadic production for the same Tomigidt individual. The source does not supply this numerical production or its target, cost and continuation rules.

GD1. A versioned production with one history

```
policy = tomigidt-field-growth-plan-act-v1
growth.format = klein-dyadic-growth-v1
growth = {format, max_epochs, cost}
```

The exact new FieldAgentManifest schema requires both routing and growth, in addition to the previous field-agent keys. Routing obeys HP1. GrowthBinding is immutable: max_epochs is a strict integer in 0..2 and cost is a strict integer in 1..127. Defaults are max_epochs=1 and cost=1. Reject Boolean/noninteger values, wrong formats and unknown or missing keys. Prior policies reject growth and preserve their original canonical schemas.

```
N_initial * 4^max_epochs <= 256
epoch k starts at 0; 0 <= k <= max_epochs
```

Validate the maximum node bound before allocating a candidate. When max_epochs is positive, the initial radius must be at least one. Other initial recipe, word, routing, observation, energy, search and cycle bounds retain their declared meanings. The immutable manifest stores the initial recipe, initial target and complete grammar binding; it is not overwritten with a later recipe.

The live state additionally retains semantic geometry epoch k, current recipe and current target. A production extends this same individual and its ordered history. It neither creates a replacement agent nor resets its energy, global cycle or event sequence. Producer epochs and interchangeable storage-index epochs are distinct from k.

Revision 8 status: VERIFIED FINITE PROFILE. GD1-GD8 preceded implementation at ec1181e; the prefix-identity refinement preceded its runtime field at 00b0e64. The separate GD capture has 546 passing tests, 107 actual-device methods and 16 conformance checks. Page 57 identifies that evidence.

This binding is a finite grammar with one production family. It does not complete arbitrary L-systems, cone/pyramid synthesis, physical growth or indefinite continuation.

Dyadic geometry and active scale

VERIFIED FINITE PROFILE | GD2 | QUOTIENT PRODUCTION

GD2. Generate a larger intrinsic quotient

```
(W,H,c,r,k) -> (2W,2H,E(c),2r,k+1)
E(u,v) = (2u,2v)
G = u*H+v; E(G) = (2u)*(2H)+2v
rho(k) = rho(0)*2^k; log-radius exponent = k
```

Apply exactly one finite generation. All new edges have unit integer weight. The new Klein quotient, oriented seams, centre and intrinsic-ball boundary are regenerated from the production. Node names remain canonical for that epoch. New names alone cannot identify a historical sample: the geometry epoch is part of its semantic context.

Quotient correspondence and transport

The doubling map respects $(u,v) \sim (u,v+H)$ and $(u,v) \sim (u+W,-v)$: doubling either representative gives equivalent new representatives. Every old cardinal edge maps to a two-edge cardinal path. A reversing old edge crosses the new reversing seam once; a nonreversing old edge crosses it zero times. Thus the composed seam XOR agrees with the original edge.

Retain the current R and η exactly through the growth mapping. The intrinsic phase $t=(-1)^\eta R \bmod 256$ is therefore preserved, as is the full mirror relation after repacking. Map the agent and centre with E . Reconstruct the new signed field from the new boundary and its shortest-path distances, using widened exact arithmetic and the existing range certificate.

```
R_next = R; eta_next = eta
G_next = E(G); B_next = phi_next(E(G))
E_energy_next = E_energy - growth.cost
```

Do not compute B_{next} by shifting or doubling the old B . The newly admitted boundary-distance certificate is authoritative. The next live pair has STEP metadata; the pre-growth pair is the EMIT pair from the completed subgoal. Recompute parity and the complete mirror.

Here active scale is the declared intrinsic dyadic generation: lengths of embedded edge paths and the ball radius expand by two while the new lattice retains unit edges. The exact exponent k controls actual geometry and its operators. No geographic origin, calibrated physical unit or claim of unchanged physical resolution is imposed.

Literal field counterexample: $W=3, H=5, \text{centre}=1, \text{radius}=3, \text{old } G=0$ gives $B=-3$. After growth, $W=6, H=10, \text{centre}=2, \text{radius}=6$ and mapped $G=0$ give $B=-4$, not -6 . Generated boundary vertices change the nearest-boundary distance.

Connectivity, audited quotient cells/links, boundary separation, exact field range, local Ψ , f_8 and HP operator conformance must hold at every candidate epoch. A rejected candidate is not an admitted generation.

Subgoals, growth and energy reserve

VERIFIED FINITE PROFILE | GD3 | ONE AUTONOMOUS LIFECYCLE

GD3. Complete a subgoal, then derive the next world

Within an epoch, retain HP planning, fresh local observations, positive directional costs, full forecasts and one-edge action admission. At the freshly observed current target, REPAIR debits the manifest repair_cost and emits the same EMIT pair as FI4. The default repair cost remains five.

Condition after REPAIR	Required status and continuation
k = max_epochs	COMPLETE. This bounded growth mission has reached its final subgoal.
k < max_epochs	GROWTH_PENDING. Retain the repaired EMIT pair. Do not report COMPLETE or silently move in the old geometry.

A new complete fresh local observation frame in the old geometry is required before GROW. The fixed grammar then derives the candidate, maps the state and selects the next target by GD4. Exactly one admitted GROW event advances the geometry epoch. Growth is an internally selected lifecycle action; sensors never provide a replacement world, target or route.

Reserve future finite obligations

```
m = max_epochs-k
reserve(k) = repair_cost + m*(growth.cost+repair_cost)
known_route_cost + reserve(k) <= available_energy
```

Require this reserve before MOVE and REPAIR; a reached target has known_route_cost=0. It includes the current repair and the remaining growth/repair costs, not the still-unknown routes in future geometries. Those routes must separately fit the energy when their epochs become current. Unseen hazards retain the HP zero-hazard hypothesis.

```
At GROWTH_PENDING, require:
growth.cost + reserve(k+1) <= available_energy
```

A positive growth cost is debited exactly once; energy is never replenished by a generation. Insufficient energy prevents the action under the declared agent status rules. The existing finite max_cycles bounds the whole mission across all geometry epochs; global cycle and event ordering do not restart.

On a successful GROW, clear effective observations, retained search and active sample FIFO. The new epoch requires a new complete fresh local observation before MOVE or REPAIR. Previous observations remain in admitted history for replay; they are not silently interpreted as measurements of the new geometry.

A zero-growth binding executes one HP subgoal with no GROW events. Max_epochs selects this finite mission size; it does not claim whole-system indefinite progression.

State-selected next target

VERIFIED FINITE PROFILE | GD4 | INTRINSIC DETERMINISTIC SELECTION

GD4. Choose among genuinely generated nodes

Let $a = E(\text{current_node})$ in the certified candidate quotient. New nodes are exactly those whose canonical u or v is odd. Use the independently certified candidate field and unweighted shortest-path distance from a . The selection order below is semantic and independent of f8 storage order.

```
C0 = {v : u(v) is odd or v_coordinate(v) is odd}
b = min(abs(phi_next(v)) for v in C0)
C1 = {v in C0 : abs(phi_next(v)) = b}
d = max(distance(a,v) for v in C1)
C2 = sorted({v in C1 : distance(a,v)=d}, by numeric G)
target_next = C2[t mod len(C2)]
```

$C0$ is nonempty after a valid dyadic production; subsequent finite minimization and maximization preserve a nonempty candidate set. Lexical node-name order is not used for this final tie. The live intrinsic phase t is unchanged by full mirroring and by index sign/origin changes, so equivalent mirrored states select the same canonical target.

A semantic generation changes future decisions

The new recipe, new certified ϕ , rebuilt local eigenstructure, routing model and selected target jointly define the next planning problem. The individual continues with its retained identity, mapped pair and remaining energy. The newly created nodes are eligible for subsequent motion through their genuine geometric adjacencies.

For an explicit GPU realization, target selection must consume the GPU-produced field after independent admission. No host field compiler may substitute a CPU field for that source. Host admission and deterministic target selection are permitted; their complete inputs and resulting target remain replayable.

The choice uses current packed phase and generated geometry rather than an externally supplied list of future targets. Changing the phase may select a different tied target; it cannot change the declared priority of boundary proximity or graph distance.

A storage reindex is still a refinement that preserves history and search. GROW is a semantic transition that changes the world, target and model context. These events have deliberately different admission and invalidation effects.

Literal tie: in the 8 by 10 candidate with centre 0, radius 4 and mapped current node 0, $C2 = [13, 17, 31, 39, 51, 59, 73, 77]$. Intrinsic phases 0, 1, 2, 6, 3 select targets 13, 17, 31, 77 respectively.

Two-generation reference: a 3 by 3 quotient, centre 0, radius 1, start 0 and target 4, phase 250, eta 0, energy 100 and zero hazards grows at cycles 4 and 10. New targets are 31 then 15. Cycle 25 ends at pair 86000F0E16000FF2, energy 33, in the 12 by 12 quotient. These are mathematical expected values.

Atomic growth and device mapping

VERIFIED FINITE PROFILE | GD5 | COMPLETE CANDIDATE ADMISSION

GD5. Prepare, certify, map, then commit

Prepare a complete candidate audited quotient, boundary field, local Psi/f8 structure, HP atlas, certified exported routing model, empty sample FIFO and runtime resources. The candidate must pass the existing independent geometry, field, index and all-bank/class operator certificates before it can replace the old world.

For explicit GPU execution, a device growth-mapping pass reads the old packed pair and energy with the old height, computes the mapped G, reads the certified new phi, debits growth cost and writes the candidate canonical pair/energy. Read back and independently validate this result before the serialized swap. Host seeding of a precomputed mapped pair is not an allowed fallback.

Validate the old EMIT pair, full mirror, old G/B context, preserved phase/orientation, mapped node, new B, STEP metadata and exact energy debit. The new device executor starts its per-epoch tick count at zero. This counter is diagnostic/local execution state; the agent cycle and event sequence remain global.

Outcome	Required effect
Pure precommit candidate failure	Reject with old archive, world, energy, FIFO and pending search unchanged. Dispose candidate resources.
Uncertain candidate dispatch/readback	Close the candidate and conservatively close the owning agent; retain its last admitted history.
Uncertain old-device result	Close the owner. Do not fabricate a successful mapping or acknowledge a new generation.
Complete certified swap	Install the new world and canonical state, debit once, clear current observations/search/FIFO and append exactly one GROW event.
Post-swap cleanup failure	Preserve the admitted GROW event in memory, report committed=True and close the owner. Do not report a rejected production.

A failed live session is fatal and must reopen the last durable file. If an atomic save was uncertain, that file can contain the history before or after GROW; replay and retry handling determine the admitted result. No in-memory guess overrides the durable archive.

```
M_active_pairs <= 8*C bytes
M_peak_growth >= M_old_world + M_candidate_world
```

GD7 resource accounting includes fields, topology, indexes, routing, state, grammar, journals, observations, search and all host/device metadata. Resource ownership remains explicit until cleanup completes; the active-pair FIFO bound is not a total-memory bound.

Epoch-aware replay and regeneration

VERIFIED FINITE PROFILE | GD6-GD7 | CONTEXT AND RETENTION

GD6. Preserve the context of every observation

```
event = {seq,input,decision,output,energy,geometry_epoch,growth}
receipt = {format,from_epoch,to_epoch,mapped_node,target,recipe}
```

For this policy only, geometry_epoch is the pre-cycle integer; growth is null except on GROW. Its receipt has exactly the keys above: format is klein-dyadic-growth-v1, mapped_node and target are new canonical names, and recipe is the full new recipe. Strict replay regenerates every event and receipt from the immutable initial manifest and admitted prefix.

Snapshots add exactly geometry_epoch, current_recipe and scale_exponent=k. The existing target is the current target, footprint.epoch=k, and baseline remains initial. A nonnull planning context additionally contains geometry_epoch. Earlier policies retain their schemas.

A new-policy live observation supplies geometry_epoch matching its pre-event context; validate locality in that geometry. Reject a stale/future epoch on a new event. A duplicate earlier producer sequence instead uses its original epoch/locality and returns the recorded event without another action or debit.

The supplied simulator applies its original hazard schedule only at epoch 0; generated epochs receive fresh zero-hazard frames in their current geometry. This is a declared simulation assumption. Live input may supply arbitrary admitted local hazards. Sensors never supply worlds, targets, routes or actions.

GD7. Generation-qualified identities and finite resources

```
derive_epoch(epoch,path) -> qualified DATA sample
recipe_at_epoch(epoch) -> admitted epoch recipe
```

The sample contains geometry_epoch, origin_sequence (zero initially; otherwise the original GROW sequence), prefix_sha256, initial-recipe/grammar identity, path and pair. Reconstruct its recipe from the initial recipe and admitted production prefix; only already admitted epochs are accessible. Retain original context rather than substituting the current tick or recipe.

```
prefix = events[:origin_sequence]; initial prefix = []
prefix_sha256 = SHA256(canonical_UTF8(prefix)).hexdigest()
```

The digest is 64 lowercase hexadecimal characters. Canonical encoding is json.dumps(prefix, sort_keys=True, separators=(',', ':'), ensure_ascii=True, allow_nan=False).encode('utf-8'), with no trailing newline. Include every original event through the admitting GROW event, not only its sequence number. Equal epoch, recipe, node and DATA pair do not identify the original prefix; prefix_sha256 records its digest. Reindexing, cache capacity and exact replay preserve this digest.

Historical derivation does not mutate the current FIFO or state. GROW starts an empty active FIFO; capacity changes may alter residency but not samples or decisions. Reindex preserves the current recipe, target, observations, search and identities; GROW invalidates that semantic context. Resource bounds and complete candidate peaks are on page 54.

The generation exponent, global cycle, producer epoch/sequence and storage-index epoch have separate purposes. None may silently substitute for another during replay, retries or regeneration.

Growth reference and acceptance

VERIFIED FINITE PROFILE | GD8 | INDEPENDENT VECTORS

GD8. Mathematical reference before runtime

Use the HP8 default world, initial pair, energy and epoch-0 hazard timeline; max_epochs=1, growth cost 1 and repair cost 5. The independent reference imports no solvefinite module; every MOVE route is cross-checked by hop-layer dynamic programming.

```
cycle 4 REPAIR: 06011145160111BB, energy 86
cycle 5 GROW: 01024045110240BB, energy 85
cycle 14 REPAIR:160027E906002717, energy 64
```

GROW maps node 17 to 64 in an 8 by 10 quotient and selects new target 39 (k:3:9). Full fields, missions, mirror/tie vectors and quotient checks are retained in docs/evidence/growth-v1/formal-reference.json with its independent reference-builder.py.

Required conformance

Obligation	Required evidence
Strict profile and finite bounds	Exact schemas and legacy preservation; Boolean/overflow rejection; max_epochs=0,1,2; preallocation rejection when $N \cdot 4^{\text{max_epochs}}$ exceeds 256.
Production and transport	Full quotient/cell audits, equivalent aliases, every old edge's two-edge image and seam XOR, fresh exact field and full mirror/phase preservation.
Autonomous continuation	Actual REPAIR, GROWTH_PENDING, GROW and later movement/repair in one history; literal targets and energy reserves; phase-dependent target ties; no external recipe/action supply.
Device refinement	GPU-produced fields and growth-mapped canonical state; no CPU compiler or host-pair fallback; certified fields, tree, all HP atlas copies and actual post-growth device actions.
Atomicity and durability	Pure rejection, uncertain candidate/old device, post-swap cleanup failure, committed status, save uncertainty and reopen/retry before and after durable growth.
Replay and residency	Generation-qualified historical samples, empty new FIFO, stale observation rejection, duplicate original-epoch retries, current-geometry simulation and cross-backend/process continuation.

Capacity and index variations must preserve history at every epoch. Reindex during DEFER and after growth; invalidate semantic planning only for growth or changed effective observations. Test cycle/energy boundaries without resetting the individual.

Revision 8 acceptance: VERIFIED FINITE PROFILE. The separate runtime/device capture passes all 16 named conformance checks. The independent formal vectors remain expected outputs; page 57 reports their tested implementation and retained evidence.

The full architecture still includes general production systems, global spectral traversal, physical adapters, wider temporal continuation and measured performance. Completing this finite profile does not discharge those obligations.

Measured autonomous geometry growth

REVISION 8 EVIDENCE | GD1-GD8 | SAME INDIVIDUAL

The same Tomigidt now repairs a subgoal, derives a larger Klein quotient, maps its packed state on the device and continues to an internally selected target. GD1-GD8 were committed at ec1181e before runtime coding; the original-event-prefix digest was bound at 00b0e64 before its runtime field. Neither step rewrites the initial manifest.

Capture	Measured result
Complete regression suite	546 passed; 0 skipped; 139.982 seconds.
Actual-device coverage	107 executed GPU methods. All 16 named GD conformance checks pass, including device mapping and post-growth actions.
One generation	14 cycles, final energy 64. The 20-node world becomes 80 nodes; cycle 5 maps node 17 to 64 and selects target 39.
Two generations	25 cycles, final energy 33. The 9-node world becomes 36, then 144 nodes, under the same global history.
Retained incremental search	Quantum 7: 61 cycles. Reindexing throughout generated-world DEFER preserves the canonical history.
Device payload / growth peak	120,232 / 177,504 bytes in the default final world / complete old-plus-candidate growth preparation.

Original context survives continuation

Historical samples retain geometry epoch, original admission sequence and the digest of the exact original event prefix. Capacity changes and storage rebuilds preserve that context. Sensors describe the current geometry; stale epochs are rejected and earlier duplicates use their original geometry. The retained suite covers rejected candidates, uncertain results, committed cleanup failures and durable save/reopen boundaries.

CPU/GPU histories agree for default, mirrored, zero-growth and two-generation missions. The conformance capture includes six fresh-process continuations before/after growth and during generated-world DEFER, with host field/index/routing compilers disabled on explicit GPU paths. A separate CLI capture resumes GPU to CPU to GPU and verifies read-only inspection.

```
python -m unittest discover -s tests -v
python -m examples.growth_conformance --output gd.json
```

Retained evidence: docs/evidence/growth-v1/. verification.json binds source hashes at historical commit 94f86c7, report hashes, actual counts and formal-first commits. Device figures are logical resource payloads; Python objects, drivers, journals and other overhead remain additional.

Remaining scope: general production tables and branch stacks, global eigenmodes, physical adapters, wider temporal continuation and comparative hardware measurements. This finite growth result establishes no GPU saturation, speedup, calibrated energy advantage or whole-architecture completion.

Parameterized branching organogram

VERIFIED FINITE PROFILE | OG1 | NEW NUMERICAL BINDING

Policy: tomigidt-field-organogram-plan-act-v1. The same individual uses its prior certified field to interpret finite parameterized productions. Emitted intrinsic balls define its next field on the existing Klein quotient. This changes the boundary and distances, not the quotient dimensions, edges, cocycle, departure turns or original index anchor.

OG1-OG8 retain their formal-first requirements at commit 5c76f2c. Revision 10 adds source-bound CPU/GPU implementation evidence on pages 67-68. GD and all earlier policies retain their contracts.

OG1. Strict immutable grammar schema

```
binding = {format,max_epochs,cost,symbols,axiom,rules,
           generations,limits}
format = "klein-branch-organogram-v1"
```

Field	Exact finite domain
Epochs / cost / generations	max_epochs 0..4; cost 1..127; generations 0..8.
symbols	Ordered list of 1..16 distinct {name,arity}; names match [A-Z][A-Z0-9_]{0,15}, excluding F,R,S,SCALE; arity 0..4.
axiom	Ordered list of 1..32 tokens. A token is exactly {symbol,args}; args is an ordered array. Symbol and argument count match its declaration.
rules	Ordered list of 0..64 {symbol,guards,rhs}; left symbol is a declared nonterminal. Guards number 0..8; RHS tokens number 0..32. No terminal is a rule left side.
limits	Exactly max_symbols 1..1024, max_steps 1..4096, max_balls 1..64 and max_stack 0..32.

Fixed terminal arities are F:1, +:1, -:1, [:0,]:0, R:1, S:0 and SCALE:1. Every listed key is required; undeclared keys, unknown symbols, wrong arities and Boolean or floating-point integers are invalid. Ordered lists remain immutable internally.

OG-TAPE32-v1 codes: F,+, -, [,],R,S,SCALE = 0..7
v = (a mod 256) + 2^8*floor(a/256) + 2^24*code
word = v + 2^31*(popcount(v) mod 2)

The separately tagged 32-bit instruction profile uses the sole tape argument a, or 0 when absent; scaling occurs in interpreter state. Decode a from bits 0..15 and code from 24..26; require bits 16..23 and 27..30 zero, even parity and terminal-specific bounds on device. One exact r32uint texel per instruction uses integer textureLoad in tape order, without filtering. This is not a current RP32 state or field sample. No instruction mirror is asserted: the same tape acts on full cursor pairs. Addresses and branch identities remain separate metadata.

Source basis: original pp.4-5,7,10,12,16,18; Solus pp.3-4,6; solipsism pp.13,16. These motivate grammar, field-guided generation and branch context. The selected syntax, limits, ball union and interpreter are explicit numerical choices. Cone/pyramid primitives, general graph rewriting, global Psi and clock-wrap continuation remain separate obligations.

Parameters, productions and finite tape

VERIFIED FINITE PROFILE | OG1-OG2 | SINGLE-VALUED EXPANSION

OG1. Parameters and exact predicates

Parameters and literals are signed i32. An axiom argument is a literal or exactly {context,mul,add}, with context tick, epoch, phase or field. A rule RHS argument is a literal or exactly {arg,mul,add}, with arg a valid left-side parameter index. mul is -32768..32767; add is signed i32. Evaluate multiplication/addition as mathematical integers and reject a result outside signed i32 before device dispatch.

```
tick = original admitting GROW sequence
epoch = next geometry epoch in 1..4
phase = (-1)^eta * R mod 256; field = old B
affine(x) = mul*x + add
```

A standalone stage tick is a strict positive signed i32; the owner also requires its actual sequence within max_cycles. A comparison guard is exactly {arg,op,value}, op eq, ne, lt, le, gt or ge, with signed i32 value. An XOR guard is exactly {arg,op,mask,value}, op xor_eq, with strict u32 mask/value. Every guard arg is a strict valid left-side parameter index, never Boolean or out of range.

```
xor_eq(a,mask,value) iff
((a mod 2^32) XOR mask) == value
```

OG2. Parallel rewrite and permanent addresses

Each generation reads only its previous word and arguments. For a nonterminal, select the first matching rule in list order; all guards must hold, and an empty guard list is true. An empty RHS erases that token. Unmatched symbols and terminals pass unchanged. New RHS tokens cannot be rewritten until the next generation. Each intermediate word, including the axiom, obeys max_symbols. Any nonterminal surviving the final generation rejects the stage.

```
axiom address = (i,)
replacement: append (generation,rule_index,rhs_index)
pass: append (generation,-1,0)
```

Indices i, rule_index and rhs_index are zero-based; generations begin at 1. A pass address is appended for every passing token, including terminals. Final tape order retains {address,symbol,args}; addresses are flat integer arrays in serialized form.

Before device work, preflight the complete tape. Brackets shall balance and every prefix depth is 0..max_stack; an initial POP is invalid. Track radius/scale with branch restoration. F takes $n=1..256$ and executes $n*2^{\text{scale}}$ steps; + and - take $m=1..16$; R takes radius 1..127; SCALE sets absolute exponent 0..4. Each S emits $\text{radius} * 2^{\text{scale}}$, at most 127. Total F steps obey max_steps and S count is 1..max_balls.

Finite generations, words and interpreter work prove stage termination. There is no partial grammar GROW or grammar DEFER; agent planning still supports DEFER. Because DEFER advances the logical cycle, changing the manifest search quantum can change the original GROW tick and hence a time-conditioned production. Retain that manifest/input history; different search budgets need not yield the same mission. Cache and reindex operations do not advance T and must preserve results. XOR predicates do not replace parity, mirrors or geometric certification.

Prior-field interpreter and branch context

VERIFIED FINITE PROFILE | OG3 | ORDERED PACKED EXECUTION

The complete stage uses its prior certified phi as a fixed waveguide. Begin at the actual repaired EMIT pair, changing only its opcode to STEP; local radius is 1 and scale is 0. Recompute parity and the full mirror whenever repacking. There is no independent heading state.

OG3. Field-directed F and phase operators

```
t = (-1)^eta * R mod 256; bank = floor(t/64)
d_i = gains[bank][i] * (1 + Psi_i^2)
q = d Hadamard g; choose e maximizing dot(q,e)
```

At every F substep, derive g and primitive Psi from the prior phi using PX, and q using HP. Choose among the four cardinal quotient edges. Break equal scores by the order [u+,v+,u-,v-] rotated left by bank positions. Apply the existing K8 departure-field-class increment and seam transport; destination B is the prior phi there. The updated intrinsic phase chooses the next lookup.

Emit one segment after every F substep, retaining the terminal address, current branch path, 1-based substep and complete resulting STEP pair. Segments guide later placements; only S emits a boundary primitive. A +m or -m terminal applies respectively +m or -m times turns[class(prior phi at G)] through orientation-aware phase arithmetic modulo 256. It does not move the node.

Terminal	Interpretation effect
R(radius)	Set the unscaled local radius.
SCALE(s)	Set absolute scale exponent s; semantic lengths multiply by 2^s. No destructive byte shift or global scale is implied.
S	Emit centre G, effective scaled radius, current STEP pair, terminal address and branch path.
[	Push complete cursor pair, radius, scale and enclosing branch path; append this PUSH terminal address to that path.
]	Restore exactly that saved context, including phase, orientation, field and complete mirror.

What branch restoration does and does not restore

Immutable tape arguments and the original stage context supply the rule environment. Program position, work counters, accumulated trace/segments/balls, owner energy and admitted history never rewind. Record a trace entry after every terminal, including PUSH and POP. The final hypothetical cursor does not replace the owner: OG5 admits the new field at the owner's unchanged node/phase/orientation.

A saved orientation bit alone is not the complete frame. Restoration must recover the entire pair and local shape context even after a nested reversing-seam crossing, signed turn and local scale change.

Intrinsic ball union and exact new field

VERIFIED FINITE PROFILE | OG4 | GEOMETRIC CONSTRUCTION

Every S emission declares an intrinsic ball on the unchanged unit-edge Klein quotient K . Let its canonical centre and positive integer radius be (c_i, r_i) . The primitive list is ordered and retains duplicates; geometry is determined by its union potential.

```
q(v) = min_i [d_K(v, c_i) - r_i]
sigma(v) = sign(q(v)); Z = {v : q(v)=0}
phi_new(v) = sigma(v) * min_(z in Z) d_K(v, z)
```

Reject a candidate with empty Z . Compute distance to the new zero set and apply the existing independent field certificate against the fixed graph and these signs. The union potential q is not generally the final signed distance: an individual ball boundary can become internal to the union. No old or per-ball distance may substitute for the certified result.

Theorem OG-A: valid separating sides

Each $v \rightarrow d_K(v, c_i) - r_i$ is integer-valued and 1-Lipschitz on a unit edge. Their finite minimum is also 1-Lipschitz: choose a minimizing index at either endpoint and use its edge bound in each direction. Therefore adjacent q values differ by at most 1. Adjacent vertices cannot have opposite nonzero signs. Nonempty Z and the existing $G1/G2$ certificate then give exact signed boundary distance and its Lipschitz bound.

Optional exact device metric

```
cyc_H(x) = min(x mod H, H - (x mod H))
d_K((u, v), (a, b)) = min(
  abs(u-a) + cyc_H(v-b),
  W-abs(u-a) + cyc_H(v+b))
```

Canonical representatives use $0..W-1$ and $0..H-1$. Lifts of (a, b) are $(a+kW, (-1)^k b+nH)$. Even horizontal lifts preserve the second coordinate; odd lifts reverse it. Their shortest horizontal displacements are $\text{abs}(u-a)$ and $W-\text{abs}(u-a)$. Independently minimizing vertical displacement over n gives the displayed formula. Independent BFS shall check it throughout the supported finite quotient domain before relying on this device construction.

The original Klein cells, adjacency, cocycle and orientation-cover certificates stay unchanged. The new boundary, signs, exact field, PX descriptors and HP operator tables receive their own candidate certificates. The index anchor remains `base.center`. Reject any new distance outside the existing exact signed-code domain.

These are intrinsic discrete balls and their union. The formula supplies no Euclidean sphere surface, cone, pyramid, continuum embedding or calibrated physical unit. Different stage fields genuinely change the represented internal distance rules while retaining the same quotient topology.

The same individual admits a generated world

VERIFIED FINITE PROFILE | OG5 | SUBGOAL, TARGET AND ENERGY

The initial KleinFieldRecipe remains immutable. The new policy manifest includes organogram and the existing Hadamard routing binding. After REPAIR, if the current epoch k is below max_epochs , enter GROWTH_PENDING. The next complete fresh local observation frame admits one GROW transition; observations cannot supply productions, primitive centres, targets or actions.

```
reserve(k) = repair_cost
+ (max_epochs-k)*(organogram.cost+repair_cost)
```

Apply the existing GD reserve/admission bounds using `organogram.cost`. MOVE and REPAIR must preserve the required remaining reserve. GROW requires its cost plus `reserve(k+1)`, consumes one global cycle, and debits its cost exactly once. Global cycle, energy and event order never reset. With `max_epochs=0` the initial mission finishes without a grammar stage.

OG5. Admit the field, not the hypothetical cursor

```
G_new = G_old; R_new = R_old; eta_new = eta_old
B_new = certified_phi_new(G_old)
opcode_new = STEP; energy_new = energy_old-cost
```

Recompute complete mirror/parity and validate this result against the candidate field. Clear effective observations, pending search and active FIFO after successful semantic admission. Retain baseline, manifest and admitted history. Final REPAIR at `max_epochs` completes the finite mission.

State-selected next target

```
C0 = all canonical nodes except owner G
C1 = argmin_(v in C0) abs(phi_new(v))
C2 = argmax_(v in C1) unweighted_hops(G,v)
target = sorted_numeric(C2)[intrinsic_phase mod len(C2)]
```

The topology is unchanged, so this rule uses no born-node condition. Tie-breaking is by numeric canonical G , not lexical name order. Full mirrors have the same intrinsic phase and select the same target. The new target is internal policy state; subsequent planning and actions use the certified new field and HP routing.

Observations and retries retain GD epoch semantics: new input must name the current pre-cycle geometry epoch and satisfy its locality; a duplicate original producer sequence resolves in its original geometry and returns its original result. Simulator hazards remain an explicitly selected input model. Arbitrary admitted local live hazards remain possible.

Cost is an abstract mission-resource unit, not measured physical energy or interpreter work; F steps have a separate bound. Reindex preserves semantic field, target, observations, search and identities; GROW invalidates their context. A successful integration shall demonstrate changed distances and autonomous behavior, not only a detached tape.

Original recipes, transcripts and replay

VERIFIED FINITE PROFILE | OG6 | COMPLETE DERIVATION CONTEXT

```
recipe = {format,base,organogram,routing,stages}
```

```
format = "klein-organogram-world-v1"
```

```
stage = {epoch,tick,start_pair,prefix_sha256}
```

base is the original KleinFieldRecipe; organogram and routing retain the immutable bindings. A generated recipe has 1..max_epochs stages; epoch 0 uses the base type. Stage epoch is 1..length, ticks strictly increase, and start_pair is the original full EMIT pair as 16 uppercase hex characters. Require valid parity/mirror, legal node and B equal to the preceding field. Reconstruct preceding stages before the next; never substitute current time or recipe.

```
stage.prefix_sha256 = SHA256(canonical(events before GROW))
canonical(x) = json.dumps(x, sort_keys=True,
separators=(',', ':'), ensure_ascii=True,
allow_nan=False).encode('utf-8')
```

No trailing newline; digests are 64 lowercase hex characters. Excluding this GROW avoids a circular digest through its receipt. A digest binds retained context, not authentication or a spatial seed. Grammar reads no energy or sensor arguments; these remain in the journal. Historical samples separately hash through their original admitted GROW under GD7, permit only admitted epochs, and retain full base, grammar and original context.

OG6. Canonical derivation document

```
derivation = {format,context,tape,trace,segments,balls,
              final_context}
```

```
format = "klein-organogram-derivation-v1"
```

Record	Exact keys and meaning
context / tape	context is the stage. Each tape item is {address,symbol,args}; addresses are flat integer arrays.
trace	After every terminal: {address,branch_path,pair,radius,scale}. radius is unscaled.
segments	{address,branch_path,step,pair}; step is 1-based within this F terminal's effective steps.
balls	{address,branch_path,center,radius,pair}; center is numeric G and radius is the effective scaled radius.
final_context	{branch_path,pair,radius,scale}; final branch_path is empty.

branch_path is an outermost-first array of PUSH terminal addresses. Every transcript pair is the current prior-field STEP pair, encoded as 16 uppercase hex characters. Hash the complete document with the canonical encoding above. Schema, ordering and numeric/string types are part of the digest contract. Event/receipt and snapshot schemas continue on page 64.

Device production and atomic admission

VERIFIED FINITE PROFILE | OG6-OG7 | REFINEMENT AND RESOURCE OWNERSHIP

OG6. Event, receipt and snapshot schema

event = {seq,input,decision,output,energy,
 geometry_epoch,growth}
receipt = {format,from_epoch,to_epoch,mapped_node,
 target,recipe,derivation_sha256}
geometry_epoch is the pre-cycle epoch; growth is null except on GROW. Receipt format is klein-organogram-growth-v1 and mapped_node is the unchanged canonical owner name. derivation_sha256 hashes OG6's complete document. The OG snapshot adds geometry_epoch and current_recipe; unlike GD it has no global scale_exponent because branches may use different local scales.

OG7. Produce on device, then certify

The host parses, expands and preflights immutable tapes for all stages before even base GPU allocation, using their declared contexts. This is host work, not proof that declared B matches the actual preceding field; verify that later per stage. The device derives every cursor/branch state and primitive from tape and prior device phi, then signs and distances. Rebuild PX/HP from the new device field.

Independent certificates validate actual outputs; they shall not replace device trajectories, fields, indexes or routing tables with CPU-produced results. Construction and validation shall use the certified device manifest path without an eager recipe property silently invoking a host field producer. Explicit GPU operation permits no CPU fallback.

Candidate admit_generated reads actual old device pair/energy. Require exactly one added stage and identical base, grammar and routing; for a first stage compare the owner's immutable binding. Match actual start_pair and the owner's pre-event prefix; coincident fields do not prove history. Compute OG5 state without a host seed after initialization and check every candidate output before serialized swap.

Outcome	Required ownership result
Pure precommit rejection	Preserve old world, pair, energy, observations, search, FIFO and archive; dispose the detached candidate.
Uncertain device work	Close affected candidate and owner conservatively; retain the last admitted history, without fabricated success.
Certified swap	Admit one GROW, install field/target/state, debit once and clear the prescribed semantic context.
Committed cleanup failure	Preserve the admitted event, report committed=True and close the owner. Durable reopen/retry resolves saved state.

Valid productions may yield an empty boundary or other rejected field; arbitrary grammar and finite energy/cycles do not guarantee GROW completion. Account old plus candidate host/device payloads, tapes, stacks, traces, recipes, journal and cache separately, including rejected complete-candidate peaks. The pair FIFO bound of 8C bytes is not total constant memory.

Independent organogram reference

FORMAL-FIRST REFERENCE | OG8 | EXPECTED RESULTS BEFORE RUNTIME

The independent reference in docs/evidence/organogram-v1/ expands the grammar and evaluates integer branches, quotient BFS, union redistance and lifted HP planning without importing solvefinite. A layered dynamic-programming oracle cross-checks its mission paths. These are mathematical expected results, not runtime conformance.

Standalone context and branch restoration

The retained standalone input names epoch 1, original tick 5 and repaired pair 06011145160111BB. Its explicit prefix is the digest of []; this mathematical input is not a claim that a runtime admitted sequence 5 with an empty journal. Production addresses, traces, branch paths and the complete derivation document remain in the reference.

Nested branch moment	Exact current context
Before first POP	91FE00B181FE004F; radius 2, scale 1, eta 0. Nested F visits 12,11,10,15,0 across the reversing seam.
After first POP	81020C5791020CA9; radius 1, scale 0, eta 1; enclosing branch path restored. Emitted balls are (12,1), (0,2), (2,1).

The retained default grammar mission expects 9 cycles, energy 76 and final pair 860006BA16000646 at target 6. Two epochs expect 14 cycles, energy 66 and pair 06000E2F16000ED1; successive targets are 6 and 14. These predictions use the full fixture inputs retained with the reference.

Union potential is not signed boundary distance

```
3 by 3 quotient; balls (centre,radius) = (0,2),(4,2)
q = [-2,-1,-1,-1,-2,-1,-1,-1,0]
phi = [-2,-1,-2,-2,-2,-1,-1,-1,0]
Z = {8}; q and phi differ at canonical nodes 2 and 3
```

Both fields encode the same side classification, but only the recomputed phi is exact distance to the union boundary. This literal case detects an implementation that substitutes min per-ball residual for the required redistance.

Exact instruction-carrier vectors

```
F(256) 80000100 +(16) 01000010 -(16) 02000010
[ 03000000 ] 84000000 R(127) 8500007F
S 06000000 SCALE(4) 07000004
```

Independent BFS checks all 702 supported quotients, 106,045 vertices and 19,130,481 ordered pairs against the closed metric. The builder pins reference and generator hashes. Mathematical mission prefixes are scoped to that transcript, not runtime canonical archive hashes; actual journal/source identities require separate evidence.

Organogram acceptance obligations

VERIFIED FINITE PROFILE | OG8 | REQUIRED ACCEPTANCE SCOPE

The retained reference in docs/evidence/organogram-v1/ supplies independent grammar expansion, integer interpretation and geometric calculations without importing solvefinite. Its vectors are mathematical expectations prepared before runtime implementation; a separate source-bound CPU/GPU capture must demonstrate refinement.

OG8. Required acceptance evidence

Object	Required scope
Grammar and bounds	Rule priority, guard conjunction and XOR, erasure, passing terminals, unmatched final symbols, affine overflow, malformed schema, nesting and every declared size/work limit.
Complete branch frame	Nested reversing-seam crossings and signed turns with radius/scale changes; exact POP restoration of full pair and context. Trace/counters/emissions continue without rewind.
Geometric truth	Closed-metric/BFS agreement over supported quotients; union potential versus true-distance counterexample; empty-boundary rejection; independent certificate for generated fields.
Context and identity	Original tick and pre-GROW prefix, immutable stages, canonical transcript digest, full mirrors, original-epoch retries and admitted historical regeneration.
Same individual	A changed generated field changes routes/actions; multi-epoch continuation retains energy/cycles/history; target selection and reserve bounds agree independently.
Device refinement	Actual GPU interpreter, traces, balls, signs, fields and subsequent actions; host producers/compilers disabled on explicit GPU paths; fresh-process cross-backend replay.
Atomicity and retention	Pure rejection, uncertain dispatch/readback, committed cleanup and save/reopen boundaries; cache/reindex/planner-DEFER preservation; unchanged earlier-policy hashes.

Revision 10 status: VERIFIED FINITE PROFILE. The separate implementation capture passes the required finite acceptance scope: 623 tests, 136 actual-device methods and 15 conformance checks. Measured results and limits follow on pages 67-68.

This finite organogram does not complete cone/pyramid or arbitrary graph production, global eigenmodes, WElip/clock-wrap continuation, physical adapters, computational universality or comparative hardware performance. Completing OG must be judged against the full contract, not only literal reference examples.

Measured organogram continuation

IMPLEMENTATION CAPTURE | OG1-OG8 | 26 SEPTEMBER 2026

The grammar now generates a new exact signed boundary field on the original Klein topology. The same Tomigidt admits it, keeps its actual node/phase/orientation and energy history, derives a new target and continues. The independent expectations on page 65 preceded implementation at formal commit 5c76f2c.

Measurement	Retained observed result
Complete regression	623 tests passed in 204.464 s; zero skipped. 136 actual-device methods include 29 OG methods.
Independent conformance	All 15 checks passed; capture wall time 17.037 s. Full stage tapes, traces, branch paths, balls, fields and digests agree with the independent reference at original runtime context.
Adapter	NVIDIA GeForce RTX 5070 Ti Laptop GPU; Vulkan; driver 591.59; wgpu 0.32.0; Python 3.12.14.

Measured missions agree across CPU and actual GPU

Fixture	Cycles / final energy	Final pair
Default; target 6	9 / 76	860006BA16000646
Full mirror; target 6	9 / 76	16000646860006BA
Two epochs; targets 6,14	14 / 66	06000E2F16000ED1
Zero epochs; initial target	4 / 86	06011145160111BB

With search quantum 7, DEFER advances original time: GROW occurs at sequence 8 and the mission completes in 17 cycles with energy 74. Different original ticks select different productions. Under the same quantum and admitted inputs, cache churn, reindex and CPU/GPU continuation preserve the complete canonical history.

Canonical default runtime archive SHA-256

e2834ba2cc6b4d51f2b23f274cd9f2a0db8
a4e88754ed9d2aadc76c074f134df

Join the displayed hash lines. The runtime stage-1 derivation digest is 30f0164177a5e289cb9e7b7e244960c4bef49bcc444208cb94c618295a0c2aa3. It differs from the standalone page-65 digest because it binds the actual pre-GROW journal prefix. Timing here measures correctness capture, not comparative hardware performance.

Evidence supports the declared finite OG profile. It does not establish general universality, texture-cache saturation, physical-energy advantage or indefinite autonomous continuation.

Organogram device and retention evidence

IMPLEMENTATION CAPTURE | DEVICE WORK, FAILURE AND RESOURCE SCOPE

The GPU reads OG-TAPE32 instructions from an integer texture, interprets branch state, emits balls, derives signs, reconstructs exact field distances and supplies subsequent packed actions. Host work performs strict parsing, bounded expansion, independent certification, route search and admission. Six OG producers and eleven earlier CPU compilers are disabled on the guarded GPU paths; the owner is never host-reseeded after initialization.

Boundary and continuation evidence

Coverage includes 32 nested branches and 4,096 actual F substeps; nested seam/radius/scale restoration; the literal union-redistance counterexample; and an unbranched F(1),S case whose hypothetical cursor leaves the actual owner. Rejection probes cover malformed/over-budget grammar, empty boundary, forged certificates and mismatched original context.

Pure precommit rejection preserves the old owner. Uncertain device failure closes it without admission; committed cleanup failure retains GROW. Save failures on either side of replacement, original-epoch retries and durable reopen are exercised. Six fresh-process CPU/GPU crossovers cover before GROW, after GROW and generated-world DEFER. Earlier FI/HP/GD archive hashes remain unchanged.

Explicit payloads for the default CLI continuation

Payload	Observed bytes / interpretation
Current device	61,140 = 52,824 buffers + 8,316 textures.
GROW preparation peak	118,412 device bytes, including old and candidate bundles plus grammar work. Instruction texture 76; movement texture 320; grammar scratch 3,472.
Shader / retained grammar	Private branch stack 640 logical bytes; retained grammar recipe 2,043; transcript 4,564. Peak host stage payload 11,379.
Separate active FIFO	16 pair bytes for CLI capacity 2. Current host index 1,296; routing table 1,360; field codes 80.

Logical private-stack bytes do not identify physical GPU residency. Payload accounting excludes Python objects and driver overhead; geometry, observations, planning and journals remain additional. Preparation/rebuild peaks are distinct. The sample FIFO does not bound total memory.

Recheck execution and rebuild this retained edition

```
python -m unittest discover -s tests -v
python -m examples.organogram_conformance --output og-recheck.json
python tools/build_formal_spec.py
```

Historical capture f125a76: docs/evidence/organogram-v1/ retains verification.json, full-tests.txt, conformance.json and cli-replay.json. The builder checks that commit's source inventory/hashes, retained reports, chronology, reference, device counts and archives. Rechecking another checkout does not reproduce these historical counts. Remaining work at that capture includes general primitives/graphs, global spectral choices, WElip/clock continuation, physical adapters and comparative hardware measurements.

WElip forward-time binding

APPENDIX MAP | W1-W8 | VERIFIED FINITE PROFILE

Revision 11 bound W1-W8 before implementation at 74e00f4. Revision 12 preserves those numerical definitions and adds the source-bound CPU/GPU capture on pages 82-83. Independent reference vectors remain mathematical expectations; prior FI/PX/HP/GD/OG evidence retains its original scope.

One separate W public interface owns the same existing Tomigidt field agent. It supports FI, HP, GD and OG, defaults to OG and preserves their numerical semantics and canonical agent archives. The five forward operations add an explicit clock, lifecycle, typed record and durable admission boundary around that individual.

Page	Complete section
70	W1-W2: immutable configuration, clock and request context
71	W3: ignition, advancement and local lifecycle operations
72	W4: forward admission and latest-request retry
73	W5: exact 16+16+32 carrier and state decoding
74	W5: LUS envelope, identity and separate energy
75	W6: ownership, private archive and cache witnesses
76	W6: internal R recovery and whole-operation atomicity
77	W7: actual GPU state emission and failure handling
78	W8: JSONL transport and complete response schemas
79	W8: validation order and exact error classes
80	Independent preimplementation W reference vectors
81 82 / 83	W8 acceptance obligations; measured implementation and recovery evidence

Source authority and new numerical choices

Original p.4 requires retained output-affecting inputs and clock epochs; p.8 identifies the WElip 16+16+32 carrier; p.14 separates record identity, integrity, admission and reproduction; p.15 defines the five-event forward alphabet, local invalidation and controlled R/W separation. The exact schemas, finite bounds, codec, clock cadence and retry protocol below are declared realization choices.

All old policies, live sessions and CLI commands retain their meanings. W is an API/profile boundary, not OS isolation or issuer authentication. Its finite clock and retained ledger do not establish indefinite time or constant total memory.

W configuration and independent clocks

FINITE CONTRACT | W1-W2 | STRICT INPUT AND TIME DOMAINS

```
protocol = "welip-field-agent-v1"
config = {format,agent,producer,producer_epoch,
          clock_origin,max_events,initial_capacity}
format = "welip-field-config-v1"
```

Field	Exact binding
agent	Existing FieldAgentManifest schema, with one of the four policies listed below; default OG.
producer	Strict string equal to its trimmed form, nonempty, at most 128 characters; default sensor.
producer_epoch	Strict u32; default 0. Producer namespace, distinct from clock carry.
clock_origin	Strict unsigned 48-bit integer; default 0.
max_events	Strict integer 1..1,000,000; default 1,000,000.
initial_capacity	Strict integer 1..256 complete pairs; default 2.

Before owner/device allocation require $\text{clock_origin} + \text{max_events} \leq 2^{48} - 1$. No Boolean or float integer coercion, unknown keys or malformed nested JSON is permitted. CLI profile selectors and full agent policies are:

```
field      tomigidt-field-observe-plan-act-v1
hadamard   tomigidt-field-hadamard-plan-act-v1
growth     tomigidt-field-growth-plan-act-v1
organogram tomigidt-field-organogram-plan-act-v1
```

W2. Four distinct clocks and counters

```
n = durably admitted W operations, initially 0
T_W = clock_origin + n
clock_epoch = T_W >> 16; tick16 = T_W & 65535
```

Each new operation, including IGNITE and EMIT, advances n and T_W exactly once. Only ADVANCE increments agent.cycle ; geometry_epoch remains the existing semantic generation. The device action counter is separate again. W operations never reset energy, agent history or OG's original GROW cycle, time or prefix. Records from one operation share T_W and use contiguous record_seq starting at 0. There is no epoch reuse or operation-sequence rollover.

Every request: exact common keys

```
{protocol,op,producer,producer_epoch,seq,clock_epoch,
 tick16,agent_cycle,geometry_epoch}
```

Add only page 71's operation fields. seq is $1..\text{max_events}+1$; a new operation requires $\text{seq}=n+1$ and $\text{clock}=\text{clock_origin}+\text{seq}$. clock_epoch is u32; tick16 is u16; agent_cycle is $0..\text{agent.max_cycles}$; geometry_epoch is $0..\text{policy.max_epochs}$, or 0 for FI/HP. Cycle/generation describe the current preoperation owner and are both 0 for initial IGNITE. At $n=\text{max_events}$ only exact latest retry is allowed. Static schema/source checks precede retry classification; current-state checks follow it.

The five forward operations

FINITE CONTRACT | W3 | ONE OWNER, EXPLICIT LOCAL EFFECTS

IGNITE: add {payload}

payload is uppercase, even-length hexadecimal encoding 0..4,096 bytes. Only seq=1 before ignition is valid. Retain those literal bytes with the configured baseline and initial owner clock/phase context. They do not replace the initial RP32 state or become instructions. Emit raw payload record 0 followed by actual owner-state record 1. Empty payload still emits record 0 with zero fragments.

ADVANCE: add {position,observations}

Require prior ignition, an open owner below its cycle bound and status other than COMPLETE. WAITING, SEARCH_DEFERRED, UNREACHABLE, INSUFFICIENT_ENERGY and GROWTH_PENDING may admit fresh observations. position is the current canonical path. observations contains only currently visible local paths and strict integer hazards 0..127. Incomplete frames retain existing WAITING semantics.

Delegate exactly one existing Tomigidt.step; the agent chooses its action. Preserve each FI/HP/GD/OG policy's fresh-frame, energy reserve, GROW and DEFER contracts. Emit actual postoperation owner-state record 0.

RESIZE: add {capacity}

capacity is a strict integer 1..256 pairs. Set the active FIFO capacity and evict oldest entries until it fits, preserving unaffected order. Identical capacity is a valid new operation. Emit actual owner-state record 0 without an agent step or energy debit.

INVALIDATE: add {paths,cause}

paths is a sorted, unique list of 1..256 canonical paths, all currently active at the request's current geometry epoch. cause is a nonempty string equal to its trimmed form, at most 128 characters. Validate the entire selector before removal. Remove complete DATA pairs and preserve unaffected FIFO order. Append removed paths to the eviction diagnostic in original FIFO order so a later get() records regeneration.

Retain baseline, recipes, field/index/routing, observations, pending search, owner pair, energy and admitted history. Retain cause in the W request; emit actual owner-state record 0. This is the bound local invalidation action, with no automatic invalidation for a rejected read.

EMIT: no additional fields

Emit actual current owner-state record 0 and advance W time, without an agent step, energy debit or cache access. After agent COMPLETE, EMIT/RESIZE/INVALIDATE remain available within the W budget; ADVANCE and a new IGNITE reject. Before IGNITE every other operation rejects. Every operation uses the same owner; cache controls never spawn or reseed it.

Forward boundary and latest retry

FINITE CONTRACT | W4 | NO PUBLIC HISTORICAL PAYLOAD

Explicit public alphabet and reverse-lock predicate

Only IGNITE, ADVANCE, RESIZE, INVALIDATE and EMIT are public W operations. Explicit op names READ, HISTORY, REPLAY and REGENERATE reject as BACKWARD_READ. Other names and extra selectors are invalid requests. There is no public historical archive, derive_epoch, legacy inspect/status or raw-owner accessor through W. Ready/next admission metadata is transport context, not another W operation.

Detection is explicit at this interface. Rejected requests leave admitted clock, history and cache unchanged. They do not automatically invalidate memory, change parity or authenticate the producer. Controlled private R recovery is defined on page 76.

Exact latest retry precedes dynamic context checks

Only the most recent durably admitted sequence $n > 0$ may be retried. Compare the complete request with the retained latest request using canonical JSON bytes: sorted keys, compact separators, ensure_ascii=True and finite values. Static shapes/ranges and source identity are checked first. Exact equality is checked before current position, generation, active membership, budget or terminal tests, because successful admission may already have changed them.

Request relation	Required result
$\text{seq} = n$; exact bytes	DUPLICATE, containing current admission cursor and seq only beyond the common response keys.
$\text{seq} = n$; changed bytes	CONFLICT. Do not reinterpret it against the new owner context.
$\text{seq} < n$	STALE_SEQUENCE without fetching historical payloads.
$\text{seq} > n + 1$	SEQUENCE_GAP. $\text{seq} = 0$ is always invalid.
$\text{seq} = n + 1$	Apply the new-operation admission order on page 79.

A duplicate returns no records, old event, packed pair, energy or old cache payload. It does not step, dispatch device emission, mutate cache/clock or write the archive. Current admission names in next remain transport metadata, with no past fields or pairs.

Acknowledgment loss and delivery meaning

A lost current-state observation can be replaced by a new forward EMIT at a new clock. This is not recovery of the old record. EMIT does not reproduce IGNITE's raw bytes: lost raw ignition output has no public recovery operation, although admitted bytes remain available to private R. No physical exactly-once actuation is implied.

WElip carrier and exact state decoding

FINITE CONTRACT | W5 | TYPED 16+16+32 WORDS

```
word profile = "welip-16-16-32-v1"
```

```
W64 = payload32 | (phase16 << 32) | (tick16 << 48)
phase16 = intrinsic_phase8 << 8
intrinsic_phase8 = (-R if eta else R) mod 256
```

All integer fields are strict. Canonical transport words contain exactly 16 uppercase hexadecimal digits. tick16 and phase16 are unsigned 16-bit values; payload32 is unsigned 32-bit. State phase has zero low eight bits: it represents the same angle in 1/65,536-turn units at the existing 8-bit resolution. This binding introduces no additional phase precision.

Raw bytes and finite fragmentation

For byte length L in 0..4,096, split bytes into consecutive four-byte little-endian chunks, zero-padding the final chunk. `fragment_count=(L+3)//4`; words contains exactly that many W64 values in original order. Empty L has `words=[]`. Decode exactly L bytes; never infer length by deleting zero padding.

Reject malformed word syntax or range, wrong tick/phase headers, count mismatch or nonzero final padding. The original bytes, including meaningful trailing zeros, must round-trip. Raw IGNITE bytes carry no RP32 semantics or implied parity/integrity.

Owner pair byte order and validation

```
state bytes = LE32(left_RP32) || LE32(full_mirror_RP32)
state byte_length = 8; fragment_count = 2
```

The pair display convention puts its high word first; payload encoding instead writes the left u32 little-endian, then the full-mirror u32 little-endian. Do not confuse printed pair order with byte order. Each W word carries one of those u32 payloads with the same tick/phase header.

State decoding additionally verifies each RP32 parity, the full mirror relation, legal STEP or EMIT metadata and agreement of the phase header with the left owner channel. Admission requires exact equality of the full owned pair, including orientation/opcode, plus energy and context. Mirror-swapping channels yields a different owned pair even when the representation remains legal.

Word ordering and parity are representation checks. Coordinated payload changes can create another legal record; actual owner/context equality is a separate admission check. Neither the carrier nor a hash supplies issuer authentication or a spatial seed.

LUS envelope and derivation identity

FINITE CONTRACT | W5 | COMPLETE RECORD SCHEMA

```
record = {format,word_profile,payload_profile,baseline_id,
producer,producer_epoch,clock_epoch,tick16,phase16,
operation_seq,record_seq,kind,agent_cycle,geometry_epoch,
energy,byte_length,fragment_count,words,status}
format = "welip-lus-v1"
```

Field group	Exact domain and meaning
Profiles	word_profile is welip-16-16-32-v1. payload_profile is bytes-v1 or RP32-relational-sdf-v2.
Baseline / producer	baseline_id is the immutable initial agent.world.baseline_id, using its existing recipe schema. producer and producer_epoch retain the W configuration bounds.
Time / record order	clock_epoch:u32; tick16/phase16:u16; operation_seq:1..max_events; record_seq:0..1.
Owner context	agent_cycle:0..agent.max_cycles; geometry_epoch:0..policy max_epochs, or 0 for FI/HP.
Separate energy	Actual postoperation owner energy, strict integer $0..2^{31}-1$. Explicit metadata; never hidden in SDF B or W64 padding.
Kind / status	kind is the admitted operation name. status is exactly ADMITTED.
Payload	byte_length:0..4,096; fragment_count=(L+3)//4; ordered words as page 73. State is exactly 8 bytes / 2 words.

Every key, type and profile must be exact. IGNITE has bytes-v1 record 0 and state record 1; each other operation has only state record 0. All records carry the postoperation agent cycle/generation, actual energy, the same W time and actual intrinsic phase. Raw ignition therefore shares the initial owner context without becoming agent instructions.

Identity under the retained configuration

```
ID = (enclosing profile, baseline, producer, producer_epoch,
clock_epoch, tick16, operation_seq, record_seq)
```

This identity is interpreted under the fixed retained configuration and original ignition input. Producer namespace allocation is a declared responsibility, not a theorem of global uniqueness. Retain literal original inputs; no hash seeds phase. Equality of identity, equality of bytes, successful private reproduction, representation integrity and origin/admission are separate claims.

The 64-bit carrier does not contain the whole envelope. Finite ordered uniform words carry payload fragments; explicit surrounding metadata carries the full clock epoch, energy, profiles, byte length and derivation identity.

Durable W ownership and archive

FINITE CONTRACT | W6 | COMPLETE PRIVATE STATE

One protected path and serialized owner

Acquire canonical-path OS StateLock, then the session RLock, then existing Tomigidt operation and cache/device locks in that order. Hold admission, execution, save and result construction inside one ownership boundary. Opening a new file binds configuration and creates the existing agent without implicitly igniting it. Persist genesis with an empty W ledger.

Initial capacity belongs to configuration. Reopen rejects supplied configuration mismatch or a constructor capacity override. Recover the mutable current capacity by replaying retained RESIZE operations from initial_capacity; current capacity may legitimately differ from that initial value. Backend and storage index binding may change on reopen, preserving admitted semantics.

Exact archive and operation row

```
archive = {format,config,operations,agent,expected}
format = "welip-field-session-v1"
row = {request,records,agent_pair,energy,status,cache}
cache = {capacity,active_paths,hit_count,regeneration_count,
         evicted_count,removed}
expected = {seq,clock_epoch,tick16,ignited,cache}
agent is the unchanged canonical agent archive. operations is the ordered private W ledger with at most
max_events rows. agent_pair is exactly 16 uppercase hexadecimal digits in the existing pair display
convention. Row energy and status equal the actual owner energy and mission status; row status is distinct
from the record status ADMITTED.
```

Cache witnesses and local lifecycle

active_paths contains unique canonical names in FIFO order, bounded by capacity. The three counters are strict nonnegative integers. removed lists explicit RESIZE/INVALIDATE removals in original FIFO order. Other operations use removed=[]; their ordinary sample evictions remain diagnostics. Final expected.cache always has removed=[], because it witnesses current state rather than repeating the last removal receipt.

State	Required witness
Genesis	seq=0; clock=clock_origin; ignited=false; capacity=initial_capacity; active_paths=[]; all three counters 0; removed=[].
GROW	Install the existing fresh FieldWorld: preserve capacity, reset active entries, counters and eviction history. Earlier W ledger rows remain retained.
Reopen	Replay lifecycle in original order and verify every cache witness, including capacity and current FIFO order.

Archive parsing validates all exact types/shapes and bounds. Baselines, recipe inputs, private records, observations, planner state and journal are additional to the active pair FIFO; this profile makes no total constant-memory claim.

Private recovery and operation atomicity

FINITE CONTRACT | W6 | DURABLE COMMIT AND FAILURE SCOPE

Controlled internal R behind the W boundary

Recreate the same agent at initial capacity and replay W operations in order, interleaving original ADVANCE observations with RESIZE and INVALIDATE. Verify every resulting record, owner state and cache witness, then the complete final canonical agent archive and expected W context. Restoring only the agent archive at final capacity is insufficient.

Recovery may recompute on CPU or actual GPU. It emits no public records, receipts or new durable operations. Device computation may repeat; physical exactly-once actuation is not claimed. No eager historical packed-world arena or public old-epoch read is introduced. Original agent-cycle time and GROW prefixes remain unchanged by intervening W operations.

Commit before acknowledgment

Compute the operation result and next archive, then call atomic write_json before acknowledging. Its payload flush/fsync and replacement semantics apply; this is not a guarantee against every hardware or power failure. A successful durable admission consumes one W sequence/time. Lost transport acknowledgment is resolved by exact latest retry without executing that operation again.

Failure boundary	Required outcome
Pure validation / rejected output	Remain usable only when the entire admitted semantic W preoperation state is proven unchanged.
After ADVANCE/GROW or cache mutation	Emission failure poisons the owner unless complete preoperation restoration is proven.
Save failure or uncertain replacement	Close/poison; no acknowledgment. A new owner consults disk to determine the durable prefix.
In-memory action before replacement	Reopen from the prior durable prefix. Unwritten state is not a durably admitted operation.
Committed GROW cleanup failure	Preserve existing internal GROW semantics; close the affected owner. Never acknowledge an unsaved W event.
After successful save; transport lost	Latest retry returns only the receipt/current cursor, with no action or emission dispatch.

The admitted semantic prestate includes all owner state: pair, energy, cycle, history, recipe, generation, target, decision, planning and observations; cache order/counters; W clock/ledger and configuration. Ephemeral emission scratch bytes are excluded. Emission may be validated before mutating a cache control. Unchanged GPU state across emission alone does not prove a preceding ADVANCE was unchanged.

Unexpected postmutation failure closes the owner. Closed/poisoned embedded owners refuse further requests without mutation or emission. Exact latest retry classification precedes budget, terminal and current-context tests only while a valid owner is open.

Actual device W emission

FINITE CONTRACT | W7 | CANONICAL STATE, READ-ONLY OUTPUT

For GPU FI/HP/GD/OG owners, state W words shall be produced from the canonical device [left,right,energy,ticks], including terminal EMIT owners. Upload only the requested tick16/context, never a host replacement pair or phase. The shader reads canonical state, checks parity, full mirror, legal STEP/EMIT, G/B and energy, then computes intrinsic phase16 and both W headers.

Fixed scratch interface

```
upload = [tick16, 0] # existing scratch buffer
header = (tick16 << 16) | phase16
output = [left,header,right,header,energy,device_ticks,ok,0]
# eight u32; ok=1 only for valid canonical state
```

The shader never advances or writes canonical owner state. Read complete output and canonical state; independently check exact agreement and preservation. G/B, energy and phase must match the actual certified owner context, in addition to representation validity. Validate reserved zeros and complete output shape.

Device action count is not a logical clock

device_ticks is the existing executor action counter. GROW resets it. It is neither agent.cycle nor W time. Verify and preserve it against that executor's expected counter. Read-only emission supports seeded terminal EMIT states and must not invoke an advance-only readiness check that rejects them.

Device outcome	Required handling
Complete valid output	Use the device-produced state words to assemble the typed LUS record; preserve the owner.
Failed/truncated dispatch or read	Poison and close the owner; no fabricated output or acknowledgment.
Canonical mismatch	Poison and close the owner. Propagate closure through the owning field agent, including FI.
Complete malformed emission	Nonfatal only under W6's whole-operation prestate rule; after semantic mutation, poison unless fully restored.

The host codec may wrap raw IGNITE bytes and assemble explicit LUS metadata. It may not substitute CPU-produced state words for the GPU result. Existing GPU compiler/producer guards remain active through ADVANCE, GROW and private recovery. CPU reference values certify outputs rather than replacing the device producer.

Revision 11 specified this device work before implementation. Revision 12 records the actual-device capture on pages 82-83. Cache residency, occupancy, throughput and physical-energy advantage remain unmeasured.

W transport and exact responses

FINITE CONTRACT | W8 | CURRENT ADMISSION METADATA

Provide JSONL endpoint agent `welip` with `--state`, `--config`, `--backend` and existing storage index options. Config generator agent `welip-config` uses `--profile` field/`hadamard`/`growth`/`organogram`, default `organogram`, mapping to page 70's policies. No `W inspect/status/history` command is provided.

Maximum request line length is 65,536 characters. Require unique JSON keys and finite numbers. An oversized line is fatal; do not parse its tail. EOF or transport failure closes the owner and releases its lock. COMPLETE still permits allowed controls/emission. `W` budget exhaustion disables new operations; the endpoint stays open for exact latest retries until EOF or close.

Common response and current head

```
response common = {protocol,type,producer,producer_epoch,head,next}
head = {seq,clock_epoch,tick16,agent_cycle,geometry_epoch}
next = null # exactly when the W event budget is exhausted
# otherwise:
next = {seq,clock_epoch,tick16,agent_cycle,geometry_epoch,
        position,visible_paths,active_paths,capacity,
        agent_status,allowed_ops}
```

`head` describes current durable admission. `next` sequence/clock are candidate values; cycle/generation are the current preoperation context. `position` is canonical; `visible_paths` is sorted local visibility; `active_paths` is current FIFO order. They contain names, not historical pairs or fields. `capacity` is current and `agent_status` is mission status.

`allowed_ops` preserves IGNITE,ADVANCE,RESIZE,INVALIDATE,EMIT order after filtering. Before ignition allow only IGNITE; afterward remove IGNITE. Remove ADVANCE on COMPLETE or agent-cycle exhaustion. Remove INVALIDATE when no active entry exists. Controls and emission remain available when advancement is unavailable.

Response type	Exact additional fields and delivery
READY	{restored:boolean}; once after durable open/recovery.
RESULT	{op,seq,records,cache,agent_status}; only after durable save.
DUPLICATE	{seq}; no prior records, pair, energy or event.
ERROR	{code,message,fatal}; nonfatal errors retain unchanged head/next. Fatal errors use head=null,next=null, with admitted configuration producer fields.

Startup failure before a valid owner exists terminates with `stderr`/nonzero exit and no READY. Error messages are human-readable diagnostics, not retained canonical history or archive dumps. On transport write failure the owner closes; no response is assured. Current cursor metadata is an explicit forward admission interface, not a public historical read.

W admission order and rejection classes

FINITE CONTRACT | W8 | DETERMINISTIC VALIDATION BOUNDARY

Classification order

Stage	Required checks before proceeding
1. Transport	Reject malformed/oversized transport; require protocol/op syntax.
2. Public alphabet	Deny explicit READ/HISTORY/REPLAY/REGENERATE as BACKWARD_READ; reject other unknown ops.
3. Static request	Require exact operation keys, shapes/ranges, canonical path syntax and source identity.
4. Retry/order	Handle exact latest retry or conflict, then stale sequence and gap. No historical payload retrieval.
5. New request context	Check W event budget, complete clock, current agent cycle/generation, ignition/terminal/cycle eligibility.
6. Dynamic admission	Check current position, visibility and active membership before execution.

The request sequence domain is 1..max_events+1, so a next-sequence request at the exhausted budget receives BUDGET_EXHAUSTED. No admitted sequence exceeds max_events. Budget classification precedes candidate-clock checks, including at the maximal 48-bit horizon. No JSON duplicate keys, unknown keys, Boolean-to-integer coercion or nonfinite values are admitted.

Complete nonfatal error alphabet

INVALID_REQUEST	SOURCE_MISMATCH	BACKWARD_READ
CONFLICT	STALE_SEQUENCE	SEQUENCE_GAP
BUDGET_EXHAUSTED	CLOCK_MISMATCH	CONTEXT_MISMATCH
NOT_IGNITED	ALREADY_IGNITED	AGENT_COMPLETE
AGENT_BOUND	INVALID_SELECTOR	MALFORMED_EMISSION

INVALID_REQUEST covers malformed static/protocol input; SOURCE_MISMATCH covers producer/namespace; CLOCK_MISMATCH covers the full clock; CONTEXT_MISMATCH covers cycle/generation/position; INVALID_SELECTOR covers nonlocal observations or nonactive invalidation paths. Missing ignition, repeated ignition, COMPLETE and agent cycle exhaustion use NOT_IGNITED, ALREADY_IGNITED, AGENT_COMPLETE and AGENT_BOUND respectively. MALFORMED_EMISSION is nonfatal only under W6. Every pure rejection leaves admitted clock/history/cache unchanged.

Fatal error alphabet

REQUEST_TOO_LARGE	OWNER_FAILED	STORAGE_FAILED
-------------------	--------------	----------------

Fatal owner/storage errors terminate the valid ownership period. A closed or poisoned embedded owner refuses all further requests without emission or mutation. Fatal transport errors never interpret a remaining line tail as another request. Responses follow page 78's null-head/null-next rule only when a valid admitted configuration exists.

The protocol does not turn an attempted read into automatic corruption or eviction. INVALIDATE is an explicit validated forward operation with a recorded cause. Neither denied names nor integrity checks supply global producer authentication.

Independent W reference vectors

INDEPENDENT REFERENCE | MATHEMATICAL EXPECTATIONS BEFORE RUNTIME

The independent reference in docs/evidence/welip-v1/ uses frozen OG mathematical owner transitions and separate integer codec/cache logic, with no solvefinite import or device execution. Its main fixture uses producer welip-reference, producer_epoch 7, clock_origin 65,530, max_events 64 and initial capacity 3, around the nine-cycle OG mission on page 65.

Complete lifecycle schedule

I, A1, E, R1, X, A2, A3, A4, E, A5/GROW, R4, A6, A7, X, A8, A9, E
I=IGNITE bytes DEADBEEF0123456789; A means ADVANCE to the stated agent cycle; E=EMIT; R=RESIZE to the stated capacity. The first X invalidates k:3:0 with cause release-old-local-copy; the second invalidates k:2:1 with cause rebuild-current-local-copy. The schedule expects 17 W operations and 18 records while preserving the nine agent transitions.

Exact words: clock, byte order and terminal state

IGNITE raw record0 (9 bytes; phase16=FA00):
FFFBFA00EFBEADDE FFFBFA0067452301 FFFBFA0000000089

IGNITE state record1 (same time and phase):
FFFBFA0001FE00FA FFFBFA0091FE0006

W operation6 / ADVANCE2 (epoch1, tick0000, phase1000):
00001000910010F0 0000100081001010

W operation17 / EMIT (epoch1, tick000B, phaseBA00):
000BBA0016000646 000BBA00860006BA

At operation 6, T_W=65,536, agent cycle=2 and energy=93. At operation 10, GROW reaches agent cycle 5 and geometry epoch 1; the new cache is empty with zero counters. Final owner pair is 860006BA16000646, energy 76, agent cycle 9, geometry epoch 1 and W clock (1,11). Final cache capacity is 4, with FIFO k:1:0,k:1:1,k:1:2,k:2:1; hits 0, regenerations 7, evictions 16.

Independent boundary qualifications

The reference includes raw lengths 0,1,5,9 and 4,096; 1,024 phase/orientation/STEP-or-EMIT codec cases; mirrored and capacity-8 lifecycle variants; malformed padding/count/header/parity/mirror vectors; and legal channel-swap rejection against the actual owner. The generic full-width carrier word FFFFFFFFFFFFFFFF is legal for tick=phase=65,535 and four FF bytes; state phase adds its stricter resolution rule.

Maximum-clock fixture: origin=281474976710591 and max_events=64 end at (4294967295,65535). A next sequence 65 receives BUDGET_EXHAUSTED before clock equality; no next cursor wraps. Full requests, records, retry receipts, cache witnesses and separate executor-counter expectations are retained. Historical OG runtime archive identity is labelled separately from these mathematical W transcript identities.

Main operation-transcript SHA-256: e2b7b8b47b057319f77fbdeea6146b73eba714c1a8e132b065b7c4671ad4c5e0. The builder pins the complete reference and generator; these are expected values, not measured W runtime results.

W acceptance and architectural scope

FINITE CONTRACT | W8 | REQUIRED IMPLEMENTATION EVIDENCE

A separate source-bound implementation capture shall demonstrate the complete finite contract. Passing carrier examples alone does not establish durable lifecycle, device refinement or forward-only admission. The independent reference precedes runtime work; actual implementation evidence must retain its original inputs and versions.

Obligation	Required acceptance scope
Codec and metadata	Raw lengths 0,1,5,9 and 4,096; exact padding/count/header checks; full parity/mirror; ordered owner channels; energy metadata, wrong actual owner/context and coordinated legal-but-wrong payload rejection.
Clocks / identity	All five new operations advance W time; IGNITE record sequence 0,1; 16-bit carry and 48-bit exhaustion; independent producer epoch, agent cycle, geometry epoch and device counter; no wrap reuse.
Agent projection	All four supported policies; unchanged canonical agent archive under inserted W controls; exact original OG GROW time/prefix; mirrored owner, incomplete WAIT frames, incremental DEFER and fresh recovery from nonterminal outcomes.
FIFO lifecycle	Capacity-preserving semantics, oldest eviction, complete selected-pair invalidation, unaffected order, cause retention and later regeneration; whole-selector rejection; fresh cache/counter reset at GROW; interleaved lifecycle replay.
Forward boundary	All backward names and extra selectors denied; latest exact retry before changed context; stale/conflicting/gapped requests; no historical payload, duplicate device dispatch, clock mutation or archive write.
Actual GPU refinement	Canonical device W production for all four policies and terminal EMIT states; host state-word fallback forbidden; complete canonical/scratch checks, preserved device counter and no reseeding after initialization.
Durability / ownership	One OS-held owner and serialized calls; immutable configuration/path; before/after replacement failures, unread acknowledgment, post-ADVANCE/cache emission failure, save uncertainty, EOF/oversized/transport failure and fresh CPU/GPU process recovery.
Prior profiles	FI/PX/HP/GD/OG schemas, numerical contracts and canonical reference archives remain unchanged. Existing live retry/history APIs keep their separate R semantics.

Revision 11 fixed these acceptance obligations before implementation. Revision 12 records the source-bound capture on pages 82-83, including actual-device execution, durable recovery and unchanged agent archives. Numerical definitions and reference vectors retain their original meanings.

Remaining architecture includes general primitive/graph productions, broader spectral choices, physical adapters, universality and comparative hardware measurements. This finite W binding is not indefinite time, global authentication, exactly-once physical actuation or total constant memory.

Measured W clock and continuation

IMPLEMENTATION CAPTURE | W1-W8 | 26 SEPTEMBER 2026

The source-bound capture passes 698 tests with zero skips, including 145 actual-device GPU methods, and all 14 W conformance checks. This is finite clock, carrier and same-owner continuation evidence under the numerical contract committed at 74e00f4.

Device: NVIDIA GeForce RTX 5070 Ti Laptop GPU, Vulkan. All three frozen 17-operation lifecycles execute on CPU and GPU. Each produces the exact 18 records and 37 carrier words, with identical complete private W archives across backends. All four field policies retain terminal emission.

Time and identity remain separate

Checkpoint	Observed result in the default fixture
Operation 6	W clock carries to (1,0); agent cycle 2, energy 93.
Operation 10	GROW reaches cycle 5 and geometry epoch 1; a new cache and executor counter start at zero. Global cycle and energy do not reset.
Operation 17	W clock (1,11); cycle 9, geometry epoch 1, pair 860006BA16000646 and energy 76.
Capacity variant	Capacity eight changes final cache hits/regenerations to 4/3 from 0/7, preserving every agent transition.

Actual device production

The W shader reads the current canonical pair, energy and action counter. Only tick16 and a reserved zero are uploaded. Both ordered fragments are read from device output and checked against a separate full canonical read. The host state encoder and earlier CPU geometry producers are disabled during GPU conformance, including fresh GPU recovery. Emission introduces no replacement owner or reseeding.

One exact latest retry returns a current cursor and sequence receipt, with no earlier records, state payload, device emission or save. A new EMIT can produce the current state after completion. Every cache removal preserves complete pairs and unaffected FIFO order; private recovery repeats lifecycle events alongside the original agent steps.

These results establish the declared finite W implementation. They do not measure texture-cache saturation, throughput, physical energy or an advantage over a conventional architecture. GPU execution still uses host admission, verification and durable journaling.

W recovery, evidence and remaining work

MEASURED RECOVERY | SOURCE IDENTITIES | ARCHITECTURAL CONTINUATION

Failures resolve through the durable prefix

Complete malformed device output is nonfatal only before any working-state mutation and after unchanged canonical state is established. Output failure after ADVANCE, uncertain dispatch and uncertain save close the owner. A successful replacement followed by a lost acknowledgement reconstructs as the latest duplicate; an unsaved transition reconstructs from the preceding prefix.

Three fresh endpoint processes execute GPU, CPU and GPU with distinct storage indexes. They reconstruct the complete original W ledger and FIFO witnesses around clock carry and geometry growth. Session tests additionally cover interruption during mutation/save, before/after atomic replacement, unread acknowledgement, concurrent duplicate admission, strict archive corruption rejection and transport closure.

Incomplete observations, deferred search and fresh observations after temporary energy failure retain the same individual. UNREACHABLE uses an injected planner outcome; the supported Klein graphs are connected. Clock exhaustion removes the next cursor but retains the latest exact retry. Mission or cycle completion stops ADVANCE while permitting controls and emission within the W budget.

Reproducible evidence and preservation

```
python -m unittest discover -s tests -v
python -m examples.welip_conformance --output w-recheck.json
python tools/capture_welip_evidence.py
python tools/build_formal_spec.py
```

Capture files: docs/evidence/welip-v1/verification.json, conformance.json and full-tests.txt. Source hashes use LF-normalized bytes; the capture checks identical source inventories before and after execution. The formal builder verifies those sources, reports, formal chronology, frozen reference, full lifecycle rows, device evidence and archive identities.

Canonical W archive SHA-256: 9dd6e452c43b37482d6b780786b76f40a80e41846d67feb8c9af6a2e10b8b9c1

Unchanged historical OG agent archive SHA-256: e2834ba2cc6b4d51f2b23f274cd9f2a0db8a4e88754ed9d2aad76c074f134df

Next source obligation

Original specification p.7 requires a geometry-to-field map for every declared primitive. The present parameterized organogram generates intrinsic balls; directional pyramid/cone boundaries and general graph production remain unfinished. A further realization must explicitly bind its geometry, phase/Psi relation, seam behavior and exact redistancing before implementation. Broader spectral choices, physical adapters, universality and comparative hardware evidence also remain open.

The three original PDFs and separate Tom/Jitske ELI5 booklet remain byte-identical. This measured milestone advances the architecture; it does not establish the full paradigm or exactly-once physical actuation.

Directional geometry: taper sections

APPENDIX MAP | DP1-DP10 | FORMAL ONLY

This contract supplies a new directional geometry-to-field map before runtime implementation. It preserves the original three source PDFs, all previous numerical profiles and the separate ELI5 booklet. The independent reference proves specific arithmetic expectations; it does not count as executed CPU/GPU conformance.

Section	Definition and obligation	Page
DP1	Source scope, new identities and exact manifest	85
DP2	Parameterized grammar and exact tape	86
DP2	Instruction decoding and rejection	87
DP3	Prior-field shaft and complete branch context	88
DP4	Projected footprint and transported frame	89
DP5	Union boundary, exact distance and proof	90
DP6	Finite work and checked arithmetic	91
DP7	Original recipes and canonical derivations	92
DP8	Same individual, target, energy and epochs	93
DP8	Explicit W version and continuation	94
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DP10	Independent numerical reference	96
DP10	Required runtime acceptance and remaining scope	97

Historical implementation baseline: 1ea9320, with 698 passing tests, 145 actual-device GPU methods and 14 W conformance checks. Those counts apply to the prior W/OG implementation only. No directional runtime result is asserted by revision 13.

Source scope and explicit new profile

FORMAL ONLY | DP1 | A DIRECTIONAL BOUNDARY FAMILY

Original p.5 describes a two-dimensional radial sweep. Original p.7 requires each primitive to declare a metric, boundary, units, sign and geometry-to-field map, and relates pyramid/cone slope and extent to a local shaft or Psi. Solus pp.3-4 and solipsism p.13 discuss side views. They supply no unique cone/pyramid equation or three-dimensional metric. DP chooses the following finite axial section.

$$\text{TAPER}(h,p,q): 0 \leq s \leq h \cdot 2^{\text{scale}}, q \cdot \text{abs}(t) \leq p \cdot s$$

Here s and t count local unit edges; p/q is transverse extent per axial step. SCALE multiplies extent, not slope. The apex is the current grammar cursor. A field/phase-selected shaft orients the section. A closed cap occurs at its finite extent. Projection onto the existing Klein quotient can wrap and self-overlap. Rasterization, boundary convention, bounds and versions are declared realization choices.

A cone and an appropriately aligned pyramid can have the same triangular axial section. This is one directional section family, not a claim of two distinct three-dimensional solids. Full volumetric cone/pyramid geometry remains an architectural obligation.

```
policy = "tomigdt-field-taper-plan-act-v1"
binding = "klein-taper-organogram-v1"
recipe = "klein-taper-world-v1"
derivation = "klein-taper-derivation-v1"
receipt = "klein-taper-growth-v1"
instruction_profile = "TP-TAPE32-v1"
```

Exact manifest extension

Retain FI/HP common manifest keys identity, target, world, initial_node, initial_phase, initial_orientation, initial_energy, repair_cost, max_search_expansions, max_hops, max_cycles, policy, word_profile and perspective. Add exactly routing and taper for this policy. taper is DP2's immutable binding; routing retains HP. No growth or organogram key is accepted. word_profile remains RP32-relational-sdf-v2 and perspective remains local-observation-v1.

world is the original KleinFieldRecipe with W,H at least 3 and $N=W \cdot H$ at most 256. Existing manifest domains, departure turns, exact signed-code range, word mirror/parity and finite mission bounds remain in force. Every listed key is required; extras and Boolean/float substitutions for integers are rejected.

Coordinates are canonical labels for the declared quotient, not a physical origin. Index sign, phase origin and tree row order cannot select geometry. DP does not change graph dimension, add a global scale, prove universal computation or measure bandwidth reduction.

Parameterized taper grammar

FORMAL ONLY | DP2 | COMPLETE FINITE INPUT

```
taper = {format,max_epochs,cost,symbols,axiom,rules,
        generations,limits}
format = "klein-taper-organogram-v1"
limits = {max_symbols,max_steps,max_primitives,
        max_stack,max_primitive_sites}
```

Inherit OG1-OG2's exact symbol, axiom, rule, affine-expression, signed-i32, guard, generation and permanent-address rules on pages 58-59. Add reserved terminal TAPER with arity 3. The fixed arities are F:1, +:1, -:1, [:0,]:0, R:1, S:0, SCALE:1, TAPER:3. Nonterminal declarations cannot reuse any terminal. max_epochs remains 0..4, cost 1..127 and generations 0..8.

Input	Exact interpretation
TAPER(h,p,q)	Strict integers h,p,q in 1..65535; gcd(p,q)=1. h is unscaled extent; p/q is slope. DP6 adds effective-work and arithmetic checks.
F, +, -, R, SCALE	Existing bounds: F 1..256; signed-turn magnitude 1..16; R 1..127; SCALE 0..4. F length and primitive extent/radius multiply by 2^scale. S effective radius must remain at most 127.
max_symbols / max_steps	Strict 1..1024 logical symbols at every generation / 1..4096 total effective F substeps.
max_primitives / max_stack	Strict 1..64 ball or taper emissions / 0..32 saved frames. Every admitted final tape emits at least one primitive.
max_primitive_sites	Strict 1..16777216 (2^24); default example budget 1048576. DP6 counts complete tested rectangles, including rejected sites, and N sites per ball.

Single-valued rewrite and preflight

Read the previous generation only; first matching rule wins, empty RHS erases, unmatched symbols pass. Reject a final nonterminal. Preserve ordered immutable arguments and OG addresses for all passes and replacements. Check every intermediate word and every final instruction. Validate brackets and restore radius/scale during preflight; reject underflow or a nonempty final stack.

TAPER emits at the current cursor without advancing it or consuming a turn. S emits a closed intrinsic ball. Their union semantics are DP5, including for a tape containing only S. No inequality residual or collection of old OG spheres substitutes for the directional footprint.

This new profile deliberately changes primitive composition. OG-TAPE32-v1, klein-branch-organogram-v1 and existing OG archives retain their original minimum-margin boundary. A profile tag must be checked before decoding or certifying data.

Typed instruction words and decoding

FORMAL ONLY | DP2 | TP-TAPE32-v1

```
raw = operand + 2^24*code
word = raw + 2^31*(popcount(raw) mod 2)
operand = word & 0xFFFF; code = (word >> 24) & 15
require (word & 0x70FF0000)==0 and even popcount
```

Code(s)	Logical instruction and required words
0..7	F,+,-,[,],R,S,SCALE in that order. One word; operand is the argument or exactly zero for [,], S. Numerical old-word encodings remain the same under this new profile.
8,9,10	One TAPER(h,p,q) is exactly three consecutive words: code 8 carries h, code 9 carries p, code 10 carries q. Each independently has even parity and zero reserved bits.
11..15	Always invalid. Codes 9 and 10 cannot begin a logical instruction.

Decode in order. A code 8 requires both continuation words with exactly the prescribed codes, then advances by three texels. Reject truncation, wrong order, isolated continuations, invalid operands, wrong arity, nonzero no-argument operands, reserved bits and parity errors. Decoding shall yield exactly the declared logical tape and arguments; no trailing ignored words are allowed.

```
offset(0)=0
offset(i+1)=offset(i)+(3 if symbol_i==TAPER else 1)
texels = logical_instructions + 2*taper_count <= 1152
```

The offset map is a deterministic derivative of the logical tape. Trace and branch identities use logical terminal addresses, never continuation-word offsets. The bound follows from at most 1024 logical symbols and at most 64 primitives. Fetch exact r32uint texels with integer textureLoad in tape order and without filtering.

Carrier type is part of the meaning

The four-bit instruction opcode is a property of TP-TAPE32-v1 only. It changes neither RP32 state layout nor the eight old OG opcodes. These instruction words are not state pairs, have no spatial node or signed distance lane and assert no full-mirror relation. The same immutable tape acts on mirrored cursor pairs.

All integer domains, preflight counts and decoded terminal identities shall also be checked on device before interpretation. Host parsing alone does not certify an uploaded instruction texture. A malformed candidate cannot advance the live owner or alter the durable history.

The prior field chooses the shaft

FORMAL ONLY | DP3 | CANONICAL SELECTION, TRANSPORTED FRAME

Use one prior certified field for the complete stage. Begin at the original repaired EMIT pair, repacked as STEP, with radius 1, scale 0 and empty branch path. OG3's F, phase turns, complete PUSH/POP and per-terminal trace rules remain in force. The emitted geometry does not feed back into this same stage's cursor decisions.

```
tau = (-1)^eta * R mod 256; bank=floor(tau/64)
g=(phi(u+)-phi(u-), phi(v+)-phi(v-))
Psi=g/gcd(abs(g_u),abs(g_v)), or (1,0) if g=(0,0)
z_i=gains[bank][i]*(1+Psi_i^2)*g_i
```

At a TAPER terminal, compute exactly the OG F direction at that canonical node: maximize $z \cdot e$ over $e=(1,0),(0,1),(-1,0),(0,-1)$. Break ties with this list rotated left by bank. Store the selected direction code 0..3 as shaft. Do not take the F step, update phase, modify radius/scale or change the owner. The current pair remains a prior-field STEP pair.

```
e = selected canonical shaft
f = (-1)^eta * (-e_v, e_u)
```

The pair (e,f) is the local axial/transverse frame at emission. Mirror the complete RP32 pair: R changes sign and eta toggles, so tau and the chosen e are unchanged while f changes sign. Because the footprint includes both signs of t, its projected occupancy is identical. Transcript pairs remain full mirrors, not byte-identical left views.

Selection and chart transport are different operations

Select e once using the canonical quotient chart and its specified tie priority. An equivalent covering-space representative must transport that already selected e and f as DP4 declares. Re-running the unchanged cyclic tie priority in an arbitrarily reflected chart is invalid: at zero gradient, bank 1 would select v+ again instead of the transported v-. The rule does not claim to be invariant under such an untransported re-selection.

PUSH/POP restores the complete pair, radius, scale and enclosing branch path, including a nested seam crossing. No independent persistent heading is added: the prior field and restored pair determine the next shaft. Program position, emitted primitives, transcript, work counters, energy and history never rewind.

Changing a declared initial phase, routing gain, original tick or grammar parameter may change geometry. Changing cache capacity or f8 storage descriptors must not. These are distinct semantic and storage operations.

Project the complete local footprint

FORMAL ONLY | DP4 | COVERING SPACE AND EDGE TRANSPORT

Let quotient dimensions be W, H and canonical apex be (a, b) , with a in $0..W-1$, b in $0..H-1$. Use $L = h \cdot 2^{\text{scale}}$ for the effective axial extent and reduced p/q for slope. The local closed lattice footprint is finite:

```
D = {(s,t) in Z^2 : 0<=s<=L, q*abs(t)<=p*s}
(X,Y) = (a,b) + s*e + t*f
k=floor(X/W); u=X-k*W
P(X,Y) = (u, ((-1)^k*Y) mod H)
```

Floor division and modulo are mathematical, including negative X or Y ; modulo returns $0..H-1$. Numeric node ID is $u \cdot H + v$. Occupancy is the set of all $P(X, Y)$ for (s, t) in D . Deduplicate projected nodes by set union. Do not keep only a nearest lift, clamp a wrap, fill an axis-aligned bounding box, or replace the section with balls.

Theorem DP-A: equivalent representatives agree

```
A_(m,n)(X,Y)=(X+m*W, (-1)^m*Y+n*H)
J_m=diag(1, (-1)^m)
apex'=A_(m,n)(apex); e'=J_m*e; f'=J_m*f
```

The transformed local point is exactly $A_{(m,n)}(\text{apex} + s \cdot e + t \cdot f)$. Its horizontal quotient is $k+m$; reducing the vertical component gives $(-1)^k \cdot Y$ plus an integer multiple of H . Therefore $P(A_{(m,n)}(X, Y)) = P(X, Y)$, for every integer m, n . This proves covariance when the selected frame is transported as well as the apex.

Equivalent local edge construction

Take s unit steps along e , then $\text{abs}(t)$ along $\text{sign}(t) \cdot f$. At each reversing u seam, negate the v component of both retained frame vectors before continuing. Projecting every intermediate lift yields the same endpoint as the formula. This supplies an independent edge-walk checker and relates the coordinate formula to the declared local transport.

Every vertical column at fixed s includes $t=0$; each accepted t has its neighboring t toward zero. The axial $t=0$ chain connects these columns. Its projection is connected, although different primitives may form a disconnected union. Multiple wraps and self-overlap are admissible if the final union has a boundary.

A physical rotation, arbitrary change of graph or distinct three-dimensional cone/pyramid surface is not a deck change. Those extensions require their own metric and transport declarations. No stored $f8$ axis or row order enters this footprint.

Union first, then exact signed distance

FORMAL ONLY | DP5 | INNER VERTEX BOUNDARY

Each S contributes the closed intrinsic ball $\{v: d_K(v, \text{center}) \leq \text{radius}\}$. Each TAPER contributes DP4's projected closed lattice set. Let X be the union of these occupied vertices on the fixed connected undirected unit-edge graph K . Duplicated primitive records remain in the transcript; occupancy ignores multiplicity.

```
Z = {v in X : some neighbor of v is not in X}
sigma(v) = 0 if v in Z, -1 if v in X \ Z, +1 otherwise
phi_new(v) = sigma(v) * min_{z in Z} d_K(v, z)
```

Reject empty X or X equal to all vertices; equivalently this connected finite graph would have no Z . A thin nonempty footprint is legal: all its occupied vertices may be boundary and it may have no negative interior. Reject any exact field outside $-127..127$. Distances use graph unit edges; slope and inequality residual are not distance units.

Theorem DP-B: separation and exact field

An occupied vertex adjacent to an unoccupied one is in Z by definition, so no edge joins negative and positive signs. Distance to nonempty Z is integer-valued and 1-Lipschitz. On an edge with equal signs its signed difference is at most one; on an edge incident to Z the other endpoint is at distance at most one; opposite nonzero signs cannot occur. Thus ϕ_{new} is an exact signed boundary distance and is 1-Lipschitz on every edge.

A boundary hidden by another primitive is removed before redistancing. The new S -only profile can therefore differ from old OG at cut loci or overlaps. This is deliberate, declared by the new policy/binding/recipe versions. Neither old minimum sphere margins nor per-primitive signed distances define DP's final signs.

A new field family, not renamed old balls

On an 8 by 8 quotient, apex (1,3), shaft $u+$, extent 3 and slope 1 produce 16 occupied vertices. The interior is {19,26,27,28}. Boundary vertex 38 has no negative neighbor. For any old OG ball union with positive integer radii, a zero-margin vertex has distance r to a minimizing ball center; stepping toward that center gives distance $r-1$ and a strictly negative union margin. Every old zero therefore has a negative neighbor. This DP field cannot equal any old OG ball-union field.

This witness distinguishes the defined discrete field families. It is not a proof of general computational power or physical superiority. Independent reference data retain the complete occupancy, boundary, signs, field and quotient neighbors for the witness.

Bound work before constructing geometry

FORMAL ONLY | DP6 | FINITE, CHECKED INTEGER ARITHMETIC

Preflight the rewritten tape with exact mathematical integer arithmetic. Restore radius and scale on POP. For each TAPER let $L=h*2^{scale}$ and $B=floor(p*L/q)$. B may be zero. Bound the full rectangle of tested lattice sites, not only occupied or distinct projected vertices:

```
K_taper=(L+1)*(2*B+1)
K_ball=N
K_total=sum K_primitive <= max_primitive_sites <= 2^24
```

Enumerate $s=0..L$ and $t=-B..B$ for this accounting; accept the point iff $q*abs(t) \leq p*s$. The valid endpoints could equivalently be generated column by column, but the declared rectangle bound remains the admission rule. A ball tests all N quotient vertices. Zero-length tapes and no-primitive results are rejected.

Check	Required range or rule
Operands / local context	h,p,q 1..65535; $gcd(p,q)=1$; scale 0..4; existing R/S/F bounds remain unchanged.
Products	L and $p*L$ are positive and at most 2147483647; $q*B$ is nonnegative and at most that value.
Lifted coordinates	$W-1+L+B$ and $H-1+L+B$ at most 2147483647. This bounds the absolute canonical-apex coordinates for every cardinal e,f and every tested rectangle site.
Aggregate work	Primitive count 1..max_primitives; effective F steps $\leq max_steps$; stack high water $\leq max_stack$; each generation $\leq max_symbols$; encoded texels ≤ 1152 .
Geometry acceptance	The occupied union is proper and nonempty; exact new signed distance and every derived field/routing/index certificate are valid.

No arithmetic wrap as a control path

Reject a failing product or sum before dispatch/allocation for candidate construction. On device, check nonnegative products with $a \leq floor(limit/b)$ before multiplying; check sums with $a \leq limit-b$. Derive B only after $p*L$ is safe. Check $2*B+1$ and K against the remaining site budget before evaluating them. Use division/remainder implementing mathematical floor and nonnegative modulo when projecting negative coordinates.

Finite grammar generations, bounded F work and finite primitive sites establish termination of this stage. Geometry rejection never becomes a partial GROW or grammar DEFER. The agent's separate bounded planner retains DEFER. Resource budgets express executable limits, not physical energy measurements.

Shader redistance and derived atlas/index preparation remain finite under $N \leq 256$. Report actual peak resource payloads and allocations when implemented; a work count alone is not a memory or speed claim.

Recipes retain original shape context

FORMAL ONLY | DP7 | REPRODUCIBLE DERIVATION

```
recipe = {format,base,taper,routing,stages}
format = "klein-taper-world-v1"
stage = {epoch,tick,start_pair,prefix_sha256}
derivation = {format,context,tape,trace,segments,primitives,
              final_context}
format = "klein-taper-derivation-v1"
```

Apply OG6's canonical JSON and stage domains: original base, immutable taper/routing, stages 1..max_epochs, contiguous epochs, strictly increasing original positive ticks, complete valid EMIT start pairs and lowercase SHA-256 of events before this GROW. Epoch zero uses the original base type. Validate each start pair against its reconstructed prior field. Never substitute current time, mutable grammar or current observations.

Record	Exact keys
context / tape	Original stage / ordered {address, symbol, args}. Logical addresses retain OG replacement/pass rules.
trace	{address, branch_path, pair, radius, scale} after every terminal.
segments	{address, branch_path, step, pair} after each effective F substep.
ball primitive	{kind, address, branch_path, center, radius, pair}; kind is ball, center is numeric node, radius is effective.
taper primitive	{kind, address, branch_path, apex, height, numerator, denominator, shaft, pair}; kind is taper; apex is numeric node, height=L, numerator=p, denominator=q, shaft=0..3.
final_context	{branch_path, pair, radius, scale}; branch_path is empty after a balanced tape.

All transcript pairs are complete current prior-field STEP pairs as 16 uppercase hex characters. branch_path lists enclosing PUSH addresses outermost first. Keep primitive order and duplicates. radius in trace/final_context is unscaled. The transverse frame is derived from shaft and pair orientation. Its mirrored sign need not be an extra field. Reject unknown record keys, wrong kinds or types and inconsistent descriptor values.

```
derivation_sha256=SHA256(canonical(derivation))
```

A descriptor is a checked output, not an independently editable input. Reconstruct the tape and cursor, certify every descriptor and derive the occupancy from it. A retained hash binds the original transcript; it is neither authentication nor a geometry seed. The certificate must bind the exact immutable recipe and original contexts.

Recorded historical field samples retain the original admission sequence and event-prefix identity under GD/OG rules. Cache eviction discards residency, not recipe dependencies. Rebuilds in a fresh process reconstruct every necessary stage with its original context.

Admit geometry in the same individual

FORMAL ONLY | DP8 | EPOCHS, SUBGOALS AND RESOURCE STATE

Apply the existing OG5 lifecycle with taper.cost and taper.max_epochs. REPAIR below the final epoch enters GROWTH_PENDING. The next complete fresh local observation frame admits one GROW transition. Sensor frames supply local observations, not shape coordinates, shafts, targets or actions. The hypothesis cursor is not a replacement individual.

```
reserve(k)=repair_cost
+(max_epochs-k)*(taper.cost+repair_cost)
G_new=G_old; R_new=R_old; eta_new=eta_old
B_new=certified phi_new(G_old); opcode_new=STEP
energy_new=energy_old-taper.cost
```

Apply the inherited reserve bounds before admission. A successful GROW increments the global cycle once and epoch once; energy is debited once. Repack both mirrors and parity. Topology, dimensions, node names, departure turns and baseline index anchor are unchanged. The live owner continues over the new field.

```
C0=all nodes except owner G
C1=argmin_(v in C0) abs(phi_new(v))
C2=argmax_(v in C1) unweighted_hops(G,v)
new_target=sorted_numeric(C2)[tau mod len(C2)]
```

This is the OG state-selected target rule. It does not use the final grammar cursor, primitive order, cache or index row order. Planning and action costs subsequently use the certified new field, PX and HP descriptors. With max_epochs=0, complete the initial mission without generating geometry.

```
event = {seq,input,decision,output,energy,
         geometry_epoch,growth}
receipt = {format,from_epoch,to_epoch,mapped_node,
          target,recipe,derivation_sha256}
receipt.format = "klein-taper-growth-v1"
```

geometry_epoch is the pre-cycle epoch; growth is null except at GROW. mapped_node is the unchanged canonical owner name. Snapshot adds geometry_epoch and current_recipe, with no global scale_exponent. Preserve the existing input identity, status, journal, energy and pending-search schemas under the new manifest policy. Reject a receipt or recipe from another production profile.

Successful GROW clears effective observations, pending planner state and active FIFO, and resets the replacement world's cache counters as in OG/W. The complete journal and global cycle/energy remain. Cache capacity/reindex operations preserve semantic state and do not advance cycles. Old observations and retries retain their original epoch, locality and admitting sequence; original duplicates cannot be reinterpreted against a later shape.

Same-owner conformance requires a complete changing-field mission, not merely matching detached primitive vectors. Final repair terminates only the declared finite mission; it does not complete the broader architecture objective.

A declared W version for the new geometry

FORMAL ONLY | DP8 | FORWARD CONTINUATION

protocol = "welip-field-agent-v2"
config.format = "welip-field-config-v2"
archive.format = "welip-field-session-v2"
Version 2 config retains exactly W1's keys format, agent, producer, producer_epoch, clock_origin, max_events and initial_capacity. For this binding its agent policy must be tomigidt-field-taper-plan-act-v1. Version 1 retains its existing four-policy domain and formats. Reject a cross-version protocol/config/archive combination before owner allocation or file replacement; never silently upgrade an old archive.

Inherited W1-W8 behavior	Exact DP extension
Five operations and clock	IGNITE, ADVANCE, RESIZE, INVALIDATE, EMIT; same strict clock origin, event bound, epoch carry, operation/record sequence, current cursor and retry rules. No historical-read operation is added.
Words and records	Retain welip-16-16-32-v1, welip-lus-v1, bytes-v1 and RP32-relational-sdf-v2. Record schema and packing equations are unchanged; no geometry discriminator is inserted into payload lanes.
Typed identity	The unchanged record baseline_id is only the original base recipe label. Scope record comparison and admission to the retained protocol and exact canonical config, including DP manifest. Bare words or baseline_id alone do not identify the geometry profile.
Private recovery	Replay original DP geometry contexts and the full W operation/cache sequence before READY. Compare typed canonical rows and expected state. Resume CPU or GPU with independent storage index choices.
Atomicity and retries	Retain W1-W8 durability, latest-only exact retry and uncertain-outcome failure rules. No second emission or action is admitted for a duplicate.

The canonical GPU W emitter reads the actual admitted owner pair, separate energy and device cycle counter, including after GROW. Host-supplied current owner words or CPU state encoding cannot substitute. Clock/tick headers remain the W projection; they do not supply geometry or replace original agent GROW tick.

Endpoint agent welip dispatches explicitly on config/archive version. agent welip-config --profile taper creates v2; existing profile names and the default organogram continue to create v1. Version 1 shall explicitly reject taper even if a shared agent-manifest parser recognizes it. Keep the other W arguments and error/result fields, replacing the protocol value where applicable. Never compare or reuse records across different protocol/config namespaces merely because payload words, producer, epoch or clock coincide.

A failure after possible state/cache mutation or durable replacement poisons the live endpoint; hold its ownership lock until close and recover from the saved prefix in a fresh session. A provably complete rejection before mutation preserves the live owner. Exactly-once physical actuation is not established by this software protocol.

Device construction and independent admission

FORMAL ONLY | DP9 | PRODUCTION, CERTIFICATES, OWNERSHIP

The host may strictly parse, expand and preflight immutable tapes. Explicit GPU execution shall derive cursor motion, phase, shaft selection, primitive descriptors, occupancy, boundary, signs and exact distance on the device from retained prior certified geometry and actual original state. Uploading host-produced centers, shafts, occupancy or final distances is not a conforming GPU producer.

A finite construction with bounded storage

An implementation may enumerate the bounded rectangle sites in parallel, test each inequality with checked integer arithmetic, project accepted sites and OR occupancy bits. Ball membership may use the OG metric already checked against quotient BFS. Synchronize completion of the entire occupied union before computing its inner boundary; then redistance and build the candidate field, PX, HP and f8 resources. Equivalent algorithms must satisfy the same observable contract and work admission.

The derivative tape-offset map and site scheduling do not alter logical order or descriptor identities. Allocate only bounded resource arrays. A full per-site transcript is not required; retain the canonical per-terminal, per-segment and per-primitive derivation. Count and report actual live/peak device resource payloads when implementing this stage.

An independent certificate for the new profile

A sealed DP certificate shall independently reconstruct original grammar/context, trajectory and branch restores; recompute each shaft score and tie from its apex pair and prior field; check ordered descriptors and bounds; recompute occupancy using a separate construction, detecting both missing and extra sites; compare inner boundary/signs; and certify exact graph distances, seam transport, pair mirror/parity, derived PX/HP/index data and recipe identity. Do not use the old OG minimum-margin certifier for DP.

The checker may run on the host, but it cannot become the production path. CPU and GPU producers and the checker must not share the new geometry-producing helper. Test by disabling CPU producer/compiler paths during actual GPU construction, regeneration, owner GROW and private W recovery. Complete readback/certificate rejection and unknown device completion are distinct outcomes.

Transactional candidate ownership

Keep the old live owner, history, energy, field, atlas/index, cache and pending search until the complete candidate is certified and admission succeeds. Reject a complete invalid candidate and release all its resources. On success, install all matching resources together and retire the old set once. If state mutation, dispatch completion, cleanup after commit or saving is uncertain, retain the inherited fatal/poison/recovery semantics; do not resume from an assumed old state.

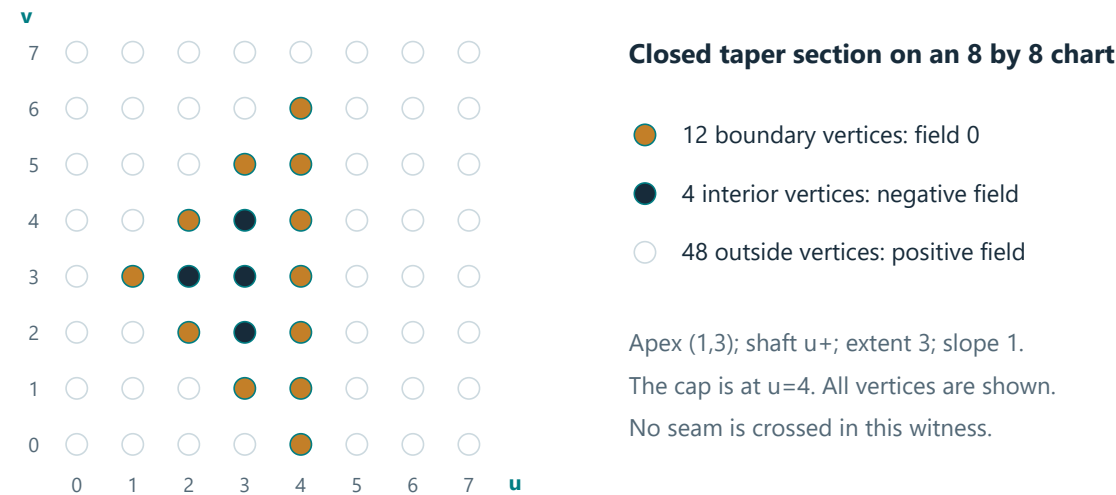
Exercise actual GPU seams, multiple wraps, original-context reconstruction, DEFER continuation, allocation failures and durable reopen boundaries. Texture residency alone supplies no proof of GPU saturation, latency improvement, throughput, measured energy advantage or elimination of physical memory traffic.

Independent directional reference

FORMAL ONLY | DP10 | LITERAL EXPECTATIONS BEFORE RUNTIME

docs/evidence/directional-v1/formal-reference.json and its reference-builder.py retain the independently computed grammar, tape, primitive, field, owner and W expectations. The generator imports pinned independent OG/GD/W arithmetic helpers and no solvefinite module. Its events are mathematical reference events; their prefix digests are not claims of existing runtime archives.

Distinct geometry and seam witness



Direct covering projection is checked against a separate transported edge walk, including negative coordinates, multiple wraps, reflected representatives and full-pair mirrors. Overlapping primitives must remove hidden boundaries; narrow positive slopes may have no negative interior; whole-quotient occupancy rejects. Each retained vector carries its complete expected occupancy, signs and exact field.

Whole-history expectations

Mission	Cycles	Energy	Final full pair
One epoch	11	69	16000534060005CC
Full mirror	11	69	060005CC16000534
Two epochs	17	55	0600111E960011E2
Zero epochs	4	86	06011145160111BB

The mixed default stage emits two tapers and one ball: 19 logical instructions, 23 texels, 7 F substeps and 41 primitive sites. Its Wv2 lifecycle contains 19 operations, 20 records and 41 words, ending at clock epoch 1, tick 13. Original tick/phase/parameter mutations, branch restoration, malformed inputs and limits have separate literal vectors.

The builder verifies source identities, helper identities and retained reference hashes before including this appendix. The historical 1ea9320 W capture is bound to its own 106 source/reference identities; new runtime changes cannot inherit its 698-test or hardware claims.

Runtime acceptance and remaining obligations

FORMAL ONLY | DP10 | IMPLEMENTATION STILL REQUIRED

Required evidence	Conformance scope
Exact construction	CPU and actual-device GPU agree with frozen tape, descriptors, occupancy, boundary, signs, exact distances and mirrors. Cover all shaft directions, both orientations, caps, thin sections, seams, repeated wraps and mixed/overlapping primitives.
Strict input and work	Reject schema/type/arity/rational/tape errors, i32 overflow, excess symbols/steps/primitives/sites/stack, underflow, incomplete triples, empty/full occupancy and bad field certificates before admission. Boundary cases at each limit are exercised.
State changes matter	Demonstrate slope, extent, original phase and original tick changes that affect geometry. Verify full branch restores after nested reversing seams and scaled primitives. Prove f8 descriptor/capacity changes cannot alter semantic output.
Same-owner continuation	Match complete one-/two-epoch, mirrored and zero-epoch missions, generated-world planning including DEFER, target/energy transitions and historical regeneration under original contexts.
Wv2 recovery	Frozen carry/cache/GROW/terminal projection; fresh GPU to CPU to GPU endpoint continuation with different indexes; strict v1/v2 mismatch rejection, duplicate no-op and uncertain/save-failure recovery.
Producer independence	Run actual GPU paths with CPU geometry, field, index, routing and state-emission producers disabled. Independently validate output and resource ownership; inject complete and uncertain candidate failures.
Historical compatibility	All old policies, instruction schemas, recipe/archive identities and Wv1 lifecycles retain their contracts. Complete suite and source-bound capture have zero unexplained skips.

After implementation, retain full commands, test counts, device identity, source inventories before/after capture, original formal commit, archived histories and allocation evidence. Update measured status in this same PDF only after the conformance evidence exists.

The full architecture objective remains open: distinct three-dimensional primitive volumes, broader graph and production families, global spectral choices, physical adapters, wider continuation, computational universality and comparative hardware evidence. The source claims about bypassing bottlenecks and saturating hardware remain hypotheses requiring appropriately scoped measurements.

Measured directional geometry and missions

IMPLEMENTATION CAPTURE | DP1-DP10 | 26 SEPTEMBER 2026

The new source-bound capture passes 796 tests with zero skips, including 179 actual-device GPU methods. It passes 15 geometry/mission conformance checks and 8 Wv2 checks. DP1-DP10 were committed at 1979e66 before this runtime implementation.

Device: NVIDIA GeForce RTX 5070 Ti Laptop GPU, Vulkan. The CPU and actual GPU construct the declared directional axial sections, mixed with intrinsic balls. Complete trajectories, primitive descriptors, inner boundaries and signed fields match the frozen independent reference. The retained reference is unchanged.

Literal mission	Cycles	Final energy	Final packed pair
One epoch	11	69	16000534060005CC
Full mirror	11	69	060005CC16000534
Two epochs	17	55	0600111E960011E2
Zero epochs	4	86	06011145160111BB

Device production and separate admission

The GPU reads typed instruction words, transports the branch cursor, selects each shaft from the prior field and intrinsic phase, and emits primitive descriptors. A parallel pass projects every admitted footprint site, forms the occupied union and its inner boundary, then computes exact graph distances. The independent host certificate reconstructs the complete trajectory and checks occupancy through inverse lift membership before accepting the field.

GPU conformance disables CPU geometry, field, index and routing producers. Tests cover negative coordinates and reversing seams, repeated wraps, all four shaft directions, both orientations, thin sections, overlap, slope/extent/phase/tick changes, complete branch restoration, strict decoding and finite workload limits. Signed cover coordinates use explicit floor arithmetic in the shader.

The new field remains on the original finite Klein graph. Directional 2D sections are implemented; distinct three-dimensional cone/pyramid volumes and arbitrary graph production are not established by these results.

Same individual and Wv2 recovery

MEASURED CONTINUATION | ORIGINAL CONTEXT | VERSIONED ADMISSION

Geometry changes preserve the owner

REPAIR reaches GROWTH_PENDING. One later fresh frame admits the complete new field, debits the declared cost and retains the owner's node, phase and orientation. Its signed-distance lane and next target change under the new geometry. The hypothetical grammar cursor never replaces the owner. Observation, planning and FIFO state reset; global cycle, energy and history continue.

One- and two-epoch missions, mirrored execution, zero-epoch completion, pending search and original-prefix historical samples agree across CPU and GPU. Capacity changes and index rebinding preserve the complete archive at the same planning quantum. Six fresh-process crossovers reconstruct before/after GROW and during generated-world DEFER. Different planning quanta can change original production time and are not claimed equivalent.

Three complete Wv2 lifecycle variants

Witness	Measured result
Default, mirror, capacity eight	Each executes 19 operations, produces 20 records and 41 carrier words, and reaches clock (1,13). CPU and GPU private archives agree.
Original geometry time	GROW remains agent cycle 5 at W operation 10. Replacement cache/executor counters reset without resetting the individual's cycle.
Fresh endpoint continuation	GPU, CPU and GPU processes reopen the retained prefix with distinct indexes, exact latest retries and no duplicate admitted action.
Final default emission	Ordered words 000DCC00060005CC and 000DCC0016000534; energy 69, geometry epoch 1.

Wv2 admits the taper policy under explicit config, protocol and archive versions. Wv1 retains its four prior policies. Both use the same raw carrier/state encoding; a bare record or baseline label cannot establish geometry identity. Admission retains the exact immutable configuration and protocol, and rejects mixed versions before allocating an owner.

Actual-device W emission runs with the host state encoder disabled. Pure complete rejection preserves the usable owner; uncertain device work or save closes it. Recovery resolves the durable prefix, including before/after replacement and latest retry behavior. Physical exactly-once actuation is outside this interface.

Directional evidence and next obligations

REPRODUCIBLE CAPTURE | PRESERVED CONTRACTS | ARCHITECTURAL SCOPE

Evidence belongs to these sources

The capture records every runtime, shader, test, example and independent reference identity, checks the source inventory before and after execution, and requires the complete suite with zero skips. The formal builder verifies report hashes, formal chronology, mission states, cross-backend archives, W lifecycle literals and protected source PDFs before displaying these results.

```
python tools/capture_taper_evidence.py
python -m examples.taper_conformance --output dp-recheck.json
python -m examples.taper_welip_conformance --output ww2-recheck.json
python tools/build_formal_spec.py
```

Capture directory: docs/evidence/directional-v1/. verification.json identifies full-tests.txt, conformance.json and welip-conformance.json. The earlier formal-reference.json, geometry-audit.json and revision-13 preservation record remain historical. This edition adds its own document-preservation audit.

Canonical default agent archive SHA-256: 94b4979c5c5d006b2ef589ffa027475d46d039e04e977cab7ec8245672822c4e

Canonical default Wv2 archive SHA-256: 0e121009810d616adc79e1fd4175e28386cfcfadcf7f7ef709c57b9a855d52d8a

Resource accounting has a defined scope

The actual GPU pipeline records instruction, movement, occupancy and scratch payloads, bounded private stack payload, retained recipe/transcript sizes and old/candidate coexistence. The maximum primitive-site budget bounds charged dispatch work; duplicate sites may project onto the same vertex. These are logical payload counts, not total driver/Python memory or physical energy measurements.

The broader architecture continues

This milestone realizes one source-motivated geometry family and its autonomous continuation. Distinct three-dimensional primitive volumes, more general graph productions, global spectral choices, physical adapters, broader continuation and computational universality still require definitions and evidence. Comparative throughput, texture-cache behavior and physical memory bottlenecks still need controlled hardware measurements.

The original three PDFs and separate ELI5 booklet remain byte-identical, including pink Jitske and green-blue Tom. All prior page bodies 4-97 and numerical contracts are retained; historical counts are never attributed to the new runtime.

Three-dimensional volume contract

APPENDIX MAP | VP1-VP16 | FORMAL ONLY

This appendix binds a genuine third spatial component into the same architecture: quotient topology, solid geometry, signed field, Psi, f8, routing, grammar, one advancing individual and W continuation. It is a proposed numerical realization of the source corpus, with independently checked finite expectations. Runtime implementation has not yet been admitted or measured.

Section	Contract	Page
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Earlier pages retain their original status and evidence dates. The 796-test DP capture on pages 98-100 belongs to commit 2943578; it is not evidence that this new volume runtime exists.

Source scope and versioned profiles

FORMAL ONLY | VP1 | EXPLICIT REALIZATION CHOICES

The original source requires explicit metric, boundary, sign, units and geometry-to-field rules for primitives; Solus says operators possess shape, boundary and volume and illustrates sphere, cone and pyramid sections. Those sections guide this binding but do not uniquely determine it. K times a circle, sampled Euclidean solid membership and graph redistance are the declared choices here.

The original two-dimensional radial phase sweep remains the RP32 phase mechanism. Three-dimensional space does not add a second phase angle. The source discussion of a four-dimensional Klein bottle supplies no defined dimensional-collapse map, so this product construction establishes no such claim.

Role	Exact profile identifier
Agent / packed state	tomigidt-field-volume-plan-act-v1 RP32-relational-sdf-v3
Field / base recipe	relational-sdf-v3 klein-volume-sphere-world-v1
Grammar / generated recipe	klein-volume-organogram-v1 klein-volume-world-v1
Derivation / growth receipt	klein-volume-derivation-v1 klein-volume-growth-v1
Routing / index	hadamard-klein-volume-routing-v1 f8-klein-volume-sdf-v1
Index snapshot / tape	f8-klein-volume-index-snapshot-v1 VP-TAPE32-v1

The agent manifest inherits the common FI/HP admission fields and adds exactly routing and volume. It has no growth, organogram or taper key. The volume binding has exactly format, max_epochs, cost, symbols, axiom, rules, generations and limits. Routing has exactly format and gains; the index binding has exactly format, epoch, psi_sign and phase_origin.

The field manifest retains the common field keys and seams, with topology exactly {kind: "klein-volume", width, height, depth}. Nodes use the canonical VP2 paths. All old integer/type/arity, graph and unit validation remains required. Generated fields use unit 1/1 and u+ for each of the three baseline operator routes. Agent navigation uses all six neighbors.

Every declared key is required; unknown keys, aliases, Boolean/float replacements for integers and implicit profile conversion are rejected. Original source PDFs, earlier profiles and the Tom/Jitske ELI5 booklet remain unchanged.

Cubical quotient and orientation cover

FORMAL ONLY | VP2 | KLEIN TIMES CIRCLE

Let W, H, D be strict integers at least 3, with $N = WHD$ at most 256. Canonical coordinates obey $0 \leq u < W$, $0 \leq v < H$, $0 \leq w < D$; $G = (uH + v)D + w$ and path $v:u:v:w$. Decimal path components have no leading zeros. Coordinates name temporary charts, not a physical external origin.

```
k = floor(X/W); u = X - kW
P(X,Y,Z) = (u, ((-1)^k Y) mod H, Z mod D)
Lift(u,v,w) = (u+mW, (-1)^m v+nH, w+lD)
```

All moduli return nonnegative remainders. The six directional stencil slots are $u+, u-, v+, v-, w+, w-$. Project each unit covering-space step, retaining its seam reflection. Neighbors stored for search are instead numerically sorted by G . These two orders must not be interchanged.

The complex contains every unit covering cube and its projected faces, edges and vertices. Canonicalize cells by their complete projected corner/incidence data: an odd reflection changes a face or cube base differently from a lone vertex. Each occupied subcomplex retains exactly the cells whose corners are all occupied.

```
V=N, E=3N, F=3N, C=N; V-E+F-C=0
Orientation cover: T(2W,H,D), with 2N vertices
(u,v,w,eta) -> (u+eta W, (-1)^eta v, w)
```

Each face has two incident cubes and every vertex link is an octahedral two-sphere. The quotient is connected, closed and nonorientable; its connected double cover is orientable. A reversing horizontal loop establishes nonorientability. The cover has exactly two preimages for each cell, with commuting face maps.

Unit graph metric

```
c_P(x)=min(x mod P, (-x) mod P)
d0=c_(2W)(u2-u1)+c_H(v2-v1)+c_D(w2-w1)
d1=c_(2W)(u2-u1+W)+c_H(v2+v1)+c_D(w2-w1)
d=min(d0,d1)
```

Equivalently compute exact six-neighbor shortest paths. The independent audit checks all 523 admissible dimension triples, including genuine cubical incidence and orientation-cover cells; Euler counts alone are insufficient.

Three distinct sampled solids

FORMAL ONLY | VP3 | FULL VOLUME MEMBERSHIP

For a canonical apex A and the ordered unit frame (e,f,h) of VP4, map local integer coordinates (s,t,r) to the covering point A+s e+t f+r h. Evaluate every admitted local member, project it with P, and take set union. Wrapping may identify several local members; no nearest-lift shortcut is allowed.

```
L = h_raw * 2^scale; B = floor(pL/q)
CONE: 0<=s<=L, |t|,|r|<=B, q^2(t^2+r^2)<=p^2 s^2
PYRAMID: 0<=s<=L, |t|,|r|<=B, q max(|t|,|r|)<=ps
SPHERE: x^2+y^2+z^2<=R^2, |x|,|y|,|z|<=R
```

The cone and square pyramid include their apex, sides and end cap. SPHERE uses all three local components around its center, with effective radius R. It is a sampled Euclidean ball before projection, distinct from the earlier intrinsic graph ball. All three produce discrete occupied vertex sets and induced cubical support.

A literal witness that separates the volumes

Use a 6 by 6 by 6 domain. Put the cone and pyramid apex at (0,3,3), shaft u+, eta=0, L=4 and slope 1/2. Use a radius-2 sphere centered at (3,3,3).

Solid	Occupied / boundary / interior	Complete occupied cubes
Cone	29 / 27 / 2	4
Pyramid	45 / 43 / 2	8
Sphere	33 / 26 / 7	8

The cone layers contain [1,1,5,9,13] vertices; pyramid layers contain [1,1,9,9,25]. Their r=0 axial sections agree, while (s,t,r)=(2,1,1) belongs only to the pyramid. Sphere layers are [1,9,13,9,1]. Its offset (1,1,1) belongs to the radius-2 sphere but not a radius-2 graph ball.

These witnesses distinguish full three-component occupancy and contain complete 3-cells. Thin admitted solids may still have only boundary vertices. The resulting signed field is graph distance to the sampled boundary, not a continuous Euclidean signed-distance function.

Select once, then transport the frame

FORMAL ONLY | VP4 | SHAFT, MIRROR AND DECK ACTION

A primitive reads the prior field and the current grammar STEP pair at its apex. Decode intrinsic phase $t = (-1)^\eta R \bmod 256$ and compute the VP8 guided vector q . Choose the shaft once in that canonical chart. The shaft code order is $u+, v+, w+, u-, v-, w-$; it differs from the six directional stencil slots.

```
e = sigma E_j, sigma in {-1,+1}, j in {0,1,2}
f = (-1)^\eta E_((j+1) mod 3)
h = sigma E_((j+2) mod 3)
det(e,f,h) = (-1)^\eta
```

Score all six cardinal shafts by $q \cdot e$. Take a maximum, breaking ties by shaft-code order rotated by $\text{floor}(6t/256)$. This makes all six axes available even for a zero guided vector. Selection does not use the index phase or the index-only Ψ sign.

Complete mirror and a changed covering chart

The complete packed mirror sends R to $-R$ modulo 256 and flips η , preserving intrinsic t . It therefore preserves the selected shaft e and h , and reverses f . The three solid inequalities are symmetric in each transverse sign, so mirrored descriptors produce identical scalar occupancy and signed fields.

```
D_m = diag(1, (-1)^m, 1)
A -> (u+mW, (-1)^m v+nH, w+lD)
(e,f,h) -> (D_m e, D_m f, D_m h)
```

When changing a deck lift, transport all three existing vectors by D_m . Do not rerun shaft selection or reconstruct the frame from transformed shaft and η . For some shafts that reconstruction changes both transverse axes by a half-turn; symmetric occupancy may coincide even though the ordered frame differs.

Equivalently trace local displacements as unit steps and transport the remaining displacement through each seam. Independent direct projection, transported walks and inverse-lift membership must agree. The scalar field is lift-independent; vector and tensor components obey their declared transport laws.

η remains one bit of reflection parity. This contract supplies an intrinsic cardinal frame for these operators. It does not encode arbitrary three-dimensional attitude, mechanical hinge dynamics, Euler angles or a physical force law.

Union boundary and exact signed field

FORMAL ONLY | VP5 | A GEOMETRIC ADMISSION OBLIGATION

Build the occupied set U by OR-union of every projected primitive membership. Ordered duplicate descriptors remain in the derivation even when their occupied vertices coincide. Reject empty U or U equal to the whole quotient before admission.

```
Z = {x in U : some six-neighbor y is not in U}
sign(x) = 0 on Z, -1 on U minus Z, +1 outside U
phi(x) = sign(x) * d_graph(x,Z)
```

Z is the inner occupied vertex boundary. Since the graph is connected and U is nonempty and proper, Z is nonempty. Every interior-to-outside path crosses Z . Adjacent signs cannot jump directly from -1 to $+1$, and exact unit-graph redistance gives $|\phi(x) - \phi(y)| \leq 1$ on each edge.

Require every signed code in $[-127, 127]$. The zero set equals Z exactly. A thin solid with no negative vertices is permitted; silently replacing its boundary with a different convention is not. Compute union first and then redistance; minimums of individual signed fields are not the specified construction.

The base field follows the same rule

The sphere base recipe projects all integer offsets in its radius cube, tests the full three-dimensional sphere inequality, and obtains its field from U and Z . It requires the same finite preflight and provenance checks. Reusing the old graph-ball radius offset would change the new profile.

A certificate checks omissions as well as additions

Independently reconstruct the prior-stage field and complete grammar trajectory. For each canonical quotient vertex, examine all inverse lifts allowed by a descriptor bound, using separately implemented local-coordinate membership. Compare this complete set with the produced occupancy, then compare Z , signs and exact distances. Verify descriptor order, apex pairs, frame selection, budget and prior-certificate identity.

Distance admission requires the exact zero set, declared signs, edge consistency and, at each nonboundary vertex, a neighbor whose absolute field value is exactly one smaller. These conditions certify exact graph distance on the finite connected domain. A separate all-pairs relaxation in the audit corroborates the literal fields.

Packed mirror/parity checks detect representation faults under their stated rules; they do not prove geometric membership, completeness, physical truth or correctness of arbitrary external observations.

A three-component local Psi operator

FORMAL ONLY | VP6 | TENSOR, SIGN AND DEGENERACY

At each canonical node form signed central differences in the directional stencil order. The scalar field has unit edge bounds, so each component is an integer in $[-2,2]$. This is a local rank-one tensor binding of the source Psi contract; it is not a global graph spectral solution.

```
g = (phi(u+)-phi(u-), phi(v+)-phi(v-),
     phi(w+)-phi(w-))
A = g g^T; lambda = g dot g, 0 <= lambda <= 12
d = gcd(|g_u|, |g_v|, |g_w|)
Psi = g/d if d>0; otherwise Psi=(1,0,0)
```

For g nonzero, $A\Psi = \lambda\Psi$ and Ψ is a primitive integer eigenvector. The selected sign agrees with g . If $g=0$, every direction is an eigenvector for eigenvalue zero; the explicit fallback resolves the degeneracy deterministically. The two-dimensional tensor, lambda bound 8 and two-component zero fallback do not apply to this profile.

Separate the geometric operator from its lookup encoding

An index binding may choose $\text{psi_sign}=+1$ or -1 for the stored eigenvector, and any phase_origin in $[0,255]$. Those choices change index records and lookup order. They cannot change the physical node identity, scalar field, movement cost, branch interpretation, intrinsic owner phase, selected subgoal or a pending search cursor.

```
g -> D_m g; Psi -> D_m Psi; A -> D_m A D_m
lambda is unchanged under the deck action
```

The zero fallback $(1,0,0)$ is unchanged under this reflection. Eigenvalue and sign conventions must be checked at seams and in zero-gradient regions. A complete packed mirror preserves scalar geometry while transporting its related views according to the original mirror contract.

Hadamard means decoded component multiplication

The routing operator multiplies three decoded integer components elementwise, as specified in VP8. It never multiplies packed RP32 words to approximate this operation. Gains, gradients, Psi, packing and phase-bank selection have separate typed domains and exact bounds.

The arithmetic reference retains full fields, g , Ψ , λ , index records, tree walks and routing penalties for the base and generated stages. These are preimplementation expectations, not a measured GPU spectral engine.

Canonical f8 index and record layout

FORMAL ONLY | VP7 | THREE COMPONENTS REQUIRE A NEW ABI

Anchor the geometric BFS at base.center. Distances use all six unit neighbors. The root is its own parent; every other parent is the smallest numerical G at distance one less. Starting from phase_origin, accumulate the departure field-class turn along each parent chain to obtain intrinsic index theta modulo 256.

```
record = (Psi_u+2, Psi_v+2, Psi_w+2,
floor(log2(d_graph+1)), theta, G,
lambda, g_u+2, g_v+2, g_w+2)
key = record[0:6]; physical record = record + (0,0)
tree row = (key[0:6], left, right); NULL = 256
```

Apply the chosen psi_sign to Psi before encoding. The ten semantic components occupy twelve u32 words, with the last two exactly zero. Each tree row occupies eight u32 words. G is slot 5, not the old slot 4. Records are retained by canonical G; sort complete keys lexicographically to construct the tree.

Use the lower-middle key of each interval, emitting root then left and right recursively in preorder. Child fields are row indices or NULL. Since G is part of every key, keys are unique. Binary traversal takes at most N.bit_length() visited rows; actual materialization must traverse and validate the resulting node, rather than treating the index as decoration.

Reindexing is an encoding operation

Binding fields are exactly format, epoch, psi_sign and phase_origin, with the inherited index epoch bounds and strict integer checks. A new epoch/sign/origin rebuilds and certifies records, tree, descriptors and lookup results. It preserves canonical geometry, owner state, observations, FIFO contents and any pending phase-aware search when its semantic context is unchanged.

Payload	Logical bytes
Padded records	48N
Tree rows	32N
Index descriptor	16
Combined index	80N + 16

Snapshots use f8-klein-volume-index-snapshot-v1. Neither a two-component snapshot nor the old five-component key/tree reader may be admitted as this profile. The index is an exact lookup structure over this finite geometry; its name alone establishes no universality theorem.

Eight phase banks and six neighbors

FORMAL ONLY | VP8 | ROUTING AND FINITE SEARCH

Routing gains are exactly eight triples of strict integers in $[-4,4]$, including zero. The defaults below retain a declared cyclic order; arbitrary gains within the bounds are accepted. Decode the owner phase before selecting the bank.

```
default gains = [(1,1,1),(-1,1,1),(-1,-1,1),(1,-1,1),
(1,1,-1),(-1,1,-1),(-1,-1,-1),(1,-1,-1)]
t = (-1)^eta R mod 256; bank = t >> 5
q_i = gain[bank,i] * (1 + Psi_i^2) * g_i
penalty(e) = max_i |q_i| - q dot e
```

Each $|q_i|$ is at most 40; each penalty is in $[0,80]$. Movement slots use numerically sorted canonical neighbors; the appropriate cardinal edge direction supplies e . Shaft selection uses the same q but its own declared code order and phase-dependent tie rule. Index psi_sign cancels in the squared factor.

```
cost(x,t,y) = 1 + |phi(y)| + hazard(y) + penalty(e_xy)
t_next = (t + turn[class(phi(x))]) mod 256
```

This is the inherited FI/HP destination-field cost, not a field-difference cost. Apply departure turns before any seam reflection in packed chart phase; intrinsic t follows the formula above. Mirror/parity repacking and the existing hazard/energy bounds still apply. All edge costs are positive.

The planner retains phase, hops and original context

Search state is (canonical G , intrinsic phase, hops), with at most 255 hops. Compare cost, then the complete tuple of canonical path strings; neighbor enumeration is numerical but path tie-breaking remains lexicographic. The retained target, observation map, epoch, energy and starting pair define the semantic context. Index-only changes do not invalidate it.

Each search quantum consumes the inherited bounded expansion budget. DEFER preserves the exact frontier, best-state map, expansion count and context, without a movement or energy debit. A new admitted input that changes that context resets the cursor. A later successful plan is checked against reserve energy and replayed through the same packed transitions.

The reference includes a quantum-7 mission and index rebuilds between its quanta. Additional DEFER cycles alter original input time and may therefore change later grammar branches. Different quantum schedules are not promised identical global histories.

Parameterized volume grammar

FORMAL ONLY | VP9 | COMPLETE ORIGINAL BRANCH CONTEXT

The profile inherits OG parameter declarations, finite integer expressions, conditions, ordered rule priorities and first-match parallel rewriting, complete derivation addresses and branch paths. Unmatched declared nonterminals follow that contract; undeclared symbols are rejected. Retain the exact rule version and original inputs, rather than regenerating from the current owner state.

Terminal	Arity and interpretation
F	1; raw length 1..256, multiplied by 2 ^{scale} . Each step consumes the prior field turn and six-axis routing.
+ / -	1; phase-turn magnitude 1..16 under the inherited signed-turn rule.
[/]	0; push / pop the complete branch context.
R / SCALE	1; radius 1..127 / scale 0..4. Effective sphere radius must remain 1..127.
SPHERE	0; emit a full sphere at the current cursor; no cursor advance.
CONE / PYRAMID	3; raw height, slope numerator and denominator, each 1..65535; reduced positive slope. Emit without advancing.

PUSH retains both packed views, radius, scale and branch path. POP restores every retained component. The interpreter never rewinds program counter, logical output order or retained history. It stores no additional mechanical heading or roll: the next primitive selects its frame from its current prior-field STEP pair.

Each stage has exact original context {epoch,tick,start_pair,prefix_sha256}: contiguous positive growth epochs, strictly increasing original ticks, the prior EMIT pair including its prior-field B, and the event digest before GROW. Initialize the cursor by repacking that pair as STEP, radius 1, scale 0 and empty branch/stack. Use one unchanged prior field for the entire stage. Rewriting reads only retained inputs and rules.

A sphere descriptor has exactly kind, address, branch_path, center, radius and pair. A cone/pyramid descriptor has exactly kind, address, branch_path, apex, height, numerator, denominator, shaft and pair. Height/radius are effective values; shaft is 0..5; pair is the descriptor-time STEP pair. Ordered duplicates remain significant provenance.

No S or TAPER alias is introduced. A SPHERE token is distinct from the historical intrinsic-ball S token. The existing complete parameter/address expression grammar is retained; the new typed terminal set changes its interpretation under the explicit volume profile.

Typed tape and finite preflight

FORMAL ONLY | VP10 | VP-TAPE32-v1

```
operand bits 0..15; opcode bits 24..27; parity bit 31
reserved mask = 0x70FF0000; total parity is even
0 F; 1 +; 2 -; 3 PUSH; 4 POP; 5 R; 6 SPHERE; 7 SCALE
CONE triple: 8/height, 9/numerator, 10/denominator
PYRAMID triple: 11/height, 12/numerator, 13/denominator
```

Opcodes 14 and 15 are invalid. Reject reserved bits, wrong parity, nonzero operands on zero-arity instructions, standalone slope words, incomplete triples and mixed CONE/PYRAMID triples. The interpreter tracks logical PC separately from texel offset; a three-word solid is one logical instruction.

Limit	Exact admitted domain
generations / max_epochs / cost	0..8 / 0..4 / 1..127
max_symbols / max_steps	1..1024 logical symbols / 1..4096 forward steps
max_primitives / max_stack	1..64 descriptors / 0..32 frames
max_primitive_sites	1..16777216; default 1048576
Tape capacity	At most 1024 logical instructions and 1152 texels

```
L=h_raw << scale; B=floor(pL/q)
cone/pyramid charged sites = (L+1)(2B+1)^2
sphere charged sites = (2R+1)^3
```

Sum charged bounding-box sites over all primitives, including duplicate projections, and require the sum at most max_primitive_sites. Test the complete cube/box budget even if a producer skips excluded cells. Radius 127 charges $255^3=16,581,375$ sites and fits the maximum single-primitive budget; charging N would undercount work.

Require $1 \leq L, pL \leq 2147483647$, $0 \leq qB \leq 2147483647$, and $P-1+L+B \leq 2147483647$ for each dimension P in {W,H,D}. Sphere coordinates obey $P-1+R \leq$ that bound. Before base owner/device allocation, preflight all retained stages: expansions, tape syntax, stack/primitive counts, operands and work. Certify produced descriptors and fields before owner installation.

These bounds reject before admission and avoid INT_MIN in lifted arithmetic. They do not authorize clamping, truncation or silently narrowing slope parameters to fit native GPU multiplication. Full and empty projected sets remain geometric rejection cases after bounded construction.

Exact wide cone membership arithmetic

FORMAL ONLY | VP11 | FOUR BASE-65536 LIMBS

Preflight bounds each of $q|t|$, $q|r|$ and ps by $2^{31}-1$. Their squares can exceed $u32$ even for very small admitted solids. The cone predicate shall therefore compare exact nonnegative multiword integers:

```
(q|t|)^2 + (q|r|)^2 <= (ps)^2
0 <= each square input <= 2147483647
maximum two-square sum = 2(2147483647)^2 < 2^63
```

Use four unsigned base-65536 limbs in least-significant-first order, or an independently equivalent exact implementation. No floating-point approximation or wrapped native- $u32$ square may determine membership. The following square algorithm uses only safe $u32$ intermediate operations.

```
x = a + 65536 b, 0<=a<=65535, 0<=b<=32767
lo = a*a
cross = 2*a*b + (lo >> 16)
upper = b*b + (cross >> 16)
square(x) = [lo & 65535, cross & 65535,
upper & 65535, upper >> 16]
```

$a*a$ is at most 4,294,836,225. $cross$ is at most 4,294,836,224, below 2^{32} ; $upper$ is below 2^{30} . The four limbs therefore encode the complete square without overflow. Add two squares limbwise with carry: each temporary digit is at most 131071. Compare limbs from most significant to least significant. Retain all four digits.

An admitted overflow witness

With raw height 2, scale 0 and reduced slope 65535/65534, the cone charges only 75 sites. The independent audit retains 40 membership decisions that a wrapped native- $u32$ implementation gets wrong. Rejecting only enormous work budgets does not remove this arithmetic obligation.

Pyramid membership needs only the already bounded $q \max(|t|, |r|)$ and ps products. For a sphere, $R \leq 127$ bounds the sum of three local squared coordinates by 48,387. Negative covering coordinates still require explicit floor division and positive remainder; a language remainder operator with different negative semantics must be adapted.

The mathematical generator and a separate audit compare limb squares, sums and geometric predicates with unbounded integer arithmetic. Runtime acceptance additionally requires actual-device comparisons on both sides of the membership boundary and at these exact operand limits.

Retained recipes and geometric admission

FORMAL ONLY | VP12 | RECONSTRUCTION WITH COMPLETE CONTEXT

The base recipe has exactly format, width, height, depth, center, radius, turns and baseline_id. center is canonical G; radius is 1..127; turns are the three inherited field-class increments. Default values are W=4,H=5,D=3, center=0, radius=1, turns=[11,53,137], baseline_id="klein-volume-world-v1".

A generated recipe has exactly format, base, volume, routing and stages. Its format is klein-volume-world-v1 and its base is the complete typed sphere recipe. Stages retain the original contexts of VP9 in order. All stages share the base dimensions; no two-dimensional archive is implicitly lifted into this geometry.

The growth receipt uses klein-volume-growth-v1 and retains from_epoch, to_epoch, mapped_node, target, recipe and derivation_sha256. The hashed derivation has exactly format, context, tape, trace, segments, primitives and final_context, with format klein-volume-derivation-v1. Preflight, encoded words, shaft certificates and union/field diagnostics are outside that document. Canonical JSON/digest rules remain unchanged.

Recursive provenance is part of admission

Begin from the certified sphere base. For each retained stage, reconstruct the original prior field, ordered rewriting, complete branch trajectory and packed pairs; certify every descriptor and the independently recomputed union field. Chain the prior certificate and exact configuration identity. A correct final array with a false derivation is rejected.

Production may enumerate local cover members. Independent admission must use a distinct inverse-lift membership construction and independently certify distances, Psi, index and routing. It must detect missing and extra occupancy, altered phase/eta/shaft, incomplete branch restore, wrong original epoch/tick/prefix, stale prior field and bad padding or typed words.

Historical samples retain their world identity

A retained historical sample after growth includes the admitted GROW in its replay prefix; the original stage context used to create that growth hashes the events before GROW. These are different boundaries. Reconstruct from the correct recipe, version and original observations, then compare the complete packed result and field value.

The producer and certifier share a specification, not a generated-array shortcut. Certificates and source identity make a result checkable within this finite contract. External observations that cannot be derived must still be retained faithfully.

One continuing individual and Wv3

FORMAL ONLY | VP13 | OWNER, ORIGINAL TIME AND NAMESPACE

The existing individual reaches its target and performs REPAIR. If another epoch is allowed, enter GROWTH_PENDING, then admit GROW on a fresh observation cycle. Debit growth cost and increment global cycle and world epoch once. Retain canonical G, R and eta; replace B from the new field and repack both views as STEP.

```
reserve = repair_cost + remaining_epochs * (volume.cost + repair_cost)
```

Choose the next target excluding the owner: minimize |phi|, then maximize graph distance from the owner, then apply the inherited intrinsic-phase tie rule over numerical G. Reset world-specific observations, planning, FIFO and counters, while preserving global energy, cycle and history. Initial creation remains distinct from this same-owner growth transition.

Namespace	Admitted volume identity
Configuration	welip-field-config-v3
Protocol	welip-field-agent-v3
Session/archive	welip-field-session-v3
State word profile	RP32-relational-sdf-v3

Wv3 admits the volume policy only. Wv1 retains its four older policies and Wv2 retains taper. Validate protocol plus exact configuration before owner allocation. baseline_id alone is merely a label; even different depth-3 and depth-4 configurations can produce identical initial state records. The independent reference retains that collision witness.

Raw LUS carrier and binary state layout are unchanged. A state record keeps word_profile=welip-16-16-32-v1; its payload_profile is RP32-relational-sdf-v3. The agent configuration word_profile is also RP32-relational-sdf-v3. Preserve all five W operations, operation order, clock carry, latest-only identical retry, durable receipt behavior and fresh-process recovery under the preceding W contract.

Field controls preserve admitted history. Recovery reconstructs the latest committed owner without rewinding admitted cycles. Once initialized, the GPU owner must advance and emit canonical state words on-device; host reseeding and host state-encoder fallback are forbidden in actual-device acceptance.

The formal Wv3 projection includes normal, complete-mirror and capacity-8 lifecycles. These establish expected arithmetic records and owner transitions; a future runtime must still prove durable CPU/GPU and fresh-process conformance to them.

Device construction and logical payloads

FORMAL ONLY | VP14 | PRODUCTION, VALIDATION AND OWNERSHIP

The declared host grammar front end may rewrite and encode the exact typed tape. The GPU shall validate and interpret that uploaded tape, produce the complete branch trace/descriptors, project occupancy, derive boundary/signs, compute distances, and construct Psi, index, routing and movement textures. Independently certify the candidate before installing it into the same owner; keep old and candidate worlds alive until admission.

Payload	Layout / logical bytes
Host geometry	8 u32/node: six directional neighbors, parent, depth; 32N
Device geometry	16 u32/node: six sorted + six directional neighbors, parent, depth, seam_flags, reserved zero; 64N
Index records/tree/descriptor	12-word record + 8-word tree row; 80N+16
Routing penalties / increments	48 u32 penalties + one u32 increment per node; 196N
Neighbor / gain arrays	6N u32 neighbors = 24N; 24 integer gains = 96
Movement/cost atlas	36 by 8N r32uint texels; 1152N bytes
Primitive movement / base operator	6 by N in shaft order = 24N; 3 by N = 12N

Device-geometry fields occur in the table order. seam_flags bits 0..5 follow sorted neighbors: set a bit exactly when that edge crosses the reversing u seam; bits 6..31 are zero. Directional slots follow VP2; primitive movement follows VP4. The atlas holds six neighbors, three field classes and two words across eight banks. Use the new index strides and node offset.

Account separately for the $4N^2$ -byte weight matrix, distance ping-pong, sign/occupancy arrays, typed tape, traces, descriptors, bounded private branch stacks, owner/search state, retained recipes, status/readback padding and simultaneous old/candidate resources. Logical payload accounting is not a total allocation or performance measurement.

The CPU implementation and GPU producers must be independently exercised against the frozen reference. Actual-device conformance shall disable CPU geometry, field, Psi/index, routing and state-emission producers as applicable. Host certification may inspect readbacks, but cannot silently replace device production.

A failed bounded construction, certificate or allocation cannot partially admit an epoch or reseed the owner. Preserve the preceding fault and retry semantics and measure their implementation. Texture use alone proves neither cache residency nor saturation nor removal of physical memory traffic.

Independent finite expectations

FORMAL ONLY | VP15 | FROZEN NUMERICAL REFERENCE

The volume reference imports no solvefinite module. It uses separately pinned earlier mathematical helpers for unchanged grammar/carrier rules. A second audit imports neither the volume producer nor runtime and independently reconstructs eight literal fields. The formal builder compares their complete occupancy, boundary, signs and distance arrays.

Mission	Cycles / energy	Final pair
Default	10 / 170	9600143F860014C1
Complete mirror	10 / 170	860014C19600143F
Two epochs	18 / 158	16002BAD06002B53
Zero epochs	4 / 185	860236C39602363D
Quantum 7	28 / 174	060020C11600203F

Default initial state: base 4 by 5 by 3, energy 200, phase 250, eta=0, target G=54 (v:3:3:0). In epoch zero, hazard 70 appears at G=6 from cycle 2. First growth is cycle 5. The stage has 19 logical instructions, 23 texels, 7 forward steps, 3 primitives, 90 charged sites and stack depth 2.

The first stage emits a height-3 slope-1/2 cone with shaft v-, a height-2 slope-1/2 pyramid with shaft v+, and a radius-1 sphere. In the reference, the owner moves along the third coordinate at cycle 8, from v:0:1:0 to v:0:1:2. A second-stage pyramid uses shaft w+. Default and second-growth targets are G=20 and G=43.

The quantum-7 mission retains 18 DEFER events and grows at original cycle 19. Rebuilding the index between its quanta preserves the exact reference history. Each of the three Wv3 lifecycles has 18 operations, 19 records and 39 carrier fragments, ending at clock epoch 1, tick 12.

The separate geometry audit passes 20 checks: all 523 admissible dimensions, 91,290 octahedral vertex links, 1,460,640 cover cells, 480 primitive cases, 111,240 transported walks, 2,400 deck/inverse-lift comparisons, 6,231 wide geometric comparisons and 10,108 preflight candidates.

Retained files: docs/evidence/volume-v1/reference-builder.py, formal-reference.json, geometry-audit.py and geometry-audit.json. Builder guards pin exact hashes. Mathematical event-prefix and W operation digests are reference identities, not measured runtime archive identities.

Acceptance and architecture maturity

FORMAL ONLY | VP16 | WHAT IS DEFINED, EXECUTED AND OPEN

Layer	Status at this edition
Architectural core	Integrated definitions for packed state, self-reference, relational geometry, derivation, one individual and conditional invariant preservation.
Existing finite realization	Implemented CPU/GPU field execution, Klein transport, local Psi/f8, Hadamard routing, planning, regeneration, branching, directional sections and Wv2 continuation.
New full 3D binding	VP1-VP16 defined with independent arithmetic and geometry audits. Runtime, device production and durable Wv3 execution remain required.
Broader research claims	Universality, global spectral choices, general graph production, physical adapters and comparative hardware advantages remain open.

Required before calling VP implemented

Implement strict typed admissions and all new strides/profiles; independently produce the full volumes and certify complete provenance; match every literal field, descriptor, tape, mission and W record; exercise all six shafts and both mirrors, seam/deck transport, degenerate Psi, reindexing, budget edges and wide arithmetic on the actual GPU.

Verify same-owner growth and subgoal selection, pending searches, eviction/regeneration, all W operations, exact retries, clock carry, malformed/wrong-profile rejection before allocation, atomic failure and fresh CPU-to-GPU/GPU-to-CPU recovery. Disable CPU production fallbacks in device checks. Run the complete existing regression suite and capture the exact source inventory before and after.

How to describe the paradigm at this stage

An experimental deterministic execution architecture in which one autonomous individual carries packed relational state, follows state-dependent geometric operators, generates its world through retained rules, and reconstructs derived state from complete original context. It is more specific than a concept sketch: substantial finite profiles are executable and tested.

The formalization is an evolving mathematical specification with independently audited numerical realizations. The 796-test capture is evidence for its declared implementation, not a machine-checked proof of the whole architecture. A bounded 256-node profile does not establish unrestricted computational universality.

“Frontier” describes the research ambition here; priority and comparative novelty require a separate prior-art assessment. “Infallible” remains conditional on the stated model, admitted inputs and faithful execution. GPU locality must still be measured against optimized baselines before claiming bottleneck elimination, cache saturation, energy advantage or broad superiority.